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Review

Advances in Flow–Structure Interaction and Multiphysics Applications: An Immersed Boundary Perspective

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Abstract

This article discusses contemporary strategies to deal with immersed boundary (IB) frameworks useful for analyzing flow–structure interaction in complex settings. It focuses on immense advancements in various fields: biology, oscillation of structures due to fluid flow, deformable materials, thermal processes, settling particles, multiphase systems, and sound propagation. The discussion also involves a review of techniques addressing moving boundary conditions at complex interfaces. Evaluating practical examples and theoretical challenges that have been addressed by these frameworks are another focus of the article. Important results highlight the integration of IB methods with adaptive mesh refinement and high-order accuracy techniques, which enormously improve computational efficiency and precision in modeling complex solid–fluid interactions. The article also describes the evolution of IB methodologies in tackling problems of energy harvesting, bio-inspiration propulsion, and thermal-fluid coupling, which extends IB methodologies broadly in many scientific and industrial areas. More importantly, by bringing together different insights and paradigms from across disciplines, the study highlights the emerging trends in IB methodologies towards solving some of the most intricate challenges within the technical and scientific domains.

Keywords: immersed boundary; flow–structure interaction; vortex-induced vibration; multiphase flow; thermal process; sound acoustics

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1. Introduction

This review deals with the historical progression of the immersed boundary method (IBM), tracing its evolution from early theoretical frameworks to the latest developments and diverse applications. With the ability to model complex fluid–structure interactions, IB methods have undergone significant refinements, including high-order accuracy schemes, adaptive mesh refinement, and hybrid approaches elaborately defining computational fluid dynamics with the aid of machine learning algorithms. These schemes helped to introduce different boundary conditions and extend their applicability across various fields ranging

from biology to engineering. Presently, simulations of IB models serve to design blood flow and analyze cellular mechanics in biological systems, whereas applications in engineering range from an assessment of vortex-induced vibration (VIV) over flexible body interaction all the way to multiphase fluid dynamic computations.

Over the last two decades, several comprehensive reviews have explored various aspects of the immersed boundary (IB) method, shedding light on its advancements and applications. Iaccarino and Verzicco (2003) [1] examined the role of IB techniques in turbulent flow simulations, emphasizing their effectiveness in handling complex geometries within turbulence modeling. Mittal and Iaccarino (2005) [2] provided a broad overview of IB methods, tracing their evolution and highlighting their significance in fluid dynamics, particularly in scenarios involving moving or deforming boundaries. Sotiropoulos and Yang (2013) [3] focused on the application of IB methods in fluid–structure interaction (FSI), particularly in aerospace engineering, addressing the challenges of accurately modeling interactions between fluids and flexible structures. Kim and Choi (2019) [4] presented a detailed analysis of advancements in IB methods for FSI, encompassing a wide range of applications from biological systems to engineering solutions. Griffith and Patankar (2019) [5] explored recent developments in IB methods for FSI, emphasizing computational efficiency improvements and their application in biological and engineering contexts. Additionally, Roy et al.'s (2020) [6] book offered an extensive discussion on IB methods, covering their theoretical foundations, numerical approaches, and applications across diverse fields, including engineering and biology. Mittal and Bhardwaj (2022) [7] concentrated on the role of IB methods in thermo-fluid simulations, particularly in heat transfer and fluid dynamics. More recently, Verzicco (2022) [8] provided a historical overview of IB methods, tracing their progression over time and discussing future directions, particularly in advancing computational techniques and expanding their applicability in complex fluid dynamics.

Despite these protracted and extensive studies, none of the studies addressed developments post-2020 in a systematic and consolidated manner, especially in terms of comparative analysis of performance across different IB approaches. This review distinctly contributes by analyzing the latest innovations, bridging the gap between practical applications and theoretical insight, and examining methodological advances in a multidisciplinary approach. In contradiction to previous studies, the emphasis here is evolving perspectives on boundary representation, solver stability, and real-time coupling frameworks. In the present review, 140 articles published after 2020 are analyzed. The literature on the immersed boundary method (IBM) highlights a notable gap in the comparative assessment and optimization of sharp versus diffuse boundary treatments in relation to fluid solvers, particularly concerning their performance in biologically relevant flows, flexible body interactions, heat transfer, sound acoustics, and particle sedimentation applications. While both sharp and diffuse boundary treatments have been explored independently in various contexts, there is a lack of systematic studies evaluating their relative strengths and limitations in these specialized domains. For instance, in biological flows and flexible body interactions, the choice between sharp and diffuse interfaces may significantly influence the accuracy of fluid–structure coupling, yet a comprehensive analysis of their effects on deformation dynamics and force transmission remains underexplored. Similarly, in heat transfer problems, the treatment of thermal boundaries—whether sharply resolved or diffusely smoothed—can critically impact temperature gradients and convective heat transfer predictions, but few studies have rigorously compared these approaches. In acoustics, the numerical treatment of boundaries can introduce spurious reflections or damping, yet the optimal boundary representation for sound wave propagation in IBM frameworks has not been thoroughly established. Furthermore, in particle-laden flows and sedimentation studies, the interplay between boundary sharpness and particle–wall interactions—such

as adhesion, rebound, or near-wall clustering—lacks detailed investigation, despite its importance in industrial and environmental applications. The absence of generalized guidelines for selecting boundary treatments across these diverse applications limits the broader adoption of IBM, particularly in interdisciplinary research, where accuracy and computational efficiency are paramount. Addressing this gap requires a unified evaluation framework that bridges theoretical insights with practical validation across multiple physical scenarios, ultimately enhancing the reliability and versatility of IBM in complex, real-world simulations.

The present review discusses the immersed boundary method (IBM) in applications like biology, vortex-induced vibration, particle sedimentation, sound acoustics, and multiphase flow. This is important because these areas involve complex fluid–structure interactions and deforming boundaries, where IBM's ability to handle stability, accuracy, and efficiency enables high-fidelity simulations. Moreover, advancements in IBM have expanded its applicability to cutting-edge fields like bio-inspired propulsion and fluid–acoustic coupling, demonstrating its versatility in addressing real-world challenges across research and industry. These applications highlight IBM's critical role in modeling intricate systems where traditional methods may struggle, making it indispensable for advancing scientific and engineering solutions. This review therefore emphasizes potential future directions such as multiscale IBM coupling strategies, optimization driven by data of interface models, and robust benchmarking protocols, which are currently lacking in the literature. These contributions aim to improve IBM's applicability in arriving at solutions for next-generation challenges in engineering and science. By tightly linking the original basis with the latest insights, the review gives a multidisciplinary interpretation, providing researchers and practitioners with the set of tools necessary to employ advanced IB techniques in solving more complex problems in science and engineering.

Literature Search Strategy and Selection Criteria

To ensure and maintain methodological transparency and mitigate selection bias, an organized and systematic literature search was conducted using major academic databases, including Scopus, Elsevier, and Google Scholar. The search encompassed articles published from January 2020 to December 2024, with the primary focus on recent developments in the immersed boundary method (IBM). Keywords used included: immersed boundary method, fluid–structure interaction, diffuse interface, sharp interface, adaptive mesh, bio-inspired propulsion, heat transfer, sound acoustics, particle sedimentation, and combinations thereof.

A total of 482 articles were initially taken into consideration. After elimination of replicas and initial screening based on titles and abstracts, 243 records were evaluated for full-text eligibility. Out of 243, 140 articles were filtered out and were included in the final review based on the following inclusion criteria:

1. Peer-reviewed journal studies or high-impact conference papers.
2. Clear alignment with the advancements in IBM post-2020.
3. Incorporation of original numerical or methodological contributions.
4. Application of IBM in at least one complex fluid–structure or multi-physics context.

Exclusion criteria comprised:

1. Editorials, short communications, or non-technical reviews.
2. Articles focused purely on meshless or ALE methods without relevance to IBM.
3. Studies using IBM terminology vaguely.

The article selection process is illustrated in a PRISMA-style flowchart in Figure 1.

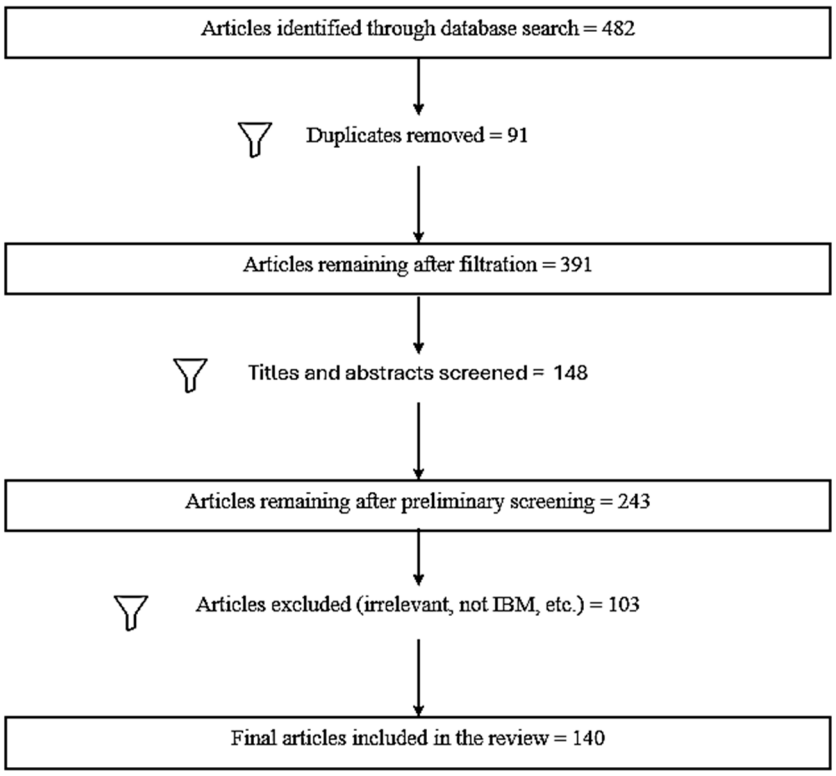


Figure 1. PRISMA-style flowchart illustrating article selection process for the review.

2. Overview of Numerical Framework

A simple overview of the implementation of the IBM is provided. The equations governing the fluid are given by a Navier–Stokes (N–S) equation (Equation (1)) and continuity equation (Equation (2)).

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla \boldsymbol{u})\right) = -\nabla p + \mu \nabla^2 \boldsymbol{u} + \boldsymbol{f} \tag{1}$$

$$\nabla \cdot \boldsymbol{u} = 0 \tag{2}$$

where:

u (**x**, **t**) is the fluid velocity (m/s) at position x (m) and time t (s). p (x, t) is pressure in Pa, ρ is density (kg/m^3), μ is dynamic viscosity (Pa. s), and **f** (**x**, **t**) is the body force per unit volume, which represents the presence of immersed structure (N/m^3).

The immersed structure is often represented as a set of discrete points, typically denoted **X** (**s**, **t**), where s is the Lagrangian coordinate of the structure. The motion of these points is governed by Newton's laws of motion (Equation (3)),

$$\rho_s \frac{\partial^2 \boldsymbol{X}}{\partial t^2} = \boldsymbol{F}_s + \boldsymbol{F}_e \tag{3}$$

where:

The structural density is given by ρ_s in (kg/m^3), $\mathbf{F_s}$ is the restoring elastic force of the structure in (N), and $\mathbf{F_e}$ is the external forces, including the fluid force acting on the structure in (N).

The fluid–structure interactions are enforced by velocity interpolation (Equation (4)) and force distribution (Equation (5)).

$$\boldsymbol{U}(\boldsymbol{s}, \boldsymbol{t}) = \int \boldsymbol{u}(\boldsymbol{x}, \boldsymbol{t}) \boldsymbol{\delta}(\boldsymbol{x} - \boldsymbol{X}(\boldsymbol{s}, \boldsymbol{t})) \boldsymbol{dx} \tag{4}$$

$$f(x,t) = \int F(s,t)\delta(x - X(s,t))ds \quad (5)$$

where:

U (**s**, **t**) is the Lagrangian velocity (m/s) and **F** (**s**, **t**) is the structural force (N).

For each time step:

Step 1: Compute structural forces $F(s,t)$ based on elasticity and external loads.

Step 2: Spread structural force to fluid grid: $f(x,t)$ via Equation (5).

Step 3: Solve Navier–Stokes equations to update u and p using Equations (1) and (2).

Step 4: Interpolate fluid velocity at Lagrangian points using Equation (4).

Step 5: Update structure positions X (s, t) using Newton's law (Equation (3)).

Step 6: Repeat for the next time step.

Force distribution and velocity interpolation are carried out using the Dirac delta function δ. In the IB method, the structure is modeled using a Lagrangian mesh and the fluid with Eulerian mesh. The N–S equations are solved on a uniform Cartesian grid using finite difference, finite volume, or spectral methods. The IB method is executed in a time-step framework. The fluid velocity and pressure are advanced in time using the N–S equations. The structural forces are determined, and the position of the IB is updated. Coupling is enforced by computing the interaction forces (f) and velocity interpolation (**U**) between the fluid and structure.

The immersed boundary (IB) method offers significant advantages over techniques such as the arbitrary Lagrangian–Eulerian (ALE) method, particularly for problems involving complex, moving, or deformable boundaries. Unlike ALE, which requires continuous mesh regeneration or deformation to track moving interfaces, the IB method uses a fixed Eulerian grid with Lagrangian markers, eliminating remeshing and simplifying computations. This makes it especially suitable for simulations with large structural deformations or fluid–structure interactions. IB methods are also more adaptable to topological changes, such as merging or splitting of interfaces, which are difficult to handle with ALE. Overall, IB methods provide greater flexibility and computational efficiency (Mittal and Iaccarino, 2005) [2].

Comparison Between Diffuse-Interface and Sharp-Interface IB

A comprehensive comparison between diffuse and sharp-interface methods used in fluid simulations is presented in Table 1 and a boundary interface shown in Figure 2. Diffuse-interface methods represent the interface as a smeared region with finite thickness, using phase-field, level-set, or VOF (volume-of-fluid) techniques (Figure 2a). These methods modify the governing equations (Equation (6)) over a narrow band and apply distributed body forces to handle interfacial dynamics. Kanchan and Maniyeri (2019) [9] demonstrated the accuracy of diffuse-interface immersed boundary method through benchmark studies that compared filament behavior across different orbit regimes—rigid, springy, C-shape, and complex—based on bending rigidity and viscous flow forcing (VFF). The orbit regimes were categorized as: rigid (VFF < 0.16), springy (0.16 < VFF < 1.31), C-shape (1.31 < VFF < 10.67), and complex (VFF > 10.67). The results closely matched the experimental data of Forgacs and Mason (1959) [10] and numerical results of Stockie and Green (1998) [11], validating the method's ability to accurately capture filament deformation and orbit transitions across a wide range of flow conditions. The diffuse-interface IBM employs a smoothed transition zone between fluid and solid domains, making it particularly well suited for modeling biological phenomena involving large deformations, intricate geometries, and multi-physics coupling. However, they may suffer from slight mass conservation issues.

Table 1. Comparison between diffuse and sharp-interface immersed boundary methods.

Feature/Aspect	Diffuse-Interface Method	Sharp-Interface Method
Interface Representation	Interface is smeared over a finite thickness (regularized)	Interface is exact, represented as a discontinuity (zero-thickness)
Governing Equation Modification	Equations are modified over a band using indicator (phase) functions	Governing equations remain valid, but interface conditions are sharply enforced
Interface Tracking	Uses phase field, level set, or volume-of-fluid (VOF) methods	Uses body-fitted mesh, ghost-cell, or cut-cell methods
Force Application	Distributed body force (e.g., penalty or smoothing force) over the interface region	Discrete delta function or sharp condition imposition at interface
Continuity Treatment	Allows for continuous transition in material properties like density or viscosity	Material properties jump discontinuously at the interface
Numerical Complexity	Easier to implement on fixed grids (Eulerian); smooth solutions	Requires special treatment for jumps, boundary reconstruction, and interpolation
Mass Conservation	Can suffer from slight mass loss/gain (especially in level-set based methods)	Better mass conservation due to explicit boundary enforcement
Utility	Flows with complex moving interfaces, topological changes (e.g., breakup, coalescence)	Problems requiring high accuracy at interface (e.g., fluid–structure interaction)

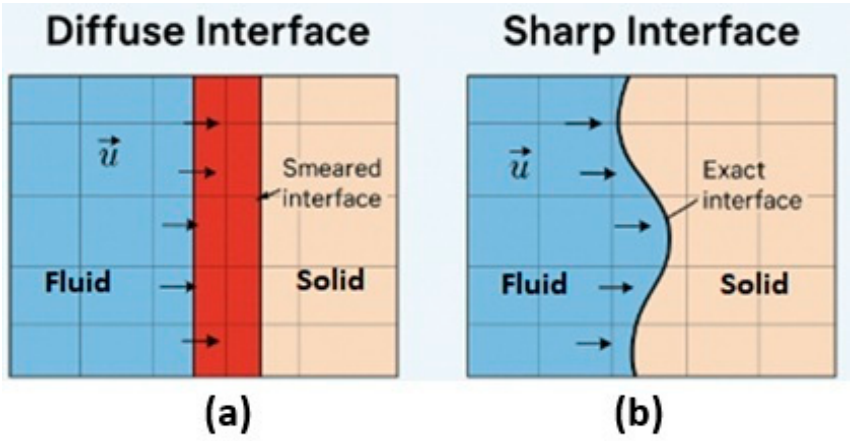


Figure 2. (**a**) Diffuse-interface methods and (**b**) Sharp-interface methods along with respective boundary interfaces.

In contrast, sharp-interface methods treat the interface as a mathematically exact, zero-thickness boundary (Figure 2b), applying sharp jump conditions and discrete forces (Equation (7)). These methods use ghost-cell or cut-cell techniques to explicitly enforce interface conditions, offering better accuracy and mass conservation. While numerically more complex, sharp-interface methods are ideal for problems requiring precise interfacial treatment, such as fluid–structure interaction or multiphase flows with clearly defined boundaries. The sharp-interface immersed boundary (IB) method (Bhardwaj and Mittal, 2012) [12] demonstrates superior performance over ALE-based methods in fluid–structure interaction (FSI), as validated by the Turek–Hron benchmark. It achieves high fidelity with benchmark results, matching the oscillation frequency exactly (f = 0.19) and predicting the plate's Y-tip displacement within ~11% (0.92 D vs. 0.83 D). The method avoids remeshing by using a fixed Cartesian grid (257 × 129) and accurately captures large, nonlinear deformations using a geometrically nonlinear structural model. It remains stable even at low density ratios ($\rho s/\rho f = 1$) and yields a lower mean drag coefficient (3.56 vs. 4.13), indicating reduced unsteady flow forces.

Modified Navier–Stokes equation with body force term:

$$\rho(\phi)\left(\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}\right)=-\nabla p+\nabla\cdot\left[\mu(\phi)\left(\nabla\boldsymbol{u}+(\nabla\boldsymbol{u})^{T}\right]+\boldsymbol{f}_{interface}\right. \quad (6)$$

Standard Navier–Stokes equation with sharp-interface enforcement:

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}\right)=-\nabla p+\mu\nabla^{2}\boldsymbol{u}+\boldsymbol{f} \quad (7)$$

With interface jump condition:
$[[\sigma\cdot\boldsymbol{n}]]=\boldsymbol{f_s}$, where $[[\cdot]]$ denotes a jump across the interface and f_s is surface force.

3. Flow–Structure Interaction Application

3.1. Biophysics and Biological Flows

IBM is extensively applied in biology to model interactions of flexible structures with surrounding fluids, such as blood flow around heart valves or the movement of cells in viscous environments. Thus, it provides the methodology of choice for the simulation of complex physiological processes like cardiovascular function, biofilm formation, and microorganism locomotion by tightly coupling deforming biological structures with incompressible fluid dynamics. Let us now consider the implications of IBM boundary treatments in this application area. Table 2 summarizes recent advancements in IBM boundary treatment with respect to fluid solvers used and applications considered.

Table 2. Recent advancements in IBM considering boundary treatment and corresponding fluid–structural solvers for biological applications.

Reference	Flow–Structural Solver	Boundary Treatment	Summary	Application Area	Year
Griffith et al., 2007 [13]	FDM	Diffuse	Adaptive, second-order accurate IBM for heart valve simulations	Cardiovascular (heart valves)	2007
Heys et al., 2008 [14]	FDM	Diffuse	Models filiform hair motion with viscous coupling	Arthropod sensory systems	2008
Kim and Lai, 2010 [15]	FDM	Diffuse	Simulates inextensible vesicles in fluid flow	Vesicle dynamics	2010
Maniyeri et al., 2012 [16]	FVM	Diffuse	Bacterial flagellum propulsion in viscous fluid	Microorganism locomotion	2012
Maniyeri and Kang, 2014 [17]	FVM	Diffuse	Bacterial flagellar bundling and tumbling	Microorganism locomotion	2014
De Rosis, 2014 [18]	LBM	Diffuse	Tandem flapping wings with phase difference effects	Insect flight dynamics	2014
Battista et al., 2018 [19]	FDM	Diffuse	Implementation of IBM as 2D software	Biological flows	2018
Fai and Rycroft, 2018 [20]	FEM	Diffuse	Improved accuracy for thin fluid layers in FSI	Vesicle migration in narrow channels	2018
Kanchan and Maniyeri, 2019 [9]	FVM	Diffuse	Buckling and recuperation of flexible filaments in shear flow	Diatom chain dynamics	2019
Kanchan and Maniyeri, 2020 [21]	FVM	Diffuse	Asymmetric filament deformation in oscillatory flow	Flexible filament dynamics	2020
Kanchan and Maniyeri, 2020 [22]	FVM	Diffuse	Self-excited oscillation of filaments in channel flow	Fluid–structure interaction	2020

Table 2. *Cont.*

Reference	Flow–Structural Solver	Boundary Treatment	Summary	Application Area	Year
Kanchan and Maniyeri, 2020 [23]	FVM	Diffuse	Multiple-filament interaction in shear flow	Filament dynamics	2020
Wang et al., 2020 [24]	FDM	Diffuse	Self-propelled flexible plate with Navier slip	Bio-inspired propulsion	2020
Meng et al., 2020 [25]	LBM	Diffuse	Dendritic growth and motion in convection	Solidification dynamics	2020
Delong et al., 2014 [26]	FVM	Diffuse	Fluctuating immersed boundary for Brownian dynamics	Static and Dynamic particle interaction	2014
Coclite et al., 2020 [27]	LBM	Diffuse	Dynamic IBM for rigid/deformable objects in 3D flow	General FSI	2020
Ong et al., 2021 [28]	FDM	Diffuse	Inextensible vesicles in 2D Stokes flow	Vesicle dynamics	2021
Ghosh, 2021 [29]	FDM	Diffuse	Fluid-induced deformation of cantilever beams	Biofilm-fluid interaction	2021
Casquero et al., 2021 [30]	FEM	Diffuse	Capsule/vesicle dynamics with B-splines/T-splines	Biological membranes	2021
Lampropoulos et al., 2021 [31]	FEM	Diffuse	Efficient hemodynamic simulation in aneurysms	Intracranial aneurysm flow	2021
Mirfendereski and Park, 2021 [32]	FDM	Diffuse	Pulsatile flow in stenotic channels	Blood flow in arteries	2021
Eldoe et al., 2022 [33]	FVM	Diffuse	Rigid filament interaction in oscillatory flow	Low-Re filament dynamics	2022
Kassen et al., 2022 [34]	FDM	Diffuse	Cell–cell interactions in whole blood	Blood cell dynamics	2022
Ntetsika and Papadopoulos, 2022 [35]	FDM	Diffuse	Filament dynamics in shear flow with ANN prediction	Flexible filament behavior	2022
Maniyeri, 2022 [36]	FVM	Diffuse	Elastic rod dynamics in fluid flow	Flagellar motion	2022
Zhu et al., 2022 [37]	LBM	Diffuse	Tea shoot deformation under negative pressure	Agricultural robotics	2022
Lai and Seol, 2022 [38]	FDM	Diffuse	Stable vesicle dynamics in 3D Navier–Stokes	Vesicle dynamics	2022
Bourantas et al., 2023 [39]	FEM	Diffuse	Efficient blood flow simulation in complex vasculature	Vascular hemodynamics	2023
Kaiser et al., 2023 [40]	FDM	Diffuse	Heart valve hemodynamics compared with 4D flow MRI	Cardiovascular flow	2023
Ladiges et al., 2022 [41]	MAC	Diffuse	Simulation of electrolytes in presence of physical boundaries	Induced charge electro-osmosis	2022
Luo et al., 2008 [42]	FDM	Sharp	Novel IBM for phonation and FSI in biological systems	Vocal fold dynamics	2008
Bhardwaj and Mittal, 2012 [12]	FDM–FEM	Sharp	Coupled IBM–FEM solver for large-scale flow-induced deformation	Flexible structures in flow	2012
Bhardwaj et al., 2013 [43]	FDM–FEM	Sharp	Blast-induced eye deformation using FSI	Biomechanics (eye trauma)	2013
Bailoor et al., 2017 [44]	FDM–FEM	Sharp	Compressible FSI with large deformations (ghost-cell method)	Blast–structure interaction	2017
Bourantas et al., 2021 [45]	FEM	Sharp	Internal flows in complex geometries (blood flow)	Cardiovascular hemodynamics	2021

Table 2. *Cont.*

Reference	Flow–Structural Solver	Boundary Treatment	Summary	Application Area	Year
Brown et al., 2022 [46]	FDM	Sharp	TAVR device simulation with patient-specific anatomy	Transcatheter aortic valve replacement	2022
Wang et al., 2022 [47]	FEM	Sharp (moving)	Heart valve flow simulation with hybrid method	Cardiovascular flow	2022
Singh and Kumar, 2023 [48]	FVM	Sharp	Tumor morphology modeling without body-conformal grids	Bioheat transfer in tumors	2023

Originally developed by Peskin for simulating blood flow in the heart, IBM has since evolved into a versatile tool for studying cardiovascular dynamics, cellular mechanics, microbial locomotion, and tissue-level biomechanics. Its ability to seamlessly handle moving boundaries while maintaining computational efficiency has made it indispensable in biomedical research, enabling high-fidelity simulations of heart valves, bacterial flagella, biofilms, and tumor growth.

In cardiovascular applications, diffuse IBM has enabled unprecedented simulations of heart valve dynamics and blood flow phenomena. Griffith et al. (2019) [5] established an adaptive second-order accurate IBM framework that successfully captured the complex interplay between blood flow and flexible valve leaflets, setting a benchmark for cardiac simulations. This work has been extended by Kaiser et al. (2023) [40], who validated diffuse IBM simulations of heart valve hemodynamics against 4D flow MRI measurements, demonstrating the method's clinical relevance. For vascular flows, Lampropoulos et al. (2021) [31] and Bourantas et al. (2023) [39] developed efficient finite element implementations of diffuse IBM that accurately simulated blood flow in aneurysms and complex vasculature while maintaining computational tractability for clinically relevant geometries. These implementations highlight diffuse IBM's ability to handle large deformations and complex boundary motions, making it indispensable for cardiovascular research. Nonetheless, this indispensability should be viewed in context, especially when relative to body-fitting methods such as the finite element method (FEM), which have exhibited strengths in clinical modeling. For instance, Griffith (2011) [49] demonstrated an ALE-based FEM framework for simulating patient-specific vascular mechanics, delivering accurate wall stress distributions vital in aneurysm evaluations. Whilst diffuse IBM is versatile and resilient under moving boundaries and large deformations, it can be prone to interface diffusion, influencing stress localization accuracy (Wang et al., 2012) [50]. In contrast, body-fitted FEM facilitates sharper interface resolution, which is beneficial in contexts like aortic dissection modeling (Kheradvar et al., 2014) [51]. However, IBM possesses a practical edge in dealing with complex moving geometries, like heart valves, with reduced complexity of mesh generation and computational stability (Lee et al., 2020) [52]. Hybrid strategies that integrate immersed fluid solvers with solid mechanics formulated using the finite element method (FEM) have also been put forward to harness the advantages of both paradigms in clinical simulations (Gao et al., 2017) [53].

The study of cellular-scale biological phenomena has particularly benefited from diffuse IBM implementations. Maniyeri et al. (2012) [16] and Maniyeri and Kang (2014) [17] pioneered the application of diffuse IBM to bacterial flagellum dynamics, capturing both propulsion mechanisms and the intricate bundling behavior of multiple flagella. Finite volume method (FVM)-based simulations revealed fundamental insights into microorganism locomotion that would be challenging to obtain experimentally. At smaller scales still, Delong et al. (2014) [26] implemented a fluctuating immersed boundary approach for Brownian dynamic simulations, enabling the study of particle interactions in biologically

relevant conditions. Kassen et al. (2022) [34] later extended these capabilities to simulate cell–cell interactions in whole blood, demonstrating diffuse IBM's versatility across multiple biological length scales.

Flexible filament and vesicle dynamics represent another area where diffuse IBM has made significant contributions. Kim and Lai (2010) [15] and Ong et al. (2021) [28] developed finite difference method (FDM)-based approaches for simulating inextensible vesicles in fluid flow, providing fundamental insights into membrane deformation physics. Kanchan and Maniyeri (2019) [9] conducted a comprehensive series of FVM-based studies on flexible filament dynamics. Various deformations or orbit classes were studied, including buckling and recuperation dynamics, as illustrated in Figures 3–5. The physics of the problem involved a "flexible structure which was massless and neutrally buoyant, suspended in a viscous incompressible fluid medium." The numerical strategy demonstrated here was based on the formal second-order IBM scheme provided in studies conducted by Kanchan and Maniyeri (2019) [9]. Further, this methodology was extended to analyze "asymmetric filament deformation in oscillatory flow" (Kanchan and Maniyeri, 2020) [21], "self-excited oscillation of flexible filament placed in channel flow" (Kanchan and Maniyeri, 2020) [22], and "multiple-filament interaction in shear flow" (Kanchan and Maniyeri, 2020) [23].

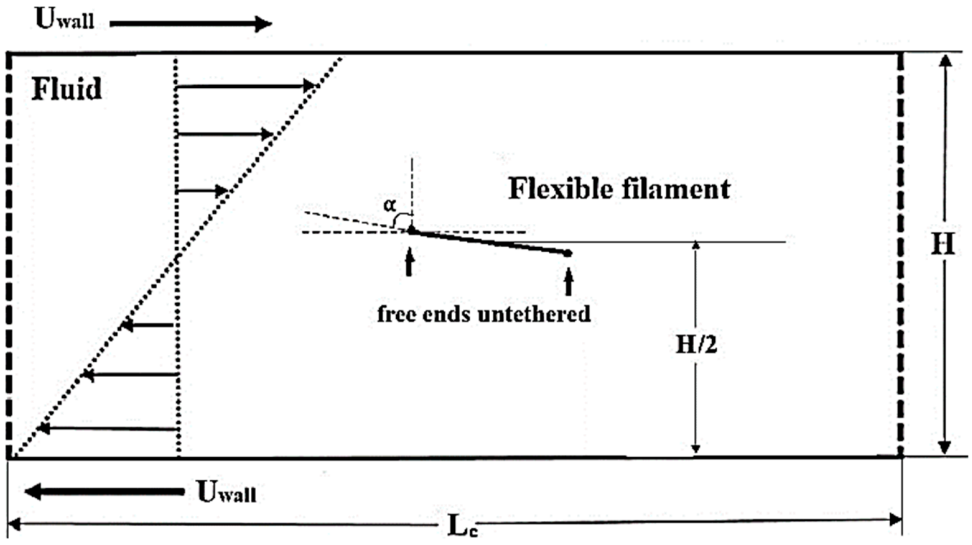


Figure 3. Schematic illustration of a flexible filament placed freely in the channel mid-plane, subjected to viscous shear flow. The filament is initially inclined at an angle of 5° (α) to induce asymmetry in its motion. Non-dimensional parameters include filament length (L), shear rate (K), and bending rigidity (K_b) (Kanchan and Maniyeri, 2019) [9].

These works have collectively advanced understanding of diatom chain dynamics and similar biological systems. Lai and Seol (2022) [38] made advancements to the "immersed boundary method by simulating 3D triangulated vesicles in unsteady Navier–Stokes flow," as illustrated in Figures 6–8. All these illustrate the versatility of IB techniques in accurately modeling FSI across a wide range of applications, encompassing cardiovascular interventions and vesicle dynamics.

The lattice Boltzmann method (LBM) has proven to be particularly synergistic with diffuse IBM for various biological applications. De Rosis (2014) [18] employed LBM-based diffuse IBM to study tandem flapping wings, revealing phase difference effects relevant to insect flight dynamics. Coclite et al. (2020) [27] developed a dynamic IBM framework within LBM that handled both rigid and deformable objects in three-dimensional flows, significantly expanding the method's applicability to general fluid–structure interaction problems. More recently, Zhu et al. (2022) [37] applied LBM-based diffuse IBM to study

tea shoot deformation under negative pressure, demonstrating the method's potential in agricultural robotic applications beyond traditional biological domains. Meng et al. (2020) [25] made significant contributions to this field through their LBM-based diffuse IBM implementation for studying dendritic growth and motion in convective flows, providing crucial insights into solidification dynamics that bridge material science and fluid mechanics applications.

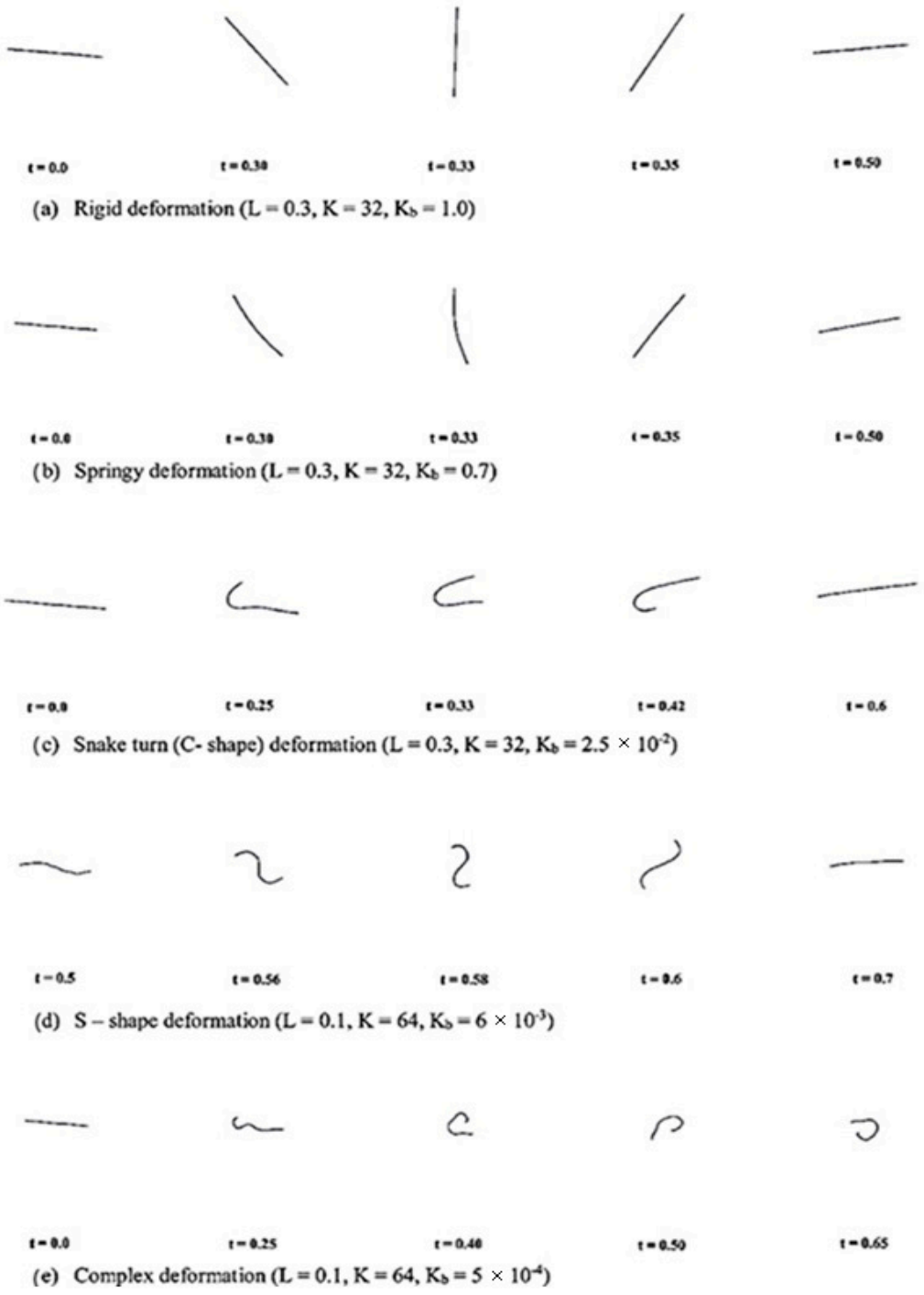


Figure 4. Deformation of a filament at different time points for various bending rigidity values (K_b), illustrating distinct deformation modes: rigid ($K_b = 1.0$), springy ($K_b = 0.7$), C-shape ($K_b = 2.5 \times 10^{-2}$), S-shape ($K_b = 6 \times 10^{-3}$), and complex deformations ($K_b = 5 \times 10^{-4}$). As bending rigidity decreases, the filament transitions from rigid rotation to springy, buckling, and eventually complex deformation modes, unable to resist hydrodynamic stresses (Kanchan and Maniyeri, 2019) [9].

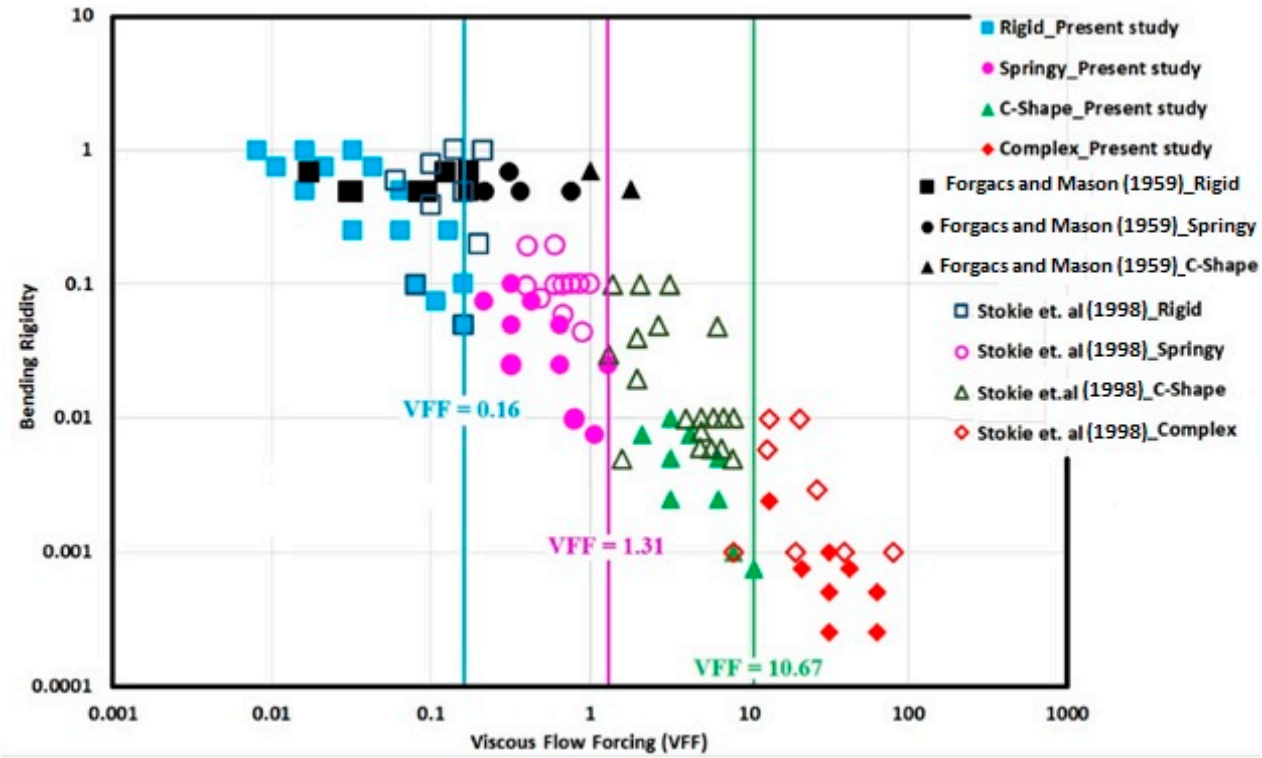


Figure 5. Classification of filament orbit regimes based on viscous flow forcing (VFF) and bending rigidity (K_b) for a fixed filament length L = 0.1 and shear rates K = 10, 16, 32. Results are compared with experimental works of Forgacs and Mason [10] and numerical studies of Stokie et al. [11]. Regimes are demarcated by solid vertical lines into rigid ($175° < \beta < 180°$), springy ($90° < \beta < 175°$), C-shape ($30° < \beta < 90°$), and complex ($\beta < 30°$) deformation classes using the exterior angle (β) (Kanchan and Maniyeri, 2019) [9].

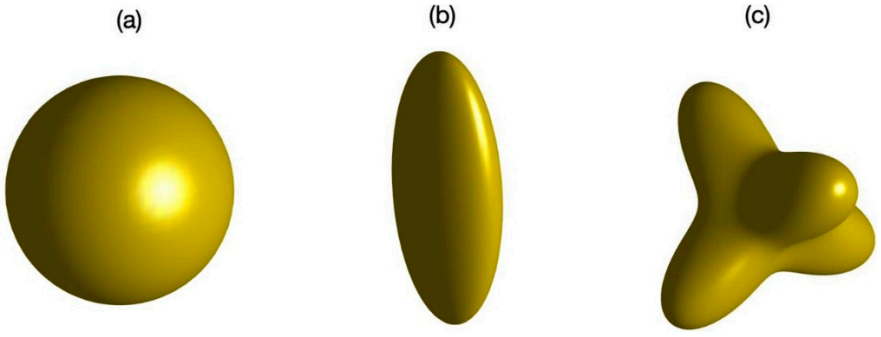


Figure 6. Visualization of three geometries: (**a**) unit sphere, (**b**) ellipsoid with aspect ratio 1:2:4, (**c**) complex surface based on the real part of the spherical harmonic Y_3^2 (Lai and Seol, 2022) [38].

Specialized numerical treatments have been developed to address challenges in specific biological systems. Fai and Rycroft (2018) [20] implemented a finite element method (FEM)-based diffuse IBM with improved accuracy for thin fluid layers, enabling precise simulation of vesicle migration in narrow channels. Casquero et al. (2021) [30] combined diffuse IBM with B-splines and T-splines for capsule and vesicle dynamic simulations, achieving superior representation of biological membrane mechanics. These specialized implementations demonstrate how diffuse IBM can be adapted to maintain accuracy in challenging scenarios involving thin fluid gaps or complex membrane mechanics.

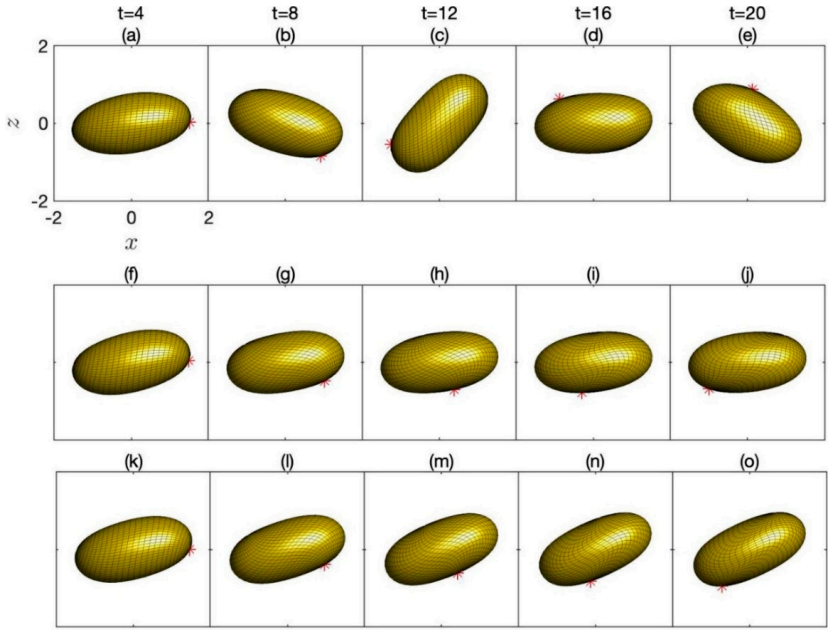


Figure 7. Snapshots of prolate vesicle motion under shear flow at different Reynolds numbers: (**a**–**e**) Re = 1, (**f**–**j**) Re = 5, (**k**–**o**) Re = 9. The other parameters are ν = 0.9, Ca = 2.5, and λ = 20. These snapshots highlight transitions from tumbling to tank-treading motion, confirming that vesicle dynamics at moderate Re (Re = O(1)) are significantly influenced by fluid inertia (Lai and Seol, 2022) [38].

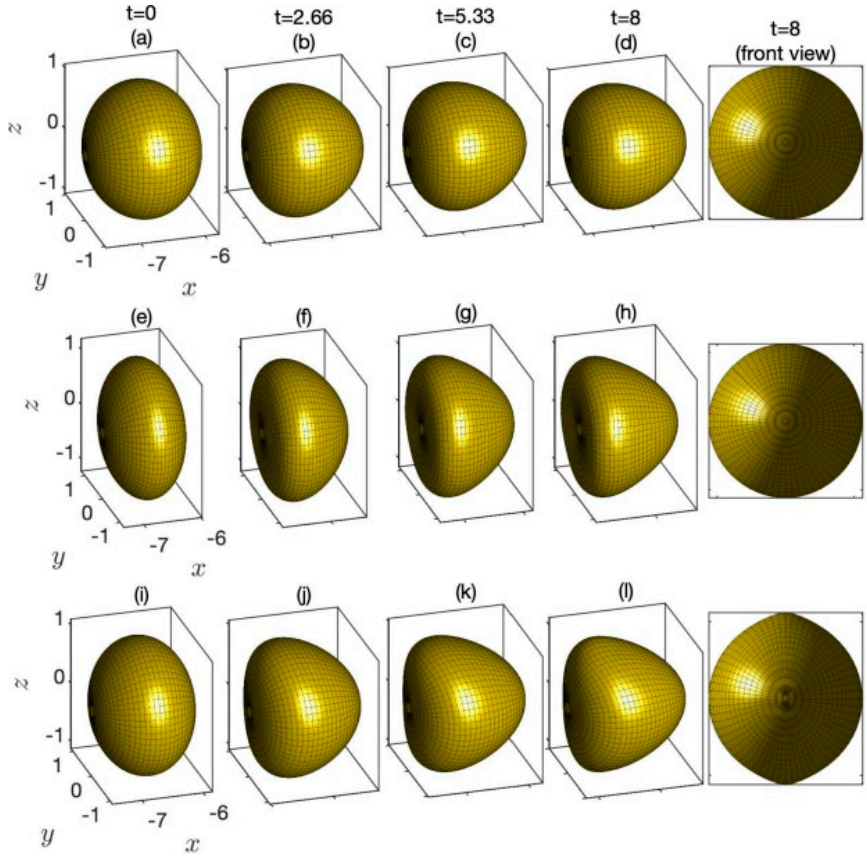


Figure 8. Snapshots of vesicle motion in Poiseuille flow for different reduced volumes (ν) and channel aspect ratios (d/e). The vesicle deforms into a bullet shape for (**a**–**d**) with ν = 0.99 and d/e = 1, a parachute shape for (**e**–**h**) with ν = 0.91 and d/e = 1, and a croissant shape for (**i**–**l**) with ν = 0.973 and d/e = 1.2 (Lai and Seol, 2022) [38].

Hemodynamic applications have driven innovations in diffuse IBM's handling of pulsatile flows and complex geometries. Mirfendereski and Park (2021) [32] developed a diffuse IBM approach for simulating pulsatile flow in stenotic channels, providing insights into blood flow through constricted arteries. Wang et al. (2020) [54] studied self-propelled flexible plates with Navier slip conditions, contributing to the understanding of bio-inspired propulsion mechanisms. These implementations showcase diffuse IBM's ability to handle both prescribed and emergent boundary motions in physiological flow conditions.

Recent advances have expanded diffuse IBM's capabilities through integration with modern computational techniques. Ntetsika and Papadopoulos (2022) [35] combined diffuse IBM simulations of filament dynamics in shear flow with artificial neural network predictions, demonstrating how machine learning can complement traditional computational fluid dynamics. Battista et al. (2018) [19] developed accessible Python and MATLAB implementations of diffuse IBM, lowering the barrier to entry for biological flow simulations. For microscale biological flows, Eldoe et al. (2022) [33] developed an FVM-based diffuse IBM approach to investigate rigid filament interactions in oscillatory flows, advancing the understanding of low-Reynolds-number filament dynamics. Maniyeri (2022) [36] extended the capabilities of diffuse IBM to elastic rod dynamics in fluid flow through an FVM implementation, providing new insights into flagellar motion and similar biological propulsion mechanisms. These developments point toward an increasingly interdisciplinary future for diffuse IBM applications.

In microfluidic and electrochemical applications, diffuse IBM has enabled new insights into complex phenomena. Ladiges et al. (2022) [41] implemented a marker-and-cell (MAC)-based diffuse IBM for simulating electrolytes near physical boundaries, particularly studying induced charge electroosmosis. This work demonstrates how diffuse IBM can be extended beyond traditional fluid–structure interaction problems to include electrochemical effects relevant to lab-on-a-chip devices and similar microsystems.

The finite difference method (FDM) has served as a foundational framework for many biomechanical sharp IBM implementations. Luo et al. (2008) [42] pioneered this approach with their novel IBM for phonation studies, successfully capturing the intricate fluid–structure interactions in vocal fold dynamics. This work demonstrated how sharp IBM could handle the complex geometry and tissue mechanics of human phonation while maintaining computational efficiency. The FDM framework was later extended by Bhardwaj and Mittal (2012) [12] through their coupled IBM–FEM solver, which enabled large-scale simulations of flow-induced deformation in flexible biological structures. This hybrid approach combined the computational efficiency of FDM for fluid dynamics with the accuracy of FEM for structural mechanics, creating a powerful tool for biomechanical simulations.

For traumatic injury modeling, sharp IBM has enabled unprecedented simulations of blast effects on biological tissues. Bhardwaj et al. (2013) [43] applied their FDM–FEM sharp IBM framework to study blast-induced eye deformation, providing critical insights into ocular trauma mechanics. Bailoor et al. (2017) [44] further advanced this capability with their compressible FSI implementation using a ghost-cell method, which accurately captured large deformations during blast–structure interactions. These studies showcased sharp IBM's ability to handle extreme deformation scenarios while maintaining precise boundary representation—a crucial requirement for reliable injury biomechanics simulations.

Cardiovascular applications have particularly benefited from sharp IBM implementations. Bourantas et al. (2021) [45] developed an FEM-based sharp IBM for simulating blood flow in complex vascular geometries, demonstrating superior accuracy in capturing hemodynamic wall shear stresses. Wang et al. (2022) [47] later implemented a moving sharp IBM within a hybrid FEM framework for heart valve flow simulations, successfully capturing both leaflet dynamics and surrounding flow features. The clinical relevance of

these approaches was further demonstrated by Brown et al. (2022) [46], who simulated transcatheter aortic valve replacement (TAVR) devices using patient-specific anatomy in an FDM-based sharp IBM framework. These implementations collectively highlight how sharp IBM has advanced cardiovascular device development and surgical planning.

In cancer research, Singh and Kumar (2023) [48] developed a finite volume method (FVM)-based sharp IBM for tumor morphology modeling, eliminating the need for body-conformal grids while accurately capturing bioheat transfer phenomena. This approach proved particularly valuable for simulating irregular tumor geometries and their interaction with surrounding tissues, demonstrating sharp IBM's versatility across diverse biomedical applications.

Effect of Fluid Solvers on Boundary Treatment for Biological Applications

The implementation of diffuse IBM across different numerical frameworks reveals important trade-offs between accuracy, computational efficiency, and ease of implementation. Finite difference methods, as employed by Heys et al. (2008) [14] for arthropod sensory systems and Ghosh (2021) [29] for biofilm–fluid interactions, offer straightforward implementation and good performance for many biological flow problems. Finite volume methods provide robust conservation properties advantageous for filament dynamics studies, as demonstrated by Kanchan and Maniyeri's series of studies. Finite element approaches excel in handling complex geometries and material nonlinearities, making them ideal for vascular simulations like those of Lampropoulos et al. (2021) [31]. Lattice Boltzmann methods, with their natural parallelism and kinetic theory foundation, have proven particularly effective for problems involving complex boundary motions and multiphase effects. The choice of numerical solvers for sharp IBM implementations in biomechanics reflects careful consideration of application-specific requirements. FDM-based approaches, as demonstrated by Luo et al. (2008) [42] and Brown et al. (2022) [46], offer computational efficiency for problems where structured grids are appropriate. FEM implementations like those of Bourantas et al. (2021) [45] and Wang et al. (2022) [47] provide superior geometric flexibility for complex anatomical structures.

Additionally, the computational performance of these solvers is just as critical a consideration. Lattice Boltzmann and finite difference methods, owing to their local stencil operations, demonstrate excellent parallel scalability. For instance, GPU formulations of LBM have exhibited over 85% parallel efficiency in large-scale boundary-resolved simulations (Krüger et al., 2016) [55]. Finite element solvers, whilst facilitating geometric flexibility, are memory-intensive. Nevertheless, unconditionally stable time integration schemes have demonstrated stable convergence with iteration counts growing sublinearly with mesh refinement, as noted in vascular flow simulations (Cueto-Felgueroso et al., 2007) [56]. The finite volume solvers utilized by Kanchan and Maniyeri report showed grid convergence and memory-efficient scaling for filament dynamics. Hybrid schemes such as Bhardwaj and Mittal's FDM–FEM frameworks capitalize on implicit solvers for structural components whilst ensuring computational efficiency in fluid domains. Whilst not all studies document explicit performance benchmarks, available metrics on memory usage, time to solution, and solver robustness assist in informing the method of choice for a given application and available computational resources.

3.2. Vortex-Induced Vibration and Flexible Body Interaction

"Vortex-induced vibration (VIV)" and the "interaction of flexible structure and fluid flow" are critical issues in the broader field of FSI, extending from engineering to biology. The interactions come about when fluid flow excites the oscillation of flexible or slender structures, such as underwater pipelines and aircraft wings, that exhibit complex dynamic

behavior. An understanding of these processes helps optimize designs, predict structural responses, and investigate natural systems, such as fish swimming and plant motion in wind. Table 3 illustrates the recent advancements in IBM implementation with various fluid solvers and boundary treatment for solving VIV and flexible body interaction scenarios.

Table 3. Recent advancements in IBM considering boundary treatment and corresponding fluid solvers for VIV and flexible body interactions.

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Kumar et al., 2015 [57]	LBM	Diffuse	Clap-and-fling mechanism at low Re	Insect flight	2015
De Rosis, 2015 [58]	LBM	Diffuse	Tandem flapping wings near ground effect	Aerial vehicle design	2015
Wang et al., 2018 [59]	FDM	Diffuse	Oscillating hydrofoils for energy harvesting	Hydrodynamics	2018
Li et al., 2018 [60]	FDM	Diffuse	Pitching motion profiles for energy extraction	Semi-active foils	2018
Xie et al., 2019 [61]	FVM	Diffuse	VIV of a cylinder with attached filament	Vortex-induced vibrations	2019
Ma et al., 2019 [62]	RANS	Diffuse	Flow physics within blade passages	Turbomachinery	2019
Wu et al., 2019 [63]	LBM	Diffuse	Flow-induced vibration of tandem elliptical cylinders	Tandem cylinder flows	2019
Wang et al., 2019 [64]	FVM	Diffuse	Elastic bodies in viscous fluids (vegetation/turbines)	FSI in laminar/ turbulent flows	2019
Wang et al., 2020 [24]	FDM–FEM	Diffuse	Hydrodynamic interaction of flexible flags	Poiseuille flow	2020
Chen et al., 2020 [65]	FDM	Diffuse	3D flag flapping dynamics in Poiseuille flow	Flexible structures	2020
Zhao et al., 2020 [66]	FVM	Diffuse	Tsunami-like wave impact on coastal bridges	Coastal engineering	2020
Zhang et al., 2020 [67]	FDM	Diffuse	Hydrodynamics of tuna caudal keels	Bio-inspired propulsion	2020
Luo et al., 2021 [68]	FDM	Diffuse	Reduced-order FSI for biofluid systems	Biofluid dynamics	2021
Kasbaoui et al. 2021 [69]	DNS	Diffuse	Swirling von Kármán flow with moving IBM	Turbulence transitions	2021
Dong et al., 2021 [70]	SGKS	Diffuse	Chordwise deformation effects on batoid fish	Bio-inspired propulsion	2021
Ai et al., 2021 [71]	FDM	Diffuse	Internal wave prediction with IBM	Stratified flows	2021
Yan et al., 2021 [72]	LBM	Diffuse	Vortex shedding around circular cylinders	Isolated/tandem cylinders	2021
Tian et al., 2021 [73]	FEM	Diffuse	Transient FSI with energy conservation	General FSI	2021
Zhao et al., 2021 [74]	FDM	Diffuse	Drag reduction using micro floating rafts	Underwater vehicles	2021
Yaswanth and Maniyeri, 2022 [75]	FVM	Diffuse	Fluid mixing in oscillating lid-driven cavity	Mixing enhancement	2022
Mazharmanesh et al., 2022 [76]	LBM	Diffuse	Energy harvesting in inverted piezoelectric flags	Piezoelectric energy	2022
Karimnejad et al., 2022 [77]	LBM	Diffuse	Pulsating flow effects on particle settling	Particle dynamics	2022

Table 3. *Cont.*

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Mao et al., 2022 [78]	FDM	Diffuse	Drag reduction with flexible hairy coatings	Drag reduction	2022
Zhang et al., 2022 [79]	DNS	Diffuse	Hydrodynamics of tuna finlets	Bio-inspired propulsion	2022
Yu and Yu, 2022 [80]	FDM	Diffuse	Flow interactions with aquatic vegetation	Environmental flows	2022
Jin et al., 2022 [81]	FVM	Diffuse	VOF-IBM for solitary wave free surface flow	Free-surface flows	2022
Fang et al., 2022 [82]	LBM	Diffuse	Fish swimming hydrodynamics	Bio-inspired robotics	2022
Huang et al., 2022 [83]	LBM	Diffuse	FSI of fish moving through turbines	Fish-turbine interactions	2022
Xiao et al., 2022 [84]	LBM	Diffuse	Water entry/exit of rigid bodies	Free-surface FSI	2022
Mi et al., 2022 [85]	URANS	Diffuse	FSI in submerged nets	Aquaculture nets	2022
Stival et al., 2022 [86]	FVM	Diffuse	LES–IBM for wind turbine flows	Wind energy	2022
Mazharmanesh et al., 2023 [87]	LBM	Diffuse	Tandem/side-by-side piezoelectric flags in oscillating flow; identified chaotic/symmetric regimes	Energy harvesting (inverted flags)	2023
Luo and Zhang, 2023 [88]	HydroFlow	Diffuse	IBM in σ-coordinate model for wave-structure interactions	Ocean engineering	2023
Dong and Huang, 2023 [89]	LES	Diffuse	Wall-modeled LES–IBM for batoid fish swimming; revealed hairpin vortices	Bio-inspired propulsion	2023
Guo and Hou, 2023 [90]	Other (DUGKS)	Diffuse	Slip-condition IBM for AUV drag reduction	Underwater vehicles	2023
Wu and Guo, 2023 [91]	LBM	Diffuse	FS-LBM with IB for aircraft ditching; included surface tension effects	Multiphase FSI (water impact)	2023
Park et al., 2023 [92]	FDM	Diffuse	Semi-Lagrangian Navier–Stokes with reduced IBM for high-inertia/elasticity FSI	General FSI	2023
Monteiro and Mariano, 2023 [93]	Spectral	Diffuse	Fourier pseudo-spectral-IBM for airfoils/VAWTs; validated at Re = 1000	Wind energy	2023
Liu et al., 2023 [94]	LBM	Diffuse	IB–LBM with AMR/VOF for free-surface flows in ocean engineering	Ocean engineering	2023
Zhang et al., 2024 [95]	FDM	Diffuse	Perforated plate-filament system; observed MFD mode and drag reduction	Drag reduction	2024
Mao et al., 2023 [96]	FDM	Diffuse (Penalty IBM)	Snap-through dynamics of buckled filament; mode transitions with varying bending rigidity/Re	Flexible filaments in flow	2023
Borazjani and Sotiropoulos, 2009 [97]	FEM	Sharp	VIV of two elastically mounted cylinders at Re = 200	Vortex-induced vibrations	2009
Seo and Mittal, 2011 [98]	FDM–FEM	Sharp	Cut-cell IBM to reduce spurious pressure oscillations	Moving boundary flows	2011
Griffith et al., 2016 [99]	FDM–FEM	Sharp	Flow around rotationally oscillating cylinders	Energy harvesting	2016
Griffith and Leontini, 2017 [100]	FDM–FEM	Sharp	Heuristic model for VIV simulations	Vortex-induced vibrations	2017

Table 3. *Cont.*

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Khalili et al., 2018 [101]	FDM	Sharp	High-order ghost-point IBM for compressible flows	Compressible viscous flows	2018
Mishra et al., 2019 [102]	FDM–FEM	Sharp	Viscoelastic plate attached to a cylinder	Flow-induced oscillations	2019
Majumdar et al., 2020 [103]	FVM	Sharp	Dynamic transitions in plunging foil flow	Transition to chaos	2020
Narváez et al., 2020 [104]	Incompact3D	Sharp	Flow-induced vibrations of tandem cylinders	Tandem cylinder FIV	2020
Xu et al., 2020 [105]	FDM	Sharp	High-order solutions for nonlinear water waves	Free-surface flows	2020
Xin et al., 2020 [106]	CIR	Sharp	Ghost cell method for sloshing in tanks	Free-surface flows	2020
Kundu et al., 2020 [107]	FDM–FEM	Sharp	FSI solver coupling IBM–FEM with dynamic under-relaxation	General FSI	2020
Kwon et al., 2020 [108]	FVM	Sharp	Shallow water flow around cylinders	Tsunami mitigation	2020
Tsai and Lo, 2020 [109]	FDM	Sharp	Fluid–structure interaction in submerged nets	Aquaculture nets	2020
Seshadri and De, 2021 [110]	FVM	Sharp	Robust framework for moving bodies	General FSI	2021
Robaux and Benoît, 2021 [111]	FDM	Sharp	Fully nonlinear potential wave tank	Free-surface flows	2021
Badhurshah et al., 2021 [112]	FDM–FEM	Sharp	VIV of a cylinder with bistable springs	Energy harvesting	2021
Tong et al., 2021 [113]	FDM	Sharp	Nonlinear wave-structure interactions	Offshore engineering	2021
Hanssen and Greco, 2021 [114]	FDM	Sharp	Overlapping grid method for water waves	Wave-body interactions	2021
Sharma et al., 2022 [115]	FDM–FEM	Sharp	FIV of cylinders with varying cross-sections	Flow-induced vibrations	2022
Giannenas et al., 2022 [116]	Incompact3D	Sharp	Wake response to harmonic forcing	Bluff body flows	2022
Khedkar and Bhalla, 2022 [117]	FDM	Sharp	MPC-IBM for wave energy converters	Renewable energy	2022
Song et al., 2022 [118]	RANS	Sharp	3D scour model for complex structures	Coastal/scour dynamics	2022
Gómez et al., 2022 [119]	Incompact3D	Sharp	VIV in four circular cylinders	Multi-cylinder FSI	2022
Badhurshah et al., 2022 [120]	FDM–FEM	Sharp	VIV with bistable springs (energy harvesting)	Vortex-induced vibrations	2022
Ji et al., 2022 [121]	LES	Sharp	Hybrid actuator line-IB for wind turbine wakes	Wind energy	2022
Xu et al., 2023 [122]	FDM	Sharp	High-order finite difference solver with IBM for wave loads on marine structures	Offshore structures	2023
Pandey et al., 2023 [123]	FDM–FEM	Sharp	FIV of cantilevered plate; identified lock-in regimes for energy harvesting	Energy harvesting	2023
Kanchan et al., 2024 [124]	FDM–FEM	Sharp	ANN-predicted oscillations of elastic plate in flow; 6 response regions	Low-Re FSI	2024

Table 3. *Cont.*

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Chern et al., 2023 [125]	LES	Sharp	Plasma-actuated dynamic stall control on flapping NACA 0012 wing	Aerodynamics (flow control)	2023
Yang et al., 2023 [126]	LES	Sharp (ghost cell)	Modified IBM for VAWT simulations; validated with Eppler387 airfoil	Wind energy	2023
Yu et al., 2023 [127]	FDM	Sharp (IB-GHPC)	2D wave tank with pressure-velocity separation; validated for free-surface flows	Coastal engineering	2023
Sundar et al., 2023 [128]	ALE	Sharp	Moving-boundary-enabled standard Navier–Stokes based Physics Informed Neural Network	Flow past plunging foils	2023
Kolahdouz et al., 2023 [129]	FDM–FEM	Sharp (ILE)	Dirichlet–Neumann coupling for nonlinear FSI; improved volume conservation	Large-deformation FSI	2023
Zargaran et al., 2023 [130]	FVM-FEM	Sharp	Lagrangian-IBM for rotor-stator mixers; reduced numerical diffusion	Industrial mixing	2023
Li et al., 2023 [131]	FDM	Sharp (level-set)	3D sharp-interface IBM for turbomachinery flows	Hydraulic turbomachinery	2023
Joachim et al., 2023 [132]	Lethe	Sharp (Nitsche's)	Parallel NIB method for fluid mixing in stirred tanks	Industrial mixing	2023
Kou and Ferrer, 2023 [133]	FDM	Sharp (volume penalization)	Combined volume penalization–SFD for moving geometries; damped spurious waves	Moving-boundary flows	2023
Agarwal et al., 2024 [134]	FDM	Sharp (direct forcing)	IGA–FDM coupling for slender rods in flow; decoupled fluid-grid resolution	Flexible structures	2024

In bio-inspired propulsion systems, diffuse IBM has enabled unprecedented simulations of aquatic and aerial locomotion mechanisms. The lattice Boltzmann method has proven particularly synergistic with diffuse IBM for problems requiring complex boundary motion and efficient parallelization. Wu et al. (2019) [63] and Yan et al. (2021) [72] employed LBM-based diffuse IBM to study flow-induced vibrations of elliptical and circular cylinders, respectively, demonstrating the method's effectiveness for bluff body aerodynamics. More recently, Fang et al. (2022) [82] and Huang et al. (2022) [83] extended these capabilities to fish swimming hydrodynamics and fish–turbine interactions, addressing important ecological and engineering challenges in hydropower systems. The natural parallelism of LBM combined with diffuse IBM's geometric flexibility has enabled these large-scale simulations of complex biological flows.

Kumar et al. (2015) [57] pioneered this application through lattice Boltzmann method (LBM) implementation studying the clap-and-fling mechanism in insect flight, revealing crucial low-Reynolds-number aerodynamics. This work was extended by De Rosis (2015) [58], who examined tandem flapping wings in ground effect, providing insights relevant to micro-aerial vehicle design. For aquatic propulsion, Zhang et al. (2020) [67] developed finite difference method (FDM)-based diffuse IBM frameworks to analyze the hydrodynamics of tuna caudal keels and finlets, respectively, identifying key morphological features contributing to swimming efficiency. These studies collectively demonstrate how diffuse IBM's ability to handle complex moving boundaries makes it indispensable for bio-inspired design. For fundamental studies of flexible structure interactions, Wang and Tian (2020) [135] developed an innovative FDM–FEM coupled diffuse IBM approach to investigate hydrodynamic interactions between flexible flags in Poiseuille flow.

Energy harvesting applications have similarly benefited from advanced diffuse IBM implementations. Wang et al. (2018) [59] and Li et al. (2018) [60] studied oscillating and pitching hydrofoils for energy extraction using FDM-based diffuse IBM, optimizing motion profiles for maximum power generation. Mazharmanesh et al. (2022) [76] and Mazharmanesh et al. (2023) [87] implemented LBM-based diffuse IBM to investigate inverted piezoelectric flags in tandem and side-by-side configurations, identifying distinct chaotic and symmetric regimes that impact energy harvesting performance. These implementations showcase diffuse IBM's capability to capture both the fluid dynamics and structural response in coupled energy harvesting systems. Luo et al. (2021) [68] created a reduced-order FSI framework using diffuse IBM that maintained accuracy while significantly decreasing computational cost for biofluid systems.

For vortex-induced vibrations and bluff body flows, finite volume method (FVM) implementations of diffuse IBM have provided valuable insights. Xie et al. (2019) [61] studied vortex-induced vibrations of a cylinder with attached filaments, while Wang and Tian (2019) [136] examined elastic bodies in viscous flows relevant to vegetation and turbine applications. Kasbaoui et al. (2021) [69] implemented a DNS-grade diffuse IBM with moving boundaries to study transitions in swirling von Kármán flow. These FVM-based approaches leverage the method's strong conservation properties to accurately capture wake dynamics and energy transfer mechanisms in turbulent flows. The RANS approach of Ma et al. (2019) [62] provided practical solutions for industrial-scale turbomachinery problems, and Stival et al. (2022) [86] further developed this capability through their LES–IBM implementation for wind turbine flows, demonstrating how diffuse IBM can be combined with advanced turbulence modeling for renewable energy applications.

Coastal and ocean engineering problems have motivated specialized diffuse IBM implementations. Zhao et al. (2020) [66] developed an FVM-based approach for tsunami-like wave impacts on coastal bridges, addressing critical structural engineering challenges. Luo and Zhang (2023) [88] later implemented diffuse IBM in a σ-coordinate hydrodynamic model for general wave–structure interactions, while Liu et al. (2023) [94] combined LBM-based diffuse IBM with adaptive mesh refinement and volume-of-fluid methods for free-surface flows in ocean engineering. Dong et al. (2021) [70] developed a simplified gas kinetic scheme (SGKS)-based diffuse IBM to investigate chordwise deformation effects on batoid fish locomotion. These implementations demonstrate how diffuse IBM can be adapted to handle the unique challenges of environmental flows with moving boundaries and free surfaces.

Recent advances in computational methods have enabled more sophisticated diffuse IBM implementations for complex FSI problems. Tian et al. (2021) [73] developed a finite element method (FEM)-based diffuse IBM with strict energy conservation properties, providing robust solutions for general FSI applications. Park et al. (2023) [92] implemented a semi-Lagrangian Navier–Stokes solver with reduced diffuse IBM for high-inertia/elasticity FSI, while Monteiro and Mariano (2023) [93] created a Fourier pseudo-spectral diffuse IBM for airfoils and vertical-axis wind turbines. In mixing enhancement applications, Yaswanth and Maniyeri (2022) [75] employed an FVM-based diffuse IBM to study fluid mixing in oscillating lid-driven cavities. These advanced implementations push the boundaries of what is possible in FSI simulation, combining numerical innovation with physical insight.

Drag reduction strategies have been another important application area for diffuse IBM. Zhao et al. (2021) [74] studied micro floating rafts for underwater vehicle drag reduction using FDM-based diffuse IBM, while Mao et al. (2022) [78] and Mao et al. (2023) [96] investigated flexible hairy coatings and buckled filaments for similar purposes. These studies leverage diffuse IBM's ability to handle complex, deforming surface textures that would be challenging to model with traditional boundary-conforming methods. More

recently, Zhang et al. (2024) [95] analyzed a perforated plate–filament system, observing novel momentum flux distribution modes that contribute to drag reduction. Kwon et al. (2020) [108] employed a finite volume method (FVM) to study shallow water flows around cylinders, providing insights into tsunami mitigation strategies. Tsai and Lo (2020) [109] developed a sharp-interface IBM to simulate submerged aquaculture nets, addressing fluid–structure interactions in marine environments. For nonlinear wave dynamics, Tong et al. (2021) [113] proposed a high-fidelity sharp-interface solver to analyze wave–structure interactions relevant to offshore engineering. Hanssen and Greco (2021) [114] introduced an overlapping grid method for efficient simulation of wave–body interactions, enhancing accuracy in free-surface flow modeling. These studies collectively demonstrate the versatility of sharp-interface techniques in tackling complex fluid dynamic problems across engineering disciplines.

Emerging applications continue to push diffuse IBM capabilities in new directions. Dong and Huang (2023) [89] implemented a wall-modeled LES–IBM for batoid fish swimming, revealing previously unseen hairpin vortex structures. Guo and Hou (2023) [90] developed a discrete unified gas kinetic scheme (DUGKS) with slip-condition IBM for autonomous underwater vehicle drag reduction. Wu and Guo (2023) [91] created a free-surface LBM with diffuse IBM for aircraft ditching simulations, incorporating surface tension effects. These cutting-edge applications demonstrate how diffuse IBM continues to evolve to meet new engineering challenges.

Vortex-induced vibration (VIV) studies have been significantly advanced through sharp IBM implementations. Borazjani and Sotiropoulos (2009) [97] pioneered FEM-based sharp IBM simulations of two elastically mounted cylinders at Re = 200, establishing benchmark results for VIV phenomena. This work was extended by Griffith and Leontini (2017) [100], who developed an heuristic model for efficient VIV predictions using an FDM–FEM sharp IBM framework. Recent applications to energy harvesting systems include Badhurshah et al. (2021) [112], Badhurshah et al. (2022) [120], who studied the VIV of cylinders with bistable springs, demonstrating how sharp IBM can capture the complex nonlinear dynamics of these systems. For more complex configurations, Gómez et al. (2022) [119] implemented a sharp IBM in Incompact3D to investigate VIV in four circular cylinders, revealing intricate wake interaction patterns.

The choice of numerical solver for sharp IBM implementations reflects careful consideration of application requirements. Finite difference methods (FDMs) have been widely employed for their simplicity and efficiency in fundamental studies. Khalili et al. (2018) [101] developed a high-order ghost-point FDM sharp IBM for compressible viscous flows, while Xu et al. (2020) [105] and Xu et al. (2023) [122] created high-order FDM implementations for nonlinear water waves and wave loads on marine structures. These approaches demonstrate how FDM-based sharp IBM can achieve excellent accuracy for both fundamental flow physics and engineering applications.

For problems requiring geometric flexibility, finite volume methods (FVMs) have proven effective. Majumdar et al. (2020) [103] used FVM sharp IBM to study dynamic transitions in plunging foil flows, capturing the route to chaos in these systems. Seshadri and De (2021) [110] developed a robust FVM sharp IBM framework for general moving body problems, demonstrating the method's versatility. The method's adaptability to industrial flows was further showcased by Zargaran et al. (2023) [130] in rotor–stator mixer simulations (Figure 9). The method was employed for modeling rotor motion, which reduced pre-processing time and enabled the modeling of diverse rotor–stator shapes with different motion patterns, as illustrated in Figures 10 and 11.

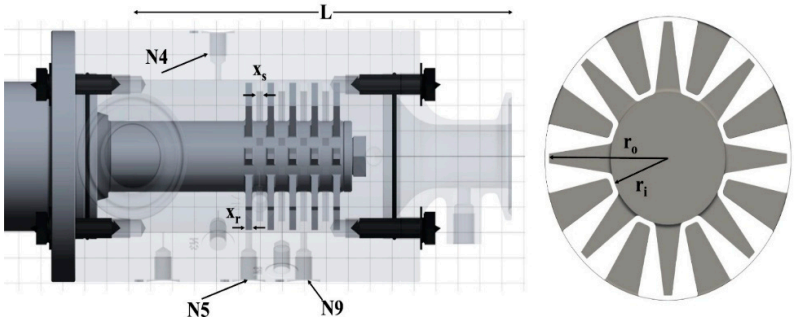


Figure 9. Industrial-intensive in-line mixer type DLM/S, designed and manufactured by INDAG Maschinenbau GmbH (Borsfleth, Germany), consists of multiple stages of star-shaped mixing elements functioning as rotor and stator. The stator elements are fixed to the housing, while the rotor elements are mounted on a shaft connected to an electrical motor. (**Left**): Top view of the mixer showing the N6 injector at the bottom. (**Right**): Magnified frontal view highlighting the rotor and stator arrangement, which ensures efficient mixing of highly viscous liquids with viscosities up to 70 kPa·s (Zargaran et al., 2023) [130].

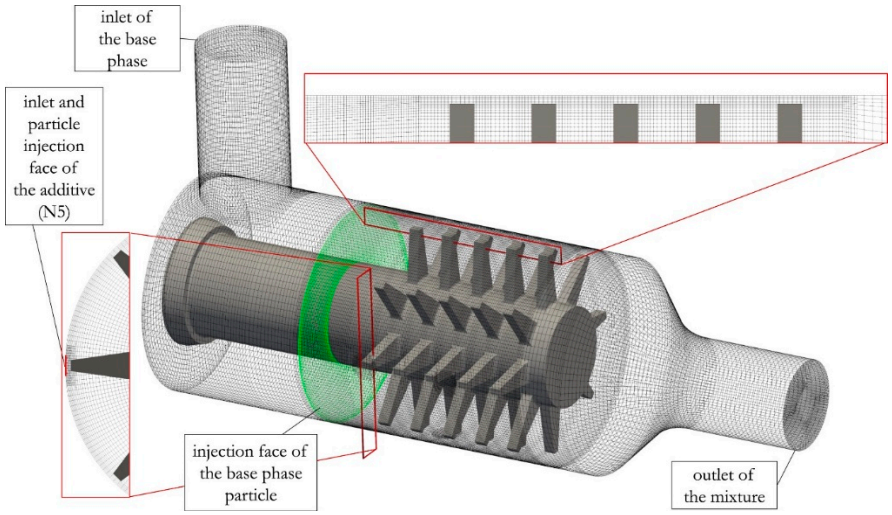


Figure 10. Isometric view of the computational grid used for the mixer simulation, featuring over 2 million fully structured hexahedral cells. The grid is refined near critical zones of high shear gradients, such as rotor–stator gaps and the additive injection location, as shown in the magnified sections. An internal face highlighted in green marks the injection location for base phase tracer particles to accelerate simulation time, particularly for highly viscous mixtures. Stator mixing elements are excluded for improved clarity (Zargaran et al., 2023) [130].

Hybrid solver approaches have expanded sharp IBM capabilities for coupled FSI problems. Seo and Mittal (2011) [98] addressed spurious pressure oscillations in moving boundary flows through their FDM–FEM cut-cell sharp IBM. Kundu et al. (2020) [107] developed an FDM–FEM sharp IBM with dynamic under-relaxation that improved stability for general FSI simulations. These hybrid implementations combine the strengths of different numerical methods to address challenging multiscale and multiphysics problems.

In renewable energy applications, sharp IBM has enabled detailed studies of energy harvesting mechanisms. Griffith et al. (2016) [99] examined flow around rotationally oscillating cylinders for energy extraction using FDM–FEM sharp IBM. Khedkar and Bhalla (2022) [117] implemented a model predictive control (MPC) sharp IBM for wave energy converters, demonstrating the method's potential for renewable energy system optimization. For wind energy applications, Ji et al. (2022) [121] and Yang et al. (2023) [126] developed LES-based sharp IBM frameworks for wind turbine wakes and vertical-axis

wind turbine (VAWT) simulations, respectively, providing insights crucial for turbine design and performance prediction.

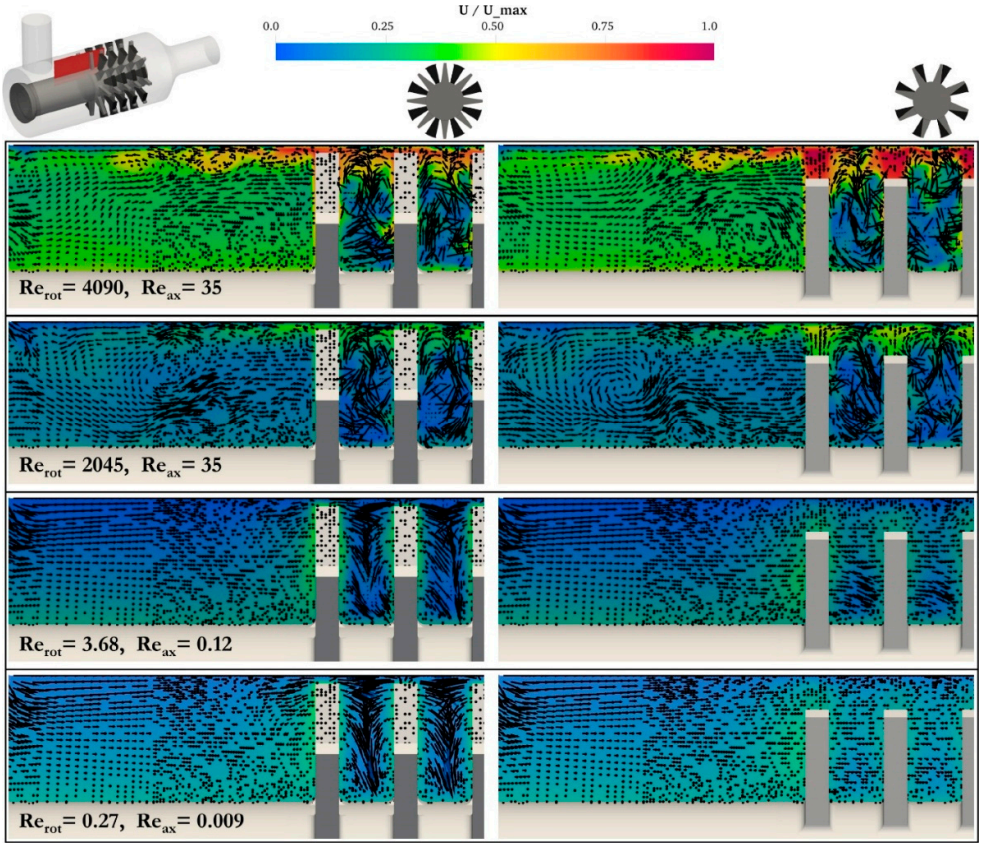
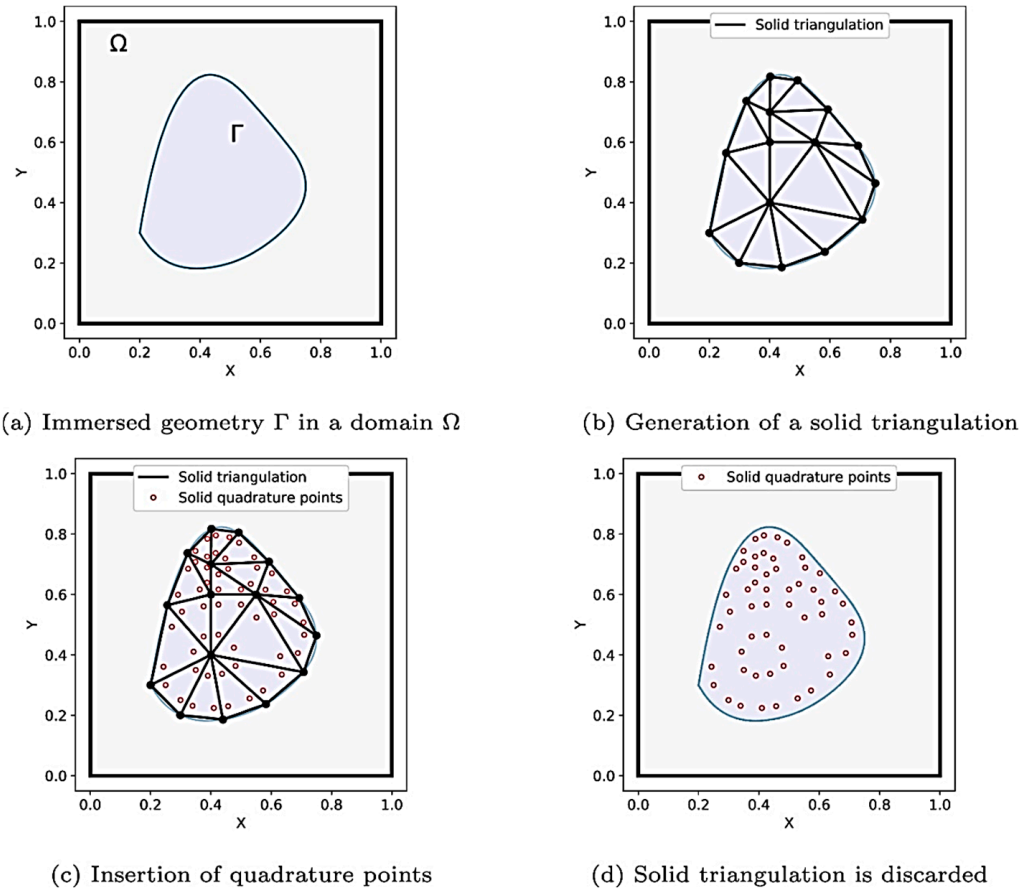


Figure 11. Velocity vectors for four different mixing scenarios at rotor angles of 0° and 30°, maintaining constant total mass flow rate. The emergence of eddies intensifies as the rotational Reynolds number (Re_{rot}) increases, visible between rotor stages and upstream of the rotor. Low Re_{rot} flows suppress eddies, limiting back-mixing. Simulations align with experimental observations in aqueous media, demonstrating longer micro-mixing times for higher viscosity flows (Zargaran et al., 2023) [130].

Free-surface flow simulations have particularly benefited from specialized sharp IBM implementations. Xin et al. (2020) [106] developed a ghost-cell sharp IBM for tank sloshing problems within a constrained interpolation profile (CIR) framework. Robaux and Benoît (2021) [111] created a fully nonlinear potential wave tank using FDM sharp IBM, while Yu et al. (2023) [127] implemented an IB–GHPC method for 2D wave tanks with pressure–velocity separation. These approaches demonstrate how sharp IBM can be adapted to handle the unique challenges of free-surface flows while maintaining interface sharpness.

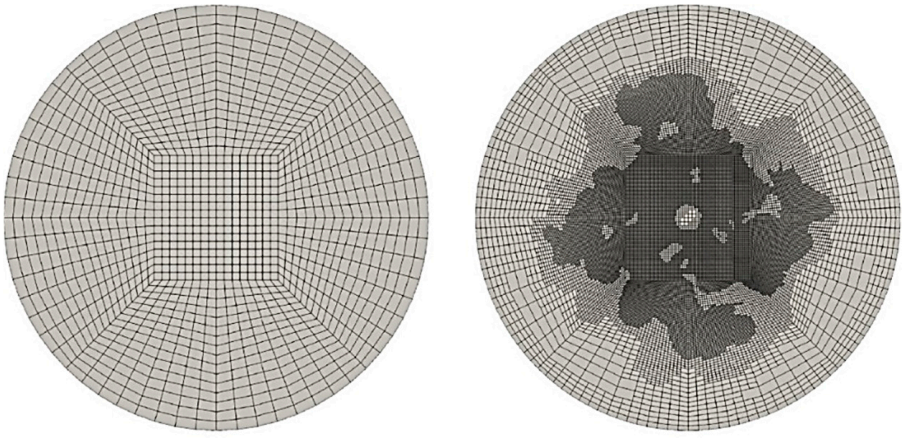
For industrial and offshore applications, sharp IBM has provided valuable design insights. Song et al. (2022) [118] developed a 3D scour model using RANS sharp IBM for complex coastal structures. Tong et al. (2021) [113] studied nonlinear wave–structure interactions for offshore engineering applications. Industrial applications have equally benefited from the method's robust handling of complex moving geometries, as evidenced by Joachim et al.'s (2023) [132] parallelized Nitsche's method for stirred tank simulations (Figure 12), which optimized mixing efficiency while resolving impeller-induced turbulence structures, as shown in Figure 13 and Figure 14, respectively.



(a) Immersed geometry Γ in a domain Ω (b) Generation of a solid triangulation

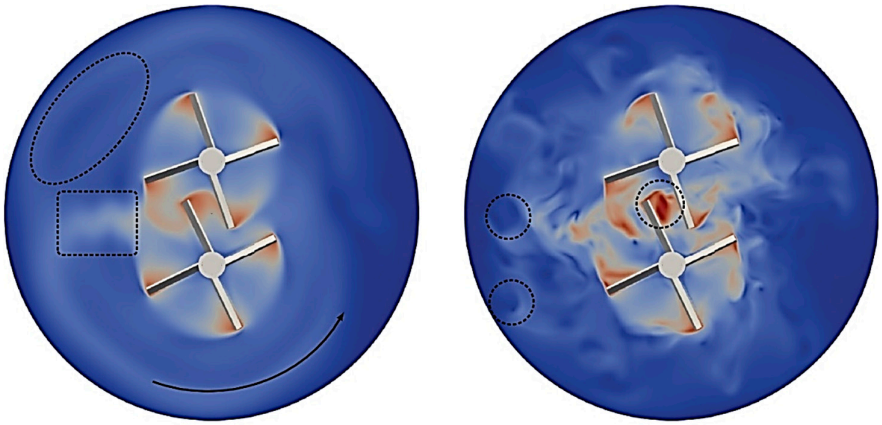
(c) Insertion of quadrature points (d) Solid triangulation is discarded

Figure 12. Process for generating the nodal immersed boundary (NIB). The immersed geometry (Γ) is discretized into a triangulation. Gauss points from the elements of this triangulation are used to create particles. Subsequently, the triangulation is discarded and the simulation is performed using particles associated with the Gauss points of the immersed triangulation (Joachim et al., 2023) [132].



(a) $t = 0\,\mathrm{s}$, initial coarse mesh (b) $t = 0.2\,\mathrm{s}$, refined mesh

Figure 13. Slice of the fluid mesh at the beginning and the end of the simulation, demonstrating the use of adaptive meshing. (**a**) Initial coarse mesh at t = 0 s with uniform refinement (p_{fluid} = 4). (**b**) Refined mesh at t = 0.2 s using adaptive refinement (p_{fluid} = 7) for Re = 1200 ($\nu = 2 \times 10^{-4}$ m^2/s). The adaptive mesh approach reduces computational costs by allowing higher refinement near the impeller while maintaining fewer cells overall (Joachim et al., 2023) [132].



(a) $\nu = 1.2 \times 10^{-3}\ \mathrm{m}^2/\mathrm{s}$ ($Re \approx 200$) (b) $\nu = 2 \times 10^{-4}\ \mathrm{m}^2/\mathrm{s}$ ($Re \approx 1000$)

Figure 14. Fluid velocity magnitude profile in the plane passing through the middle of the blades. The left image (**a**) represents flow at $\nu = 1.2 \times 10^{-3}$ m^2/s (Re $\approx$ 200) for a single impeller mixing rig, where the flow oscillates in the wake of the overlapping region between two impellers, with reduced velocity zones highlighted by black dashed ellipses. The right image (**b**) corresponds to $\nu = 2 \times 10^{-4}$ m^2/s (Re $\approx$ 1000), showing a more turbulent flow with multiple eddies propagating through the vessel, highlighted by black dashed circles. The velocity magnitude ranges from 0 (dark blue) to 7.4 m/s (tip-speed ratio of 1.25) (Joachim et al., 2023) [132].

Recent advances in sharp IB methodology have focused on improving accuracy and robustness. Kolahdouz et al. (2023) [129] developed an immersed Lagrangian–Eulerian (ILE) sharp IBM with improved volume conservation for large-deformation FSI. The "flexible-body immersed Lagrangian–Eulerian (ILE) scheme" combined immersed boundary (IB) methods with a "Dirichlet–Neumann coupling" strategy to distinguish between fluid and solid subregions with separate momentum equations, as seen in Figures 15 and 16. The proposed approach was validated against computational and experimental FSI benchmarks, demonstrating its robustness and accuracy, as observed in Figure 17.

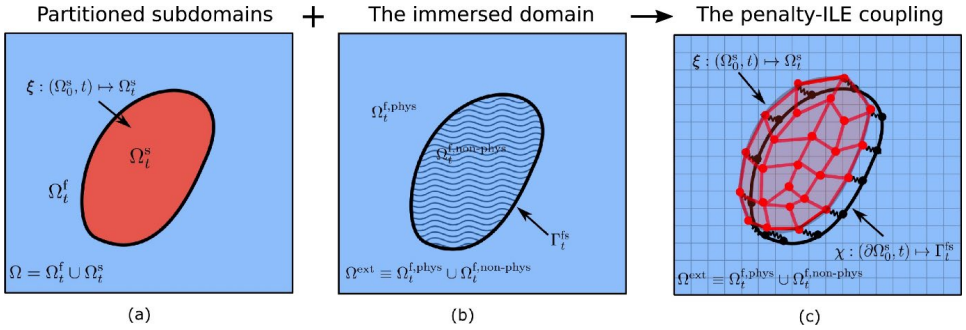


Figure 15. Aspects of (**a**) partitioned and (**b**) immersed FSI formulations are integrated in the (**c**) penalty–ILE coupling. The fluid–structure interface representation (determined by χ) conforms to the boundary of the structure (determined by ξ) in an approximate sense. The discrepancy between the two representations is exaggerated here for illustration purposes. In practice, we ensure that this maximum displacement is no greater than 0.1 of the Cartesian grid spacing. The local fluid velocity determines the motion of the surface representation Γ_t^{fs}, whereas the no-slip condition is satisfied in an approximate sense by spring-like forces that penalize displacements between the two representations of the fluid–structure interface (Kolahdouz et al., 2023) [129].

Li et al. (2023) [131] created a 3D level-set sharp IBM for turbomachinery flows that accurately handled complex blade geometries. Kou and Ferrer (2023) [133] combined volume penalization with selective frequency damping (SFD) to suppress spurious waves

in moving boundary simulations. These methodological innovations continue to push the boundaries with sharp IBM.

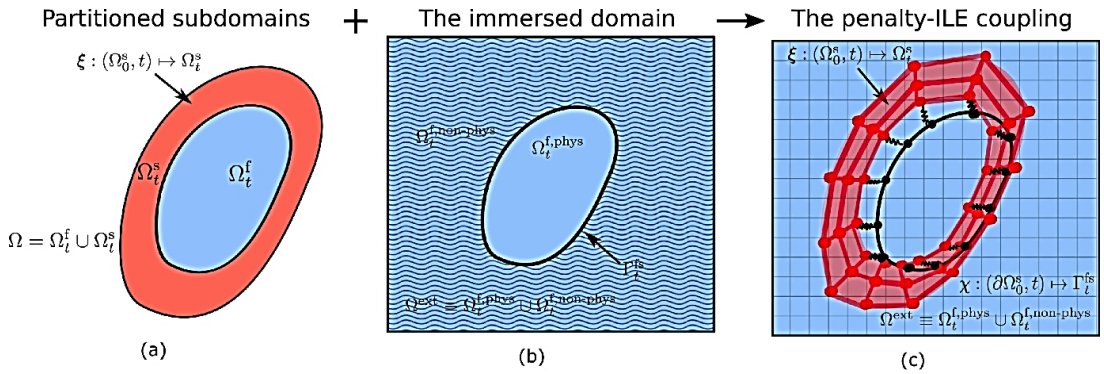


Figure 16. Illustration of the penalty–ILE coupling for compressible structures. (**a**) Partitioned formulation and (**b**) immersed FSI formulations combine to form (**c**) the penalty–ILE coupling. The fluid–structure interface (χ) approximates the inner structural boundary (ξ), with exaggerated displacement shown for clarity (Kolahdouz et al., 2023) [129].

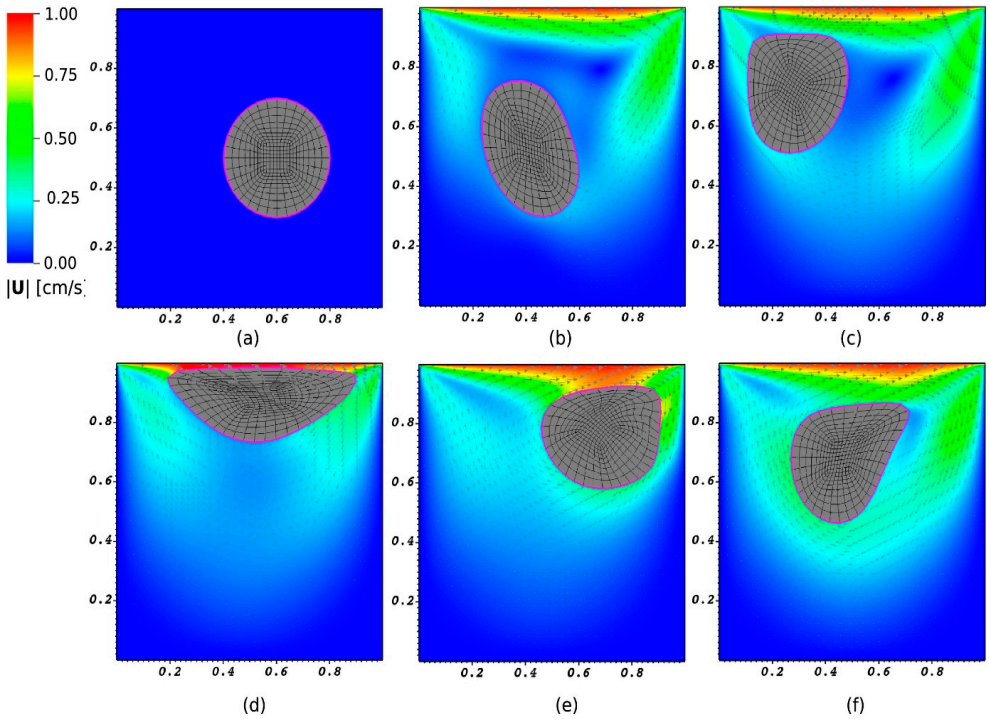


Figure 17. Snapshots of the computed velocity field and soft disk motion in a lid-driven cavity flow at (**a**) t = 0.0s, (**b**) t = 2.0 s, (**c**) t = 3.5 s, (**d**) t = 5.25 s, (**e**) t = 6.5 s, and (**f**) t = 9.0 s. The disk deformation and induced flow illustrate the strong interaction between the fluid and structure. Colors represent the velocity field, with near-contact dynamics emphasized at later times (Kolahdouz et al., 2023) [129].

For flexible structure interactions, several studies have demonstrated sharp IBM's capabilities. Mishra et al. (2019) [102] studied viscoelastic plates attached to cylinders using FDM–FEM sharp IBM. Sharma et al. (2022) [115] investigated the FIV of cylinders with varying cross sections. Pandey et al. (2023) [123] identified distinct lock-in regimes for flow-induced vibrations of elastic plates, correlating these with optimal energy extraction conditions through high-fidelity simulations that resolved both the fluid vorticity fields and structural mode shapes shown in Figures 18–20.

In computational wind engineering, Giannenas et al. (2022) [116] and Narváez et al. (2020) [104] employed Incompact3D with sharp IBM to study wake response and tandem cylinder vibrations, respectively. These spectral element-based implementations demonstrate how sharp IBM can be combined with high-order methods for accurate turbulence simulations.

Emerging applications continue to drive sharp IBM development. Chern et al. (2023) [125] implemented LES sharp IBM to study plasma-actuated dynamic stall control on flapping wings. Recent machine learning integrations by Sundar et al. (2023) [128] coupled physics informed neural networks with IBM. Kanchan et al. (2024) [124] further enhanced predictive capabilities, where artificial neural networks trained on sharp-interface IBM data identified six distinct oscillation regimes for elastic plates in low-Reynolds-number flows, enabling real-time response prediction for adaptive energy harvesting systems (Figures 21–23).

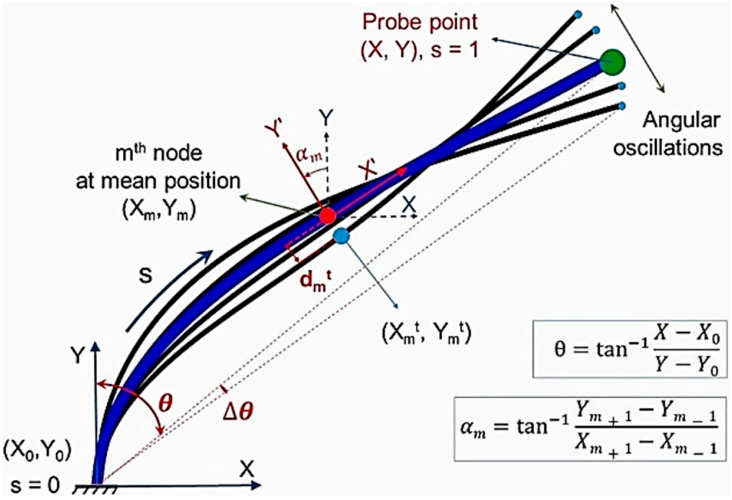


Figure 18. Schematic representation of the deformed plate showing angular oscillations used for post-processing. The figure highlights the location of the probe point and the method for computing transverse displacement relative to the plate's mean position to analyze modal contributions (Pandey et al., 2023) [123].

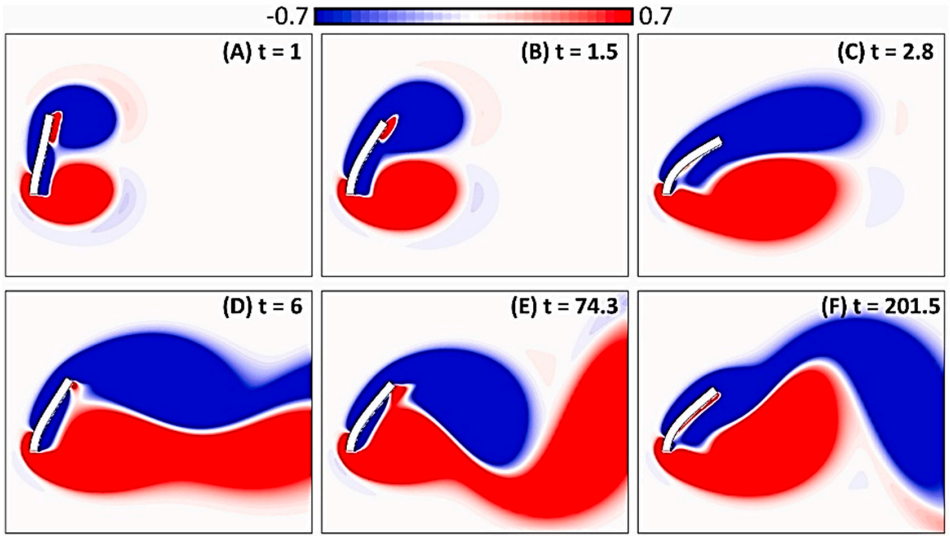


Figure 19. Instantaneous vorticity field and plate deformation at various time points for a plate with E = 1400 and M = 2, illustrating the flow-induced vibration response (Pandey et al., 2023) [123].

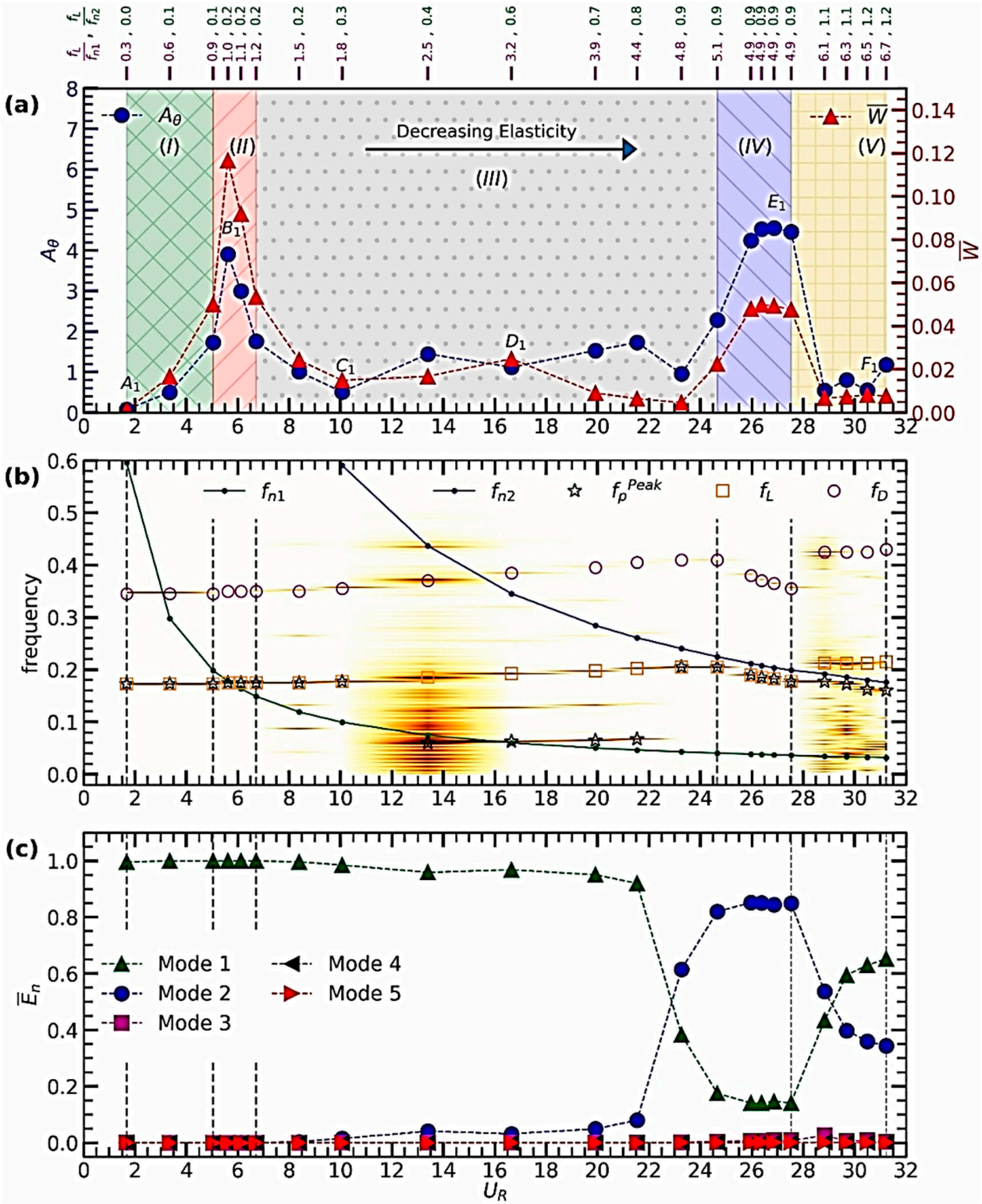


Figure 20. (**a**) Variation in plate vibration amplitude (A_θ) and dimensionless energy exchange (WWW) between fluid and solid with reduced velocity (U_R). Key regions illustrate transitions due to decreasing elasticity. (**b**) Contour plot of plate frequency (f_p) as a function of reduced velocity (U_R). The plot highlights the first (f_{n1})- and second (f_{n2})-mode natural frequencies, along with peak oscillation frequency (f_p^{Peak}), lift frequency (f_L), and drag frequency (f_D). (**c**) Contributions of initial five Euler–Bernoulli natural modes (modes 1–5) to total energy ($\overline{E}i$) with varying reduced velocity (U_R) (Pandey et al., 2023) [123].

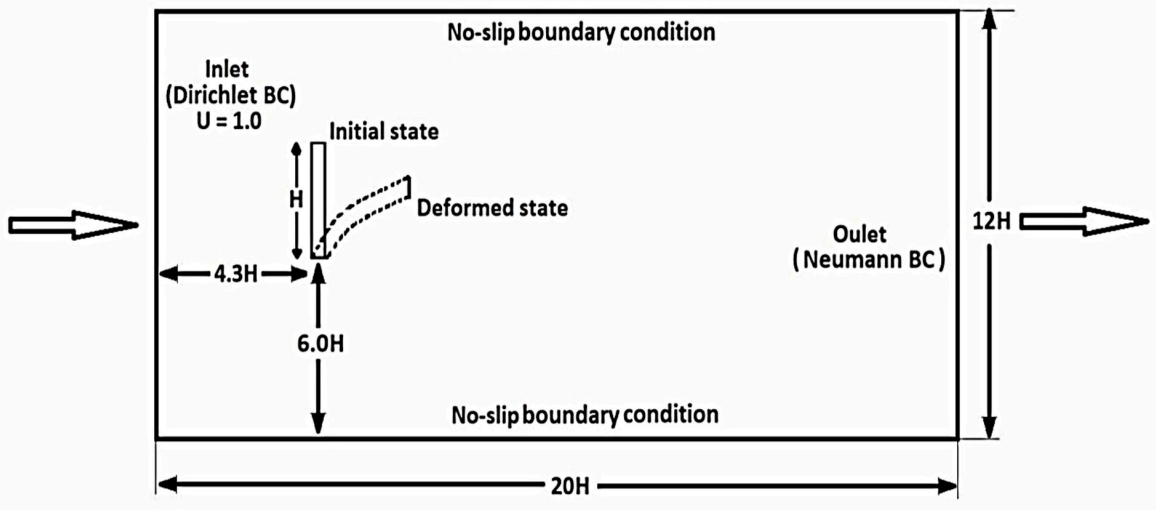


Figure 21. Computational domain consisting of a vertically held flexible plate in laminar viscous flow. The domain has dimensions of 20 H × 12 H. The plate, located at (4.6, 6.0), has a length H = 1.0 and thickness B = 0.01. Dirichlet boundary condition (U = 1.0) is applied at the inlet, and Neumann boundary condition is applied at the outlet. Top and bottom walls are subjected to no-slip conditions (Kanchan et al., 2024) [124].

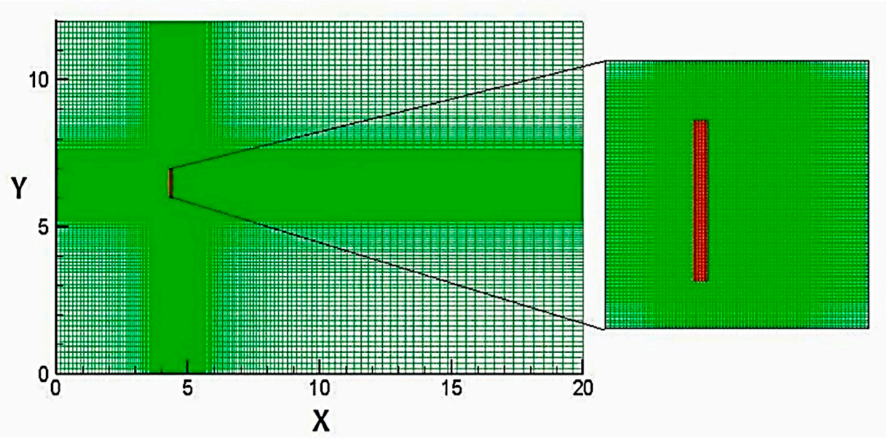


Figure 22. Computational grids (shown in green) surrounding the flexible plate (highlighted in red) on the right side of the image (Kanchan et al., 2024) [124].

Effect of Boundary Treatment on Flexible Body Interaction

The choice of numerical solver for diffuse IBM implementations reflects careful consideration of application-specific requirements. LBM approaches excel for problems with complex boundary motion and parallel scalability needs, as demonstrated by Karimnejad et al. (2022) [77] for particle dynamics and Xiao et al. (2022) [84] for water entry/exit problems. For environmental and geophysical flows, Ai et al. (2021) [71] implemented a diffuse IBM framework capable of predicting internal wave generation and propagation in stratified flows. Their FDM-based approach successfully handled the complex density variations and boundary interactions characteristic of these phenomena, opening new possibilities for oceanographic and atmospheric modeling. FDM implementations offer simplicity and efficiency for fundamental studies of flexible structures, as shown by Chen et al. (2020) [65] for flag flapping and Yu and Yu (2022) [80] for aquatic vegetation. FVMs provide robust conservation properties for turbulent and environmental flows, as exemplified by Jin et al. (2022) [81] for solitary waves and Mi et al. (2022) [85] for submerged nets. Recent advances in multiphase FSI simulations have pushed diffuse IBM capabilities to new levels. Wu and Guo (2023) [91] developed a free-surface LBM with diffuse IBM for

aircraft-ditching simulations that incorporated surface tension effects. Recent innovations like Agarwal et al.'s (2024) [134] isogeometric analysis coupling (Figures 24–26) and Liu et al.'s (2023) [94] AMR-enhanced free-surface solver point toward future directions with cutting-edge computational techniques to address multiscale, multiphysics challenges across engineering and biological domains.

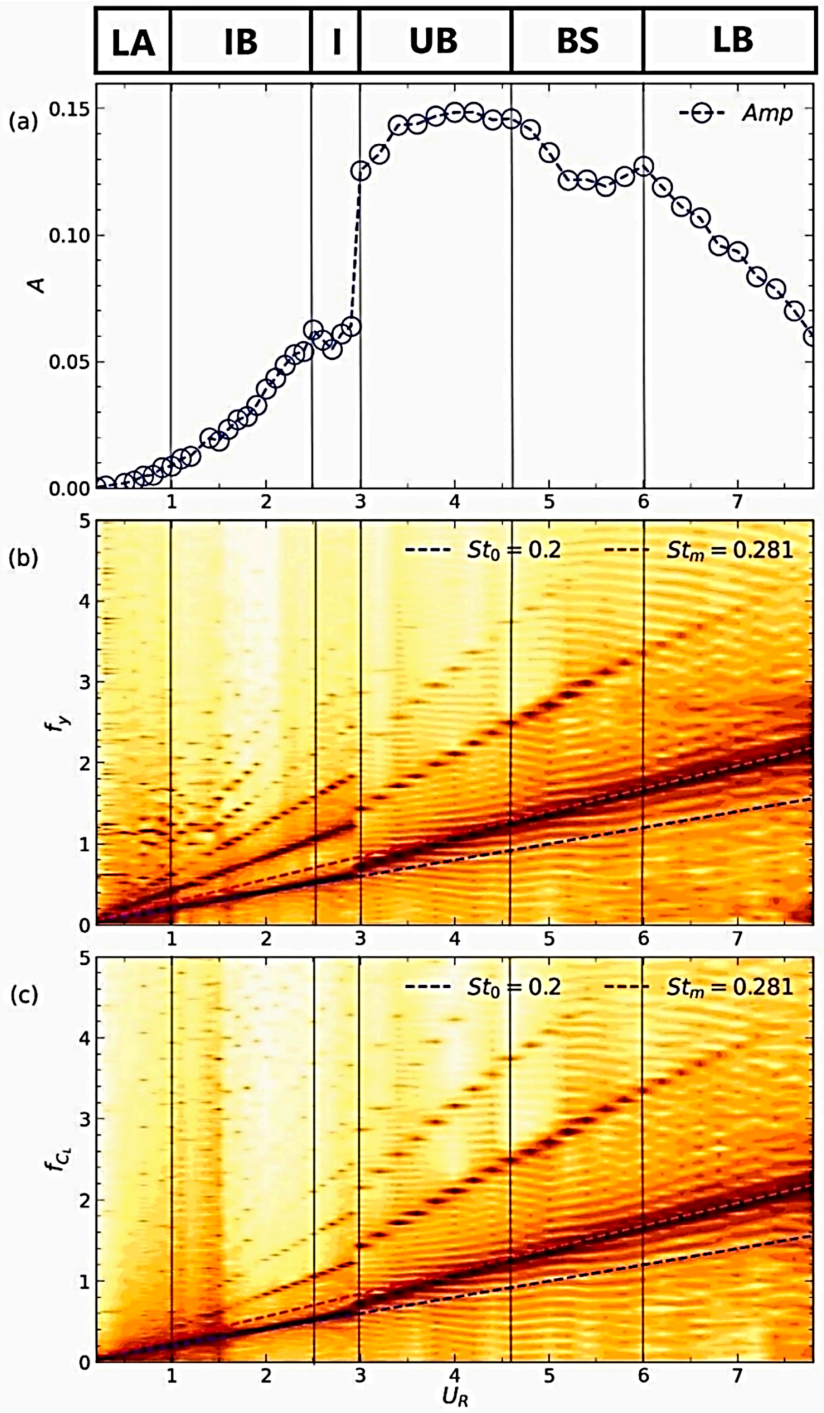


Figure 23. Flow-induced deformation response of the flexible plate, illustrating various branches from the study: low amplitude (LA), initial branch (IB), intermediate (I), upper branch (UB), bi-stable (BS), and lower branch (LB). (**a**) Normalized amplitude (A) versus reduced velocity U_r; (**b**) power spectral density (PSD) contours of vertical displacement (y) with normalized frequency and reduced velocity; (**c**) PSD contours for lift force coefficient (C_L) with normalized frequency and reduced velocity (Kanchan et al., 2024) [124].

Figure 24. 2D color contours of instantaneous spanwise vorticity in the wall-normal plane for clamped flexible filaments. Maximum positive (**left**) and negative (**right**) deflections of the filament centerline are shown (Agarwal et al., 2024) [134].

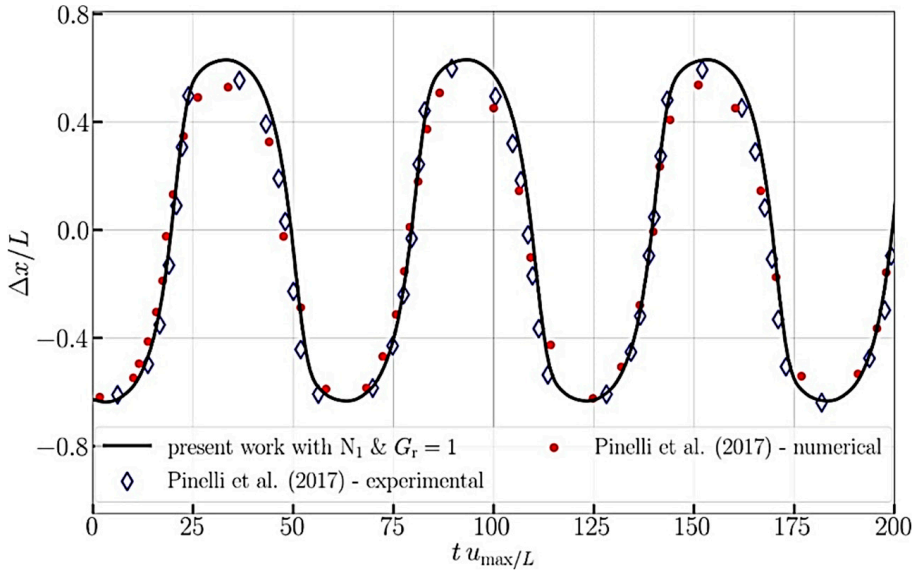
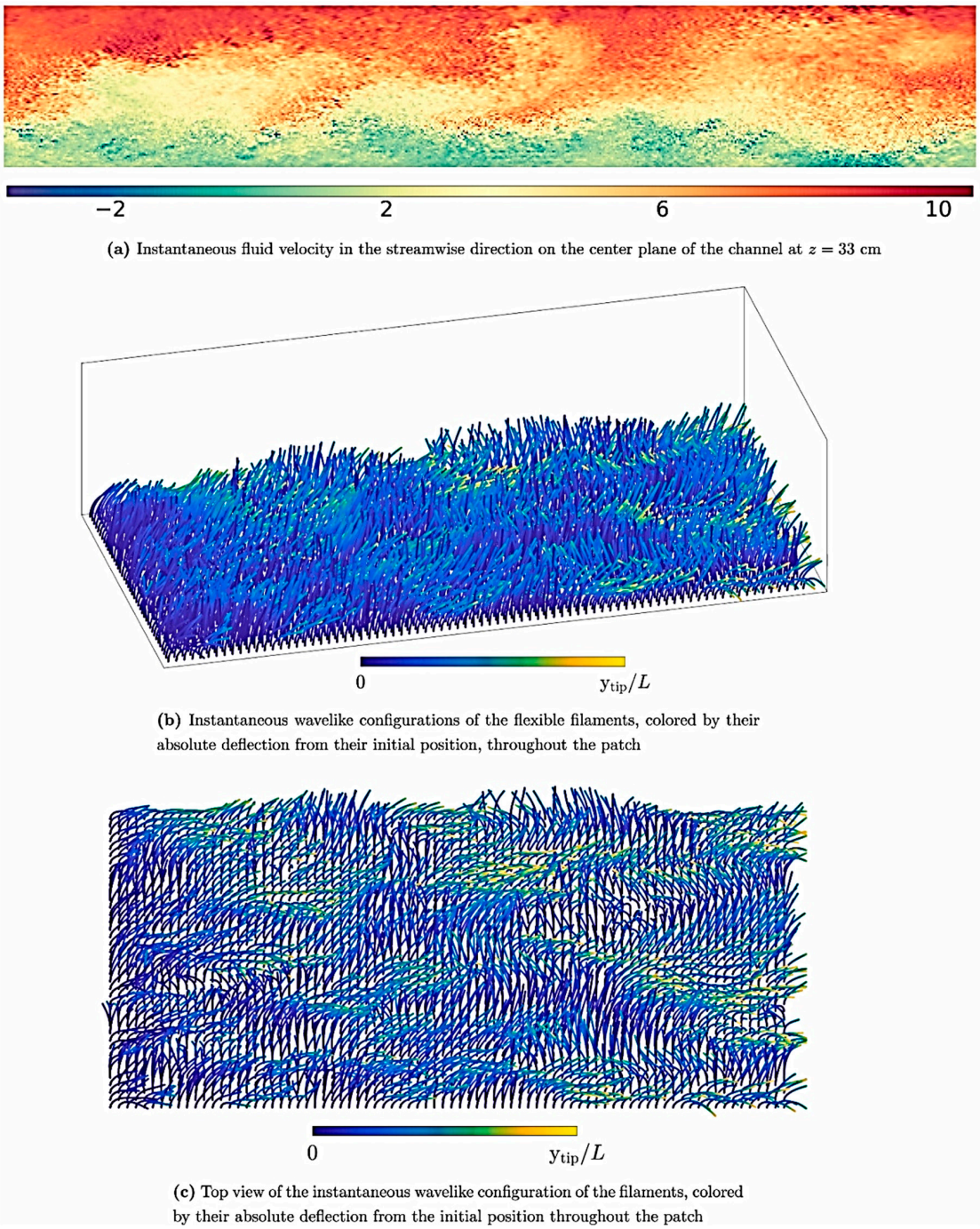


Figure 25. Comparison of the time evolution with experimental and numerical studies (Pinelli et al. [137]) of the streamwise tip displacement of the last filament in an oscillatory channel flow (Agarwal et al., 2024) [134].

Sharp interface immersed boundary methods (IBMs) significantly enhance simulations of flexible body interactions by precisely enforcing boundary conditions at fluid–structure interfaces. Unlike diffuse approaches, sharp IBM maintains distinct boundaries, enabling accurate prediction of flow-induced vibrations and large deformations critical for energy harvesting and vortex-induced vibration studies. The method's exact momentum transfer preserves phase relationships between vortex shedding and structural motion, while its geometric flexibility handles complex morphologies (Sharma et al., 2022) [115]; (Kolahdouz et al., 2023) [129]. For turbulent flows, sharp IBM maintains small-scale features affecting flexible body responses, as demonstrated in wind turbine studies (Ji et al., 2022) [121]; (Yang et al., 2023) [126]. Recent advances combine sharp IBM with data-driven approaches (Sundar et al., 2023) [128]; (Kanchan et al., 2024) [124] and multiphysics applications like plasma flow control (Chern et al., 2023) [125]. The technique's industrial relevance is shown in mixer simulations, where it reduces numerical diffusion (Zargaran et al., 2023) [130]. By preserving physical fidelity at interfaces, sharp IBM provides unmatched accuracy for analyzing flexible body dynamics across engineering applications.



(a) Instantaneous fluid velocity in the streamwise direction on the center plane of the channel at $z = 33$ cm

(b) Instantaneous wavelike configurations of the flexible filaments, colored by their absolute deflection from their initial position, throughout the patch

(c) Top view of the instantaneous wavelike configuration of the filaments, colored by their absolute deflection from the initial position throughout the patch

Figure 26. Snapshots of the streamwise velocity contour and wavelike configurations of the filaments.

4. Multiphysics Application

4.1. Heat Transfer

The IB method, or immersed boundary method, is a computational way of simulating fluid flow and heat transfer across complex moving boundaries. By embedding solid structures in a fixed Cartesian grid, the necessity of body-fitted meshes is greatly eradicated, and therefore proper solutions can be provided in applications concerning multiphase flows and thermal interactions. The wide range of its application allows for precise modeling of turbulent flow patterns, energy transport, and heat exchange in particulate systems, and is therefore very important in computational physics and engineering. Table 4 summarizes recent work carried out in the field of heat transfer and its corresponding IBM formulation.

Table 4. Recent advancements in IBM considering boundary treatment and corresponding fluid solvers for heat transfer applications.

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Ren et al., 2013 [138]	FDM	Diffuse	Dirichlet (velocity) + Neumann (heat flux) corrections for momentum/energy equations	Conjugate heat transfer	2013
Mazharmanesh et al., 2020 [139]	LBM	Diffuse	Multiple piezoelectric flags in tandem/side-by-side configurations	Energy harvesting	2020
Tong et al., 2020 [140]	LBM	Diffuse	Multiblock LBM–IBM for fouling/ash deposition in heat exchangers	Fouling dynamics	2020
Abaszadeh et al., 2022 [141]	LBM	Diffuse	IB–LBM for radiative heat transfer in 2D irregular geometries	Radiative transfer	2022
Jiang et al., 2022 [142]	LBM	Diffuse	Parallel IB–LBM for fully resolved particle-laden flows	Suspension dynamics	2022
Tao et al., 2022 [143]	LBM	Diffuse	Simplified IB–LBM for thermal flows with no-slip/temperature BCs	Thermal flows	2022
Haeri and Shrimpton, 2013 [144]	FEM	Diffuse (fictitious)	Implicit fictitious domain method for flow–heat transfer past immersed objects	Fluid-immersed object interaction	2013
Chen et al., 2020 [145]	LBM	Diffuse (IB–STLBM)	Simplified LBM with IBM for incompressible thermal flows	Thermal flows with immersed objects	2020
Xu and Choi, 2023 [146]	FDM	Diffuse (MIBPM)	Monolithic IB projection method for incompressible flows with heat transfer	Incompressible thermal flows	2023
Hosseini et al., 2021 [147]	LBM	Diffuse (MRT)	IB–LBM for elastic vortex generator (EVG) in microchannels	Microscale mixing	2021
Wu et al., 2023 [148]	LBM	Diffuse	Explicit IBM for TFSI with Neumann BCs (heat flux)	Thermal-fluid–structure interaction	2023
Chen et al., 2020 [149]	FDM	Diffuse (penalty IBM)	Fluid–structure–thermal interaction of flexible flags in heated channels	Heat transfer enhancement	2020
Wang et al., 2023 [150]	FDM–FEM	Diffuse (penalty)	IBM for rarefied gas flow with slip-modeled velocity/temperature jumps	Rarefied gas flows	2023
Zhang et al., 2016 [151]	LBM	Diffuse (PIBM)	Particulate IBM for thermal particle–fluid interactions with DEM	Particle-laden flows	2016
Wu et al., 2024 [152]	LBM	Diffuse	Implicit IBM for TFSI with Robin BCs (combined convection–radiation)	Thermal-fluid–structure interaction	2024
Pacheco et al., 2005 [153]	FVM	Sharp	Finite-volume non-staggered grid with Dirichlet/Neumann BCs for complex geometries	General heat transfer	2005
Soti et al., 2015 [154]	FDM–FEM	Sharp	Strongly coupled FSI solver for flexible structures with convective heat transfer	Energy harvesting	2015
Garg et al., 2018 [155]	FDM–FEM	Sharp	VIV of cylinder with thermal buoyancy (Boussinesq approximation)	Vortex-induced vibrations	2018
Garg et al., 2020 [156]	FDM–FEM	Sharp	Thermal buoyancy effects on wake-induced vibration (WIV) of square prisms	Bluff body heat transfer	2020
Lou et al., 2020 [157]	FVM	Sharp	Finite-volume IBM for membrane distillation with Robin BCs	Desalination	2020

Table 4. *Cont.*

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Mohammadi and Nassab, 2021 [158]	FVM	Sharp	FVM–IBM for radiative heat transfer in irregular geometries	Radiative transfer	2021
Mohammadi and Nassab, 2021 [159]	Hybrid (LBM–FVM)	Sharp	LBM–FVM–IBM for radiative–convective heat transfer	Combined heat transfer	2021
Riahi et al., 2023 [160]	FVM	Sharp	Discrete IBM for compressible flows with heat transfer	Compressible heat transfer	2023
Ahn et al., 2023 [161]	FVM	Sharp	IBM for conjugate heat transfer with melting/solidification	Phase-change heat transfer	2023
Cruz and Lamballais, 2023 [162]	FDM	Sharp	IBM for high-fidelity CHT in pipe flows with irregular geometries	Conjugate heat transfer	2023
Wang et al., 2022 [163]	FDM	Sharp (direct forcing)	Improved direct-forcing IBM for particle-laden flows with heat transfer	Multiphase heat transfer	2022
Narváez et al., 2021 [164]	InCompact3D	Sharp (dual IBM)	Dual IBM for turbulent flow with conjugate heat transfer	Turbulent heat transfer	2021
Wu et al., 2022 [165]	LBM	Sharp (EIB)	Explicit IB–RTLBFS for TFSI problems with Dirichlet BCs	Thermal-fluid–structure interaction	2022
Zhao and Yan, 2022 [166]	FEM	Sharp (EIBM)	Enriched IBM for interface-coupled multiphysics (eg, convective heat transfer)	Conjugate heat transfer	2022
Ou et al., 2022 [167]	DINO	Sharp (GCIB)	Directional ghost-cell IBM for low-Mach reacting flows	Reacting flows	2022
Xia et al., 2014 [168]	FDM	Sharp (ghost cell)	High-order ghost-cell IBM for heat transfer; reduced grid points by ~2/3	Thermal flows	2014
Tao et al., 2022 [169]	Other (DUGKS)	Sharp (ghost cell)	DUGKS–IBM for particulate flows with heat transfer	Fluid–solid heat transfer	2022
Fernandez et al., 2011 [170]	Spectral Method	Sharp (IBC)	Grid-independent 3D heat conduction in slots with time-dependent boundaries	Heat conduction	2011
Shrivastava et al., 2013 [171]	FVM	Sharp (LSIBM)	Level set-based IBM for transient CFD with moving boundaries	Dynamic boundary flows	2013
Ménez et al., 2023 [172]	FVM	Sharp/diffuse	Comparison of volume penalization and IBM for compressible flows with thermal BCs	Compressible thermal flows	2023

In thermal-fluid systems, diffuse IBM implementations have enabled breakthroughs in conjugate heat transfer simulations by simultaneously handling velocity and temperature boundary conditions. Ren et al. (2013) [138] pioneered this capability through their finite difference method (FDM) implementation incorporating both Dirichlet (velocity) and Neumann (heat flux) corrections, establishing a framework for coupled momentum and energy equations. This approach was extended by Chen et al. (2020) [145] in their lattice Boltzmann method (LBM)-based IB–STLBM, which simplified thermal flow simulations while maintaining accuracy for no-slip and temperature boundary conditions at immersed surfaces. Xu and Choi (2023) [146] further improved the numerical stability of IBM by developing a monolithic immersed boundary projection method (MIBPM) for incompressible flows with thermal transport, ensuring accurate resolution of pressure–velocity–temperature coupling in diffuse IBM frameworks, as shown in Figure 27. The approach involved a two-step approximate lower–upper decomposition technique to decouple momentum and energy equations, incorporating immersed boundary forcing, as noted in Figure 28.

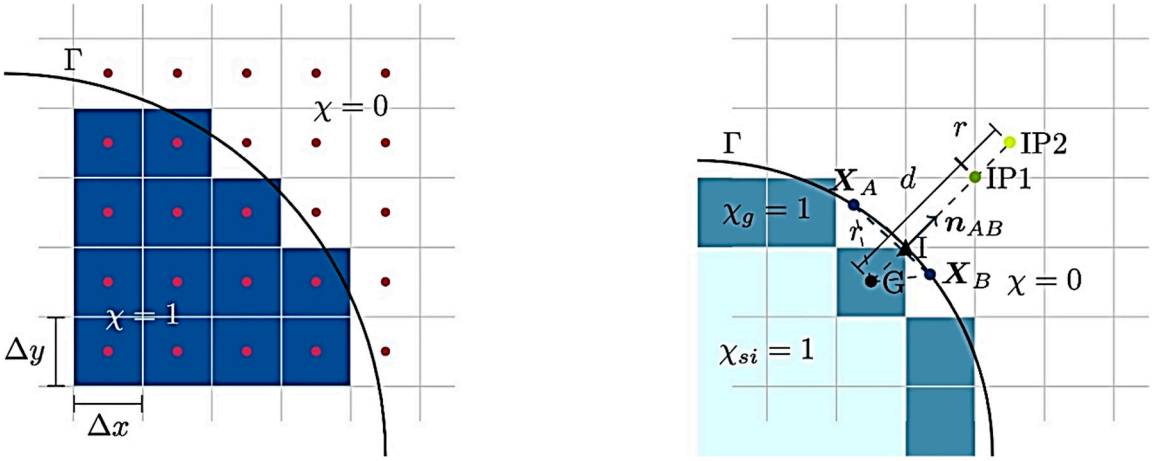
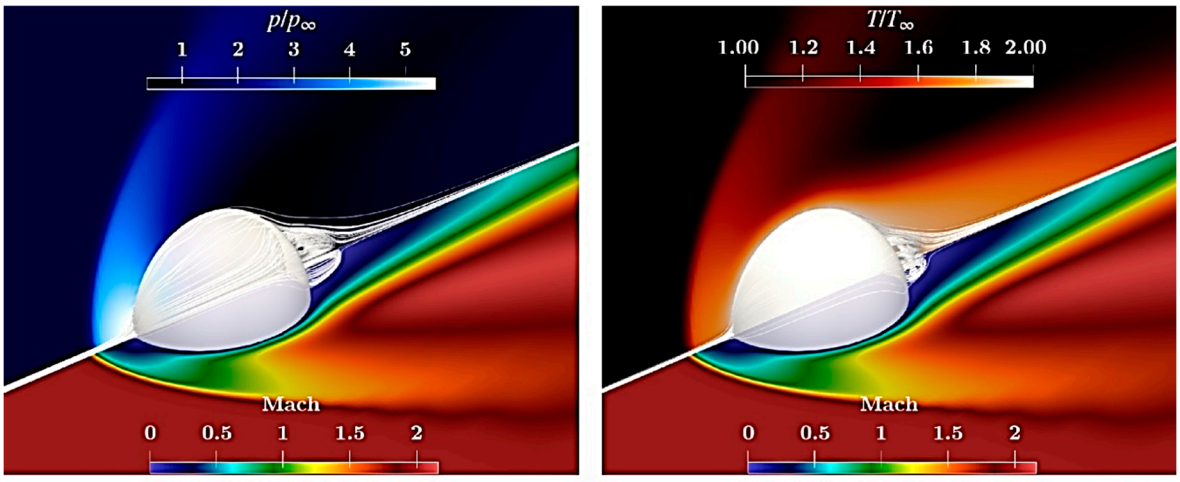


Figure 27. (**Left**) Illustration of the setup used in penalization for isothermal or T-neutral conditions. Eulerian points (cell centers), penalized cells. (**Right**) Illustration of the setup used in penalization for an adiabatic condition. Solid interior cells, ghost cells, • ghost point, ▲ interface point, first image point, second image point, Lagrangian points (Xu and Choi, 2023) [146].



(a) Unheated sphere: streamlines, pressure (x-y plane) and Mach (x-z plane) visualization

(b) Heated sphere ($T_w = 2T_\infty$): streamlines, pressure (x-y plane) and Mach (x-z plane) visualization

Figure 28. Supersonic flow past a sphere (M_∞ = 2, Re_∞ = 300) (Xu and Choi, 2023) [146].

These implementations demonstrate how diffuse IBM's ability to distribute thermal boundary effects across multiple nodes makes it particularly suitable for problems with complex heat transfer pathways.

Lattice Boltzmann methods have shown remarkable synergy with diffuse IBM for thermal and particulate flow applications, benefiting from their kinetic theory foundation and natural parallelism. Jiang et al. (2022) [142] developed a parallel IB–LBM for fully resolved particle-laden flows that maintained computational efficiency while accurately capturing two-way fluid–particle coupling. Zhang et al. (2016) [151] enhanced this capability through their particulate IBM (PIBM) incorporating the discrete element method (DEM) for thermal particle–fluid interactions, enabling detailed study of heat transfer in granular systems. For microscale applications, Hosseini et al. (2021) [147] implemented a multiple-relaxation-time (MRT) LBM with diffuse IBM to analyze elastic vortex generators in microchannels, demonstrating how the method could enhance mixing in confined geometries. These LBM-based approaches leverage diffuse IBM's ability to handle complex moving boundaries without remeshing—a crucial advantage for particulate and microfluidic systems.

Energy systems and heat-exchanger applications have driven significant innovations in diffuse IBM. Tong et al. (2020) [140] developed a multiblock LBM–IBM framework for fouling and ash deposition studies in heat exchangers, addressing the challenging combination of particulate deposition and thermal performance degradation. Mazharmanesh et al. (2020) [139] examined energy harvesting from multiple piezoelectric flags using LBM-based diffuse IBM, revealing how tandem and side-by-side configurations affect power extraction efficiency. These implementations showcase diffuse IBM's versatility in handling both thermal and structural coupling in energy systems, where conventional methods often face difficulties with evolving deposit geometries or oscillating energy harvesters. Tao et al. (2022) [143] simplified IB–LBM for thermal flows while maintaining accurate no-slip and temperature boundary conditions, reducing computational overhead for large-scale simulations.

For rarefied gas flows and microscale heat transfer, specialized diffuse IBM implementations have extended the method's applicability beyond continuum regimes. The adaptability of diffuse IBM was also evident in studies on rarefied gas flows, where Wang et al. (2023) [150] employed a penalty-based approach to account for velocity and temperature jumps in slip-modeled gas flows. This work provided valuable insights into heat transfer behavior in vacuum environments, space propulsion, and other low-pressure applications (Figures 29–31).

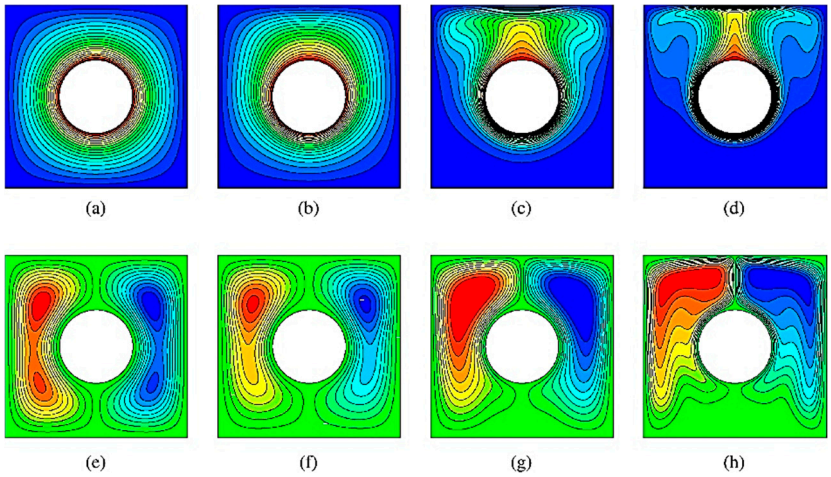


Figure 29. Temperature and vorticity contour at different Rayleigh numbers. Top (**a**–**d**): temperature with a 0.05 increment; bottom (**e**–**h**): vorticity; left to right: Ra = 10^3, 10^4, 10^5, 10^6 (Wang et al., 2023) [150].

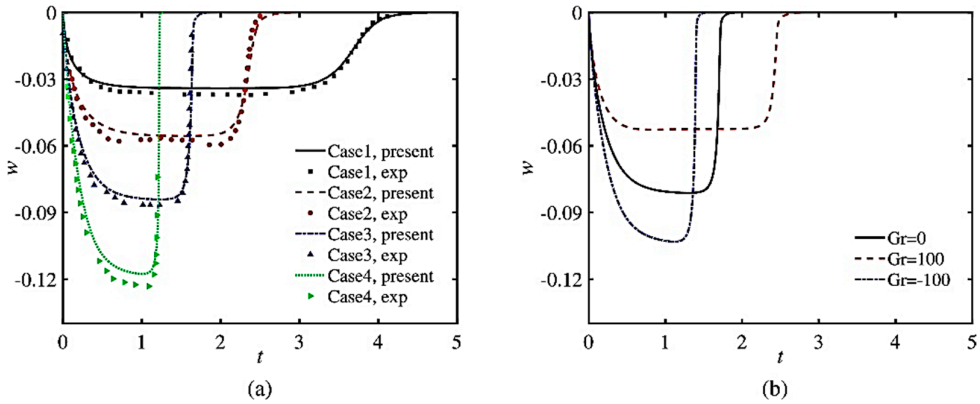


Figure 30. The settling velocities of a single particle sedimentation: (**a**) without heat transfer; (**b**) with heat transfer at different Grashof numbers (Wang et al., 2023) [150].

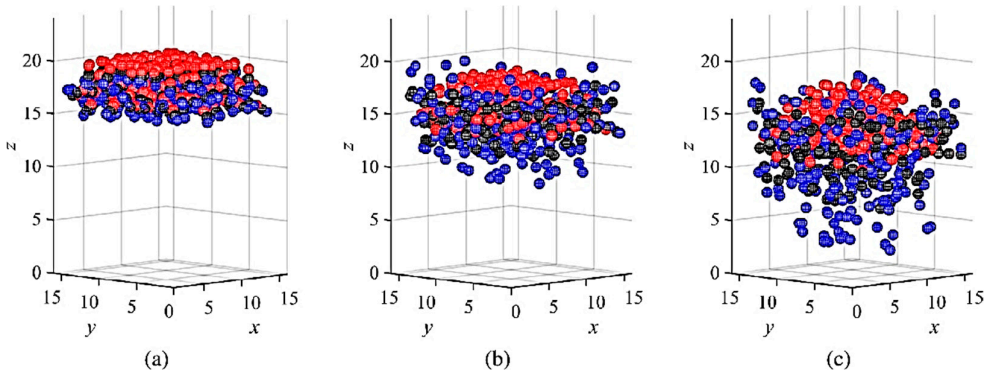


Figure 31. Particle locations at different times: (**a**) t = 10; (**b**) t = 20; (**c**) t = 30, where black, red, and blue indicate results for cases with Gr = 0, 100, and −100 (Wang et al., 2023) [150].

Advanced thermal-fluid–structure interaction (TFSI) problems have benefited from improved boundary condition treatments in diffuse IBM implementations. Wu et al. (2023) [148] developed an explicit IBM for TFSI with Neumann boundary conditions (heat flux), enabling precise control of thermal energy transfer at fluid–structure interfaces. Next, Wu et al. (2024) [152] further advanced the field by developing an implicit IBM for TFSI problems with Robin boundary conditions, incorporating combined convection and radiation effects in Figure 32. Their study demonstrated how IBM can bridge the gap between computational efficiency and physical accuracy in multiphysics simulations, as shown in Figure 33.

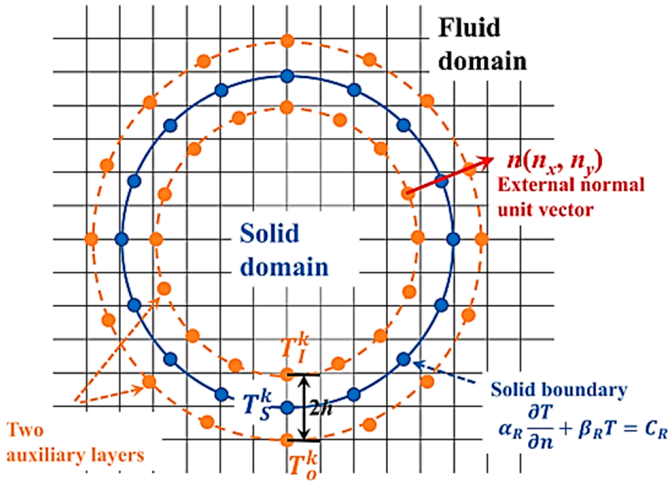


Figure 32. Illustration of the immersed boundary method for Robin boundary conditions. The two yellow circles denote the auxiliary layers, and the blue line represents the solid boundary. The Robin boundary condition is imposed on the solid boundary, and the Lagrangian points on these three layers are distributed along the external normal direction (Wu et al., 2024) [152].

Chen et al. (2020) [149] implemented a penalty-based diffuse IBM for flexible flags in heated channels, capturing the coupled fluid–structure–thermal dynamics responsible for heat transfer enhancement. These approaches demonstrate how diffuse IBM's boundary smoothing can be strategically employed to handle challenging multiphysics coupling without sacrificing solution accuracy. The finite element method (FEM) has contributed important diffuse IBM variants for thermal-flow problems. Haeri and Shrimpton (2013) [144] developed an implicit fictitious domain approach for flow and heat transfer past immersed objects, combining FEM's geometric flexibility with diffuse IBM's moving boundary capabilities. This implementation proved particularly effective for problems

requiring precise thermal boundary condition enforcement on complex geometries, where traditional methods might struggle with mesh generation.

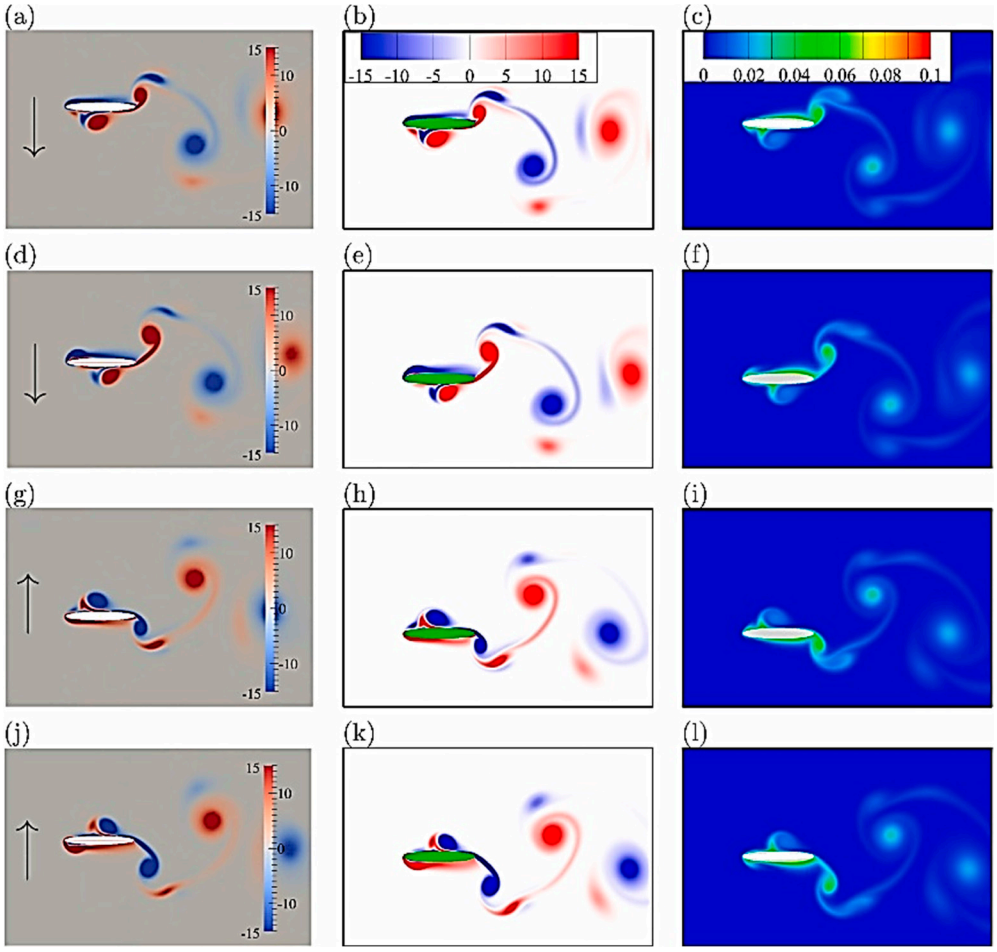


Figure 33. Comparisons between instantaneous vorticity field around a heaving airfoil provided by Martín-Alcántara et al. (left column) and the present results (middle column) at Re = 500, St_a = 0.16, and St_c = 0.5. The instantaneous isotherms around the heaving heated airfoil are presented in the right column. (**a**–**c**) $t/T_c = 1/8$, (**d**–**f**) $t/T_c = 1/4$, (**g**–**i**) $t/T_c = 5/8$, and (**j**–**l**) $t/T_c = 3/4$. The instantaneous vorticity contours obtained by the present method agree well with those of Martín-Alcántara et al., which qualitatively verify that the proposed method can accurately capture the flow structures around complex moving boundaries (Wu et al., 2024) [152].

Recent advances have focused on improving computational efficiency and physical fidelity in thermal diffuse IBM implementations. Abaszadeh et al. (2022) [141] extended diffuse IBM to radiative heat transfer in irregular geometries, demonstrating the method's potential for combined mode heat transfer problems. These developments continue to expand diffuse IBM's applicability to increasingly complex thermal-fluid phenomena. In particulate flow simulations, diffuse IBM has enabled new insights into suspension dynamics and heat transfer. Jiang et al.'s (2022) [142] fully resolved particle-laden flow simulations and Zhang et al.'s (2016) [151] thermal DEM-coupled approach have provided fundamental understanding of how particles affect heat transfer in fluid systems. These implementations leverage diffuse IBM's natural handling of moving boundaries to track particle motion and its thermal effects without the computational overhead of conforming mesh updates.

Finite volume methods (FVMs) have proven particularly effective for sharp IBM implementations in thermal systems, combining strong conservation properties with ge-

ometric flexibility. Pacheco et al. (2005) [153] established an early FVM-based sharp IBM framework for general heat transfer problems using non-staggered grids with rigorous Dirichlet/Neumann boundary condition enforcement. This approach was extended by Lou et al. (2020) [157] to membrane distillation systems incorporating Robin boundary conditions, demonstrating the method's versatility for complex thermal boundary conditions in desalination applications. For radiative heat transfer in irregular geometries, Mohammadi and Nassab (2021) [158] developed an FVM–IBM that accurately captured surface-to-surface radiation effects while maintaining sharp-interface treatment. Recent advances by Riahi et al. (2023) [160] and Ahn et al. (2023) [161] have further expanded FVM-based sharp IBM capabilities to compressible flows with heat transfer and phase-change problems involving melting/solidification, respectively, showcasing the method's adaptability across thermal regimes.

The coupling of sharp IBM with hybrid numerical frameworks has enabled innovative solutions for combined heat transfer problems. Mohammadi and Nassab (2021) [159] developed a novel LBM–FVM–IBM hybrid that leveraged the strengths of both methods for radiative–convective heat transfer simulations, with LBM handling the complex radiation physics and FVM providing robust convection treatment. This hybrid approach maintained sharp-interface resolution while efficiently handling the multiphysics nature of the problem. Similarly, Narváez et al. (2021) [164] implemented a dual IBM approach within the InCompact3D framework for turbulent heat transfer, demonstrating how multiple IBM strategies could be combined to address challenging conjugate heat transfer problems in turbulent flows.

Soti et al. (2015) [154] developed a strongly coupled FDM–FEM sharp IBM solver that accurately captured the interplay between flexible structures and convective heat transfer, enabling optimized design of thermal energy harvesters. Garg et al. (2018) [155] and Garg et al. (2020) [156] extended these capabilities to study thermal buoyancy effects on vortex-induced vibrations (VIVs) and wake-induced vibrations (WIVs), revealing how temperature gradients fundamentally alter bluff body dynamics and energy transfer mechanisms. These implementations demonstrate sharp IBM's ability to maintain solution fidelity in complex multiphysics environments where thermal, structural, and fluid dynamic effects are tightly coupled.

High-order accurate sharp IBM implementations have significantly advanced thermal flow simulations by reducing grid dependence while maintaining boundary precision. Xia et al. (2014) [168] developed a high-order ghost-cell IBM that reduced required grid points by approximately two-thirds for heat transfer problems while maintaining solution accuracy. Fernandez et al. (2011) [170] implemented a spectral method-based sharp IBM for grid-independent 3D heat conduction analysis, demonstrating exponential convergence for time-dependent boundary problems. These high-order approaches prove particularly valuable for thermal systems where boundary layer resolution is crucial, but computational efficiency remains a concern.

In reacting flows and combustion systems, sharp IBM has enabled precise treatment of thermal boundaries in complex geometries. Ou et al. (2022) [167] developed a directional ghost-cell IBM within their DINO framework that accurately handled thermal boundaries in low-Mach reacting flows, maintaining flame structure fidelity near immersed boundaries. This approach demonstrated how sharp IBM could be adapted to handle the unique challenges of reacting flows where both thermal and species boundaries require precise treatment. Similarly, Cruz and Lamballais (2023) [162] implemented a sharp IBM for high-fidelity conjugate heat transfer in pipe flows with irregular geometries, capturing the subtle interactions between flow physics and thermal transport in complex duct systems.

The treatment of multiphase heat transfer problems has particularly benefited from advanced sharp IBM implementations. Wang et al. (2022) [163] improved direct-forcing IBM for particle-laden flows with heat transfer, enabling accurate resolution of thermal interactions between discrete particles and carrier fluids. Tao et al. (2022) [169] extended these capabilities to discrete unified gas kinetic scheme (DUGKS) simulations, combining the advantages of kinetic theory with sharp-interface treatment for fluid–solid heat transfer problems. For industrial applications involving moving boundaries and transient heat transfer, sharp IBM has provided robust solutions. Shrivastava et al. (2013) [171] developed a level set-based IBM for transient CFD with moving boundaries that maintained sharp-interface treatment throughout complex motion sequences. Zhao and Yan (2022) [166] implemented an enriched IBM for interface-coupled multiphysics problems that improved accuracy for convective heat transfer applications. These approaches demonstrate sharp IBM's ability to handle the combined challenges of geometric complexity, boundary motion, and thermal transport that characterize many industrial thermal systems. Recent comparative studies have provided valuable insights into sharp IBM performance for thermal applications.

Ménez et al. (2023) [172] conducted a comparative study of volume penalization and IBM for compressible flows with thermal boundary conditions, illustrating the strengths of sharp-interface IBM in handling compressible heat transfer problems. As seen in Figure 34, the study investigated the accuracy and computational cost of both methods and extended the investigation to "three-dimensional supersonic flow configurations using the penalization method." The instantaneous vorticity structures, iso-surface, and isotherm along with a side view of flapping profile are shown in Figure 35.

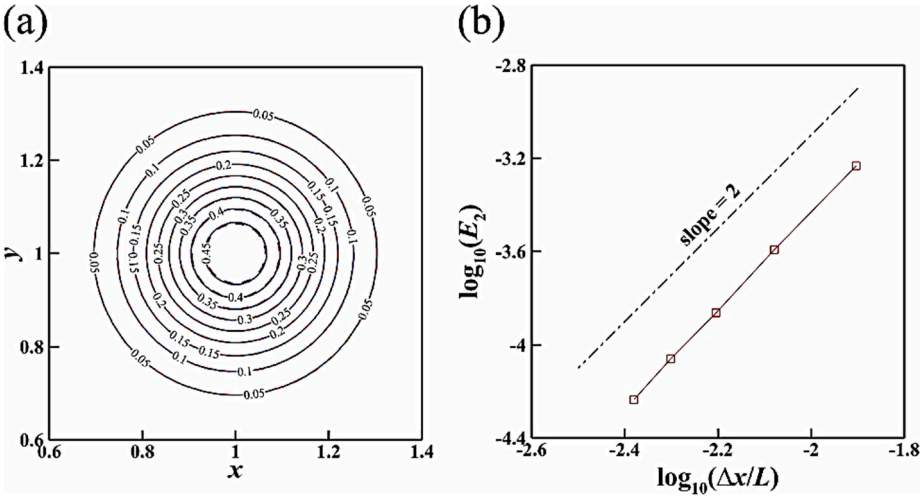


Figure 34. Numerical accuracy test of the proposed EIB-RTLBFS by the transient heat diffusion of a Gaussian hill. (**a**) Comparison of the isotherms of predicted results (red dashed line) and analytical solution (blue solid line) at $F_o = 0.005$. (**b**) Convergence of numerical error versus mesh spacing for unsteady heat diffusion of a Gaussian hill, indicating that the overall accuracy of the EIB-RTLBFS is of second order. The solidlines with symbols indicate results obtained at each interval and dashed line indicates a line having slope of 2. (Ménez et al., 2023) [172].

Their work demonstrated how sharp IBM implementations could maintain superior accuracy for thermal boundary treatment while providing computational efficiency advantages for certain classes of problems. The lattice Boltzmann method (LBM) has also contributed to sharp IBM advancements in thermal-fluid systems. Wu et al. (2022) [165] developed an explicit IB–RTLBFS method for thermal-fluid–structure interaction problems with Dirichlet boundary conditions, demonstrating how LBM's kinetic theory foundation

could be combined with sharp-interface treatment for efficient thermal simulations. This approach proved particularly effective for problems requiring frequent boundary updates or complex thermal-structural coupling.

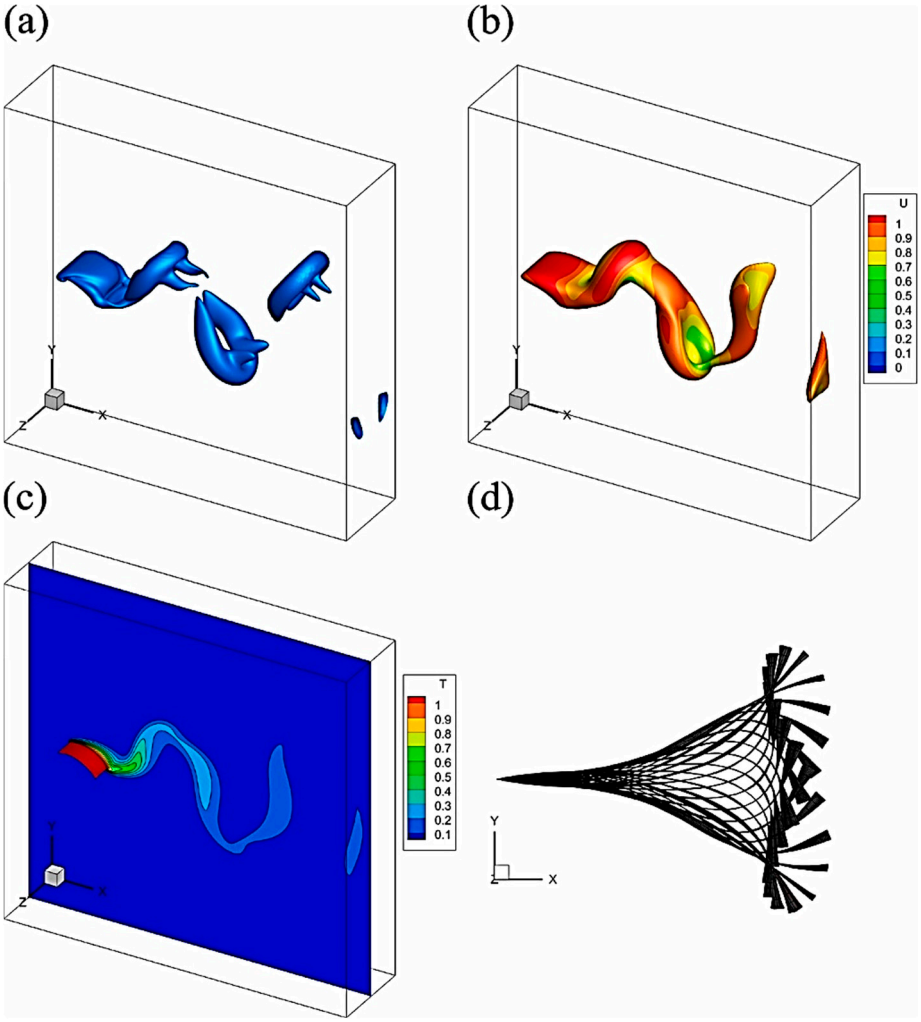


Figure 35. Instantaneous (**a**) vorticity structures (visualized by the iso-surface of Q = 0.2), (**b**) iso-surface of T = 0.1 colored by the streamwise velocity, and (**c**) isotherm on the z-mid plane for the heated flapping membrane structure at a singular moment and Re = 200; (**d**) side view of the flapping profile of the heated membrane structure (Ménez et al., 2023) [172].

4.2. Particle Sedimentation, Multiphase Flows, and Sound Acoustics

The topic of multiphysics applications encompasses areas such as particle sedimentation, multiphase flow, and sound acoustics. The particle sedimentation technique simulates the motion of solid particles dissolved in a fluid under the influence of gravity, dragging, lift, and fluid-induced forces into consideration. The IBM allows for efficient simulation of droplet dynamics, bubble formation, and fluid–structure interactions without requiring deformable meshes. In acoustics, it is used for studying aeroacoustics, wave scattering, and noise control, accurately capturing sound–fluid–structure interactions. This technique finds application in engineering, oceanography, and aerospace to improve the understanding of multiphase transport and the behavior of acoustic waves. Table 5 illustrates recent applications of IBM in multiphysics scenarios.

Table 5. Recent advancements in IBM considering boundary treatment and corresponding fluid solvers for multiphysics applications.

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Zhang et al., 2019 [173]	LBM (CLBM)	Diffuse	Coupled DEM–IB–CLBM framework for erosive particle impacts	Erosive particle flows	2019
Yang et al., 2019 [174]	FDM (DNS)	Diffuse	Eulerian method for optimal perturbations in particle-laden flows	Turbulent channel flows	2019
Wang et al., 2021 [175]	LBM	Diffuse	Polygonal DEM–LBM coupling with energy-conserving contact algorithm	Arbitrarily shaped particles	2021
Zhang et al., 2021 [176]	LBM	Diffuse	IB–LBM for coffee-ring formation in evaporating droplets	Droplet evaporation	2021
Romanus et al., 2021 [177]	LBM	Diffuse	Domain-transferring LBM–IBM for high-Re particle settling	Non-spherical particle dynamics	2021
Wang et al., 2022 [178]	LBM	Diffuse	IB–LBM for elliptical particle deposition in viscous flows	Particle deposition	2022
Fukui and Kawaguchi, 2022 [179]	LBM	Diffuse	Microscopic particle arrangement effects on suspension rheology	Narrow channel suspensions	2022
Cheng and Wachs, 2022 [180]	LBM (MRT)	Diffuse	Adaptive octree-grid IB–LBM for particle-resolved flows	Particle-laden flows	2022
Yadav and Ghosh, 2022 [181]	FDM	Diffuse	IBM for settling of permeable/impermeable planktonic particles	Biological suspensions	2022
Kawaguchi et al., 2022 [182]	LBM (RLBM)	Diffuse	Comparison of VFM and IBM for suspension flows	Suspension rheology	2022
Ghosh and Panghal, 2022 [183]	FDM	Diffuse	IBM for flexible circular particle settling	Flexible particle dynamics	2022
Chang et al., 2023 [184]	Other (PDDO)	Diffuse	Hybrid peridynamic–Eulerian–IBM for FSI	Fluid–structure interaction	2023
Panghal and Ghosh, 2023 [185]	FDM	Diffuse	IBM for flexible/permeable planktonic particle settling	Biological sedimentology	2023
Zhang et al., 2023 [186]	LBM	Diffuse	IB–LBM for nanofluid droplet freezing with particle expulsion	Nanofluid dynamics	2023
Yadav et al., 2023 [187]	FDM	Diffuse	IBM for semi-torus-shaped permeable particle settling	Permeable particle dynamics	2023
Ghosh et al., 2023 [188]	LBM	Diffuse	Stabilized IBM for light particles (density ratio $\geq$004)	Light particle suspensions	2023
Patel and Natarajan, 2018 [189]	FVM	Diffuse	Interpolation-free IBM for multiphase flows with moving bodies	Multiphase FSI	2018
Souza et al., 2022 [190]	FEM (LES)	Diffuse	AMR–VOF–IBM for turbulent multiphase FSI	Industrial multiphase flows	2022
Niu et al., 2022 [191]	LBM	Diffuse	Simple diffuse IB scheme for curved boundaries in multiphase flows	Binary/multiphase flows	2022
Wang et al., 2023 [192]	FDM	Diffuse	Energy-stable IBM for deformable membranes with non-uniform properties	Biological membranes	2023
Wang and Tian, 2019 [136]	FDM	Diffuse	IBM for flapping wing FSI and acoustics at Mach 01	Bio-inspired aerodynamics	2019
Wang and Tian, 2020 [135]	FEM	Diffuse	Sound generation by 3D flexible flapping wings during hovering	Bioacoustics	2020
Ye et al., 2020 [193]	FVM	Diffuse	Discrete-forcing IBM for ship hydrodynamics with VOF	Marine engineering	2020

Table 5. *Cont.*

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Bilbao, 2023 [194]	FDTD	Diffuse	1D IBM for impedance/acoustic barriers	Linear acoustics	2023
Bilbao, 2023 [195]	FDTD	Diffuse	3D IBM for irregular boundaries in wave-based acoustics	3D acoustic simulations	2023
Hou et al., 2023 [196]	FDTD	Diffuse	Time-domain IBM for acoustic propagation between gas bubbles	Bubble acoustics	2023
Jiang et al., 2022 [142]	LBM	Diffuse (BTDF–IBM)	Parallel FR–DNS for settling suspensions with lubrication model	Large-scale suspensions	2022
Nangia et al., 2019 [197]	FDM	Diffuse (DLM–IBM)	DLM–IBM for high-density ratio WSI with AMR	Wave-structure interaction	2019
Cheng et al., 2021 [198]	FDM	Diffuse	Semi-implicit IBM for viscous flow-induced sound	Computational aeroacoustics	2021
Zeng et al., 2022 [199]	Godunov scheme	Diffuse (DLM–IBM)	Adaptive DLM–IBM with subcycling for single/multiphase FSI	Adaptive FSI	2022
Yan et al., 2022 [200]	LBM (MLBFS)	Diffuse	IB–MLBFS with flux correction for multiphase FSI	Multiphase FSI	2022
Hori et al., 2022 [201]	SAC	Diffuse (implicit)	Eulerian-based IBM with implicit lubrication for particle suspensions	Dense suspensions	2022
Liu et al., 2017 [202]	DNS (FDM)	Diffuse (point-particle)	Combined DNS, IBM, and DPM for sediment particle distribution in turbulent boundary layers	Sediment transport	2017
Wang et al., 2017 [203]	FDM (WENO)	Diffuse	Compressible multiphase FSI with ANCF structural solver	Explosive/impact dynamics	2017
Cheng et al., 2017 [204]	FDM	Diffuse (smoothed IB)	Compressible IBM for flow-induced noise using influence matrices	Aeroacoustics	2017
Zhang et al., 2013 [205]	FDM	Diffuse (VOF-IBM)	Two-phase IB–VOF model for ocean engineering problems	Free-surface flows	2013
Bürchner et al., 2023 [206]	FEM (FCM)	Diffuse (γ-scaling)	FWI with γ-scaling IBM for crack detection in NDT	Non-destructive testing	2023
Fazli et al., 2023 [207]	FDM	Diffuse	IBM for yield-pseudoplastic particulate flows	Non-Newtonian suspensions	2023
Chéron et al., 2023 [208]	FVM	Hybrid (sharp–diffuse)	Hybrid IBM for dense particle-laden flows with non-symmetrical operators	Dense suspensions	2023
Ido et al., 2017 [209]	LBM	Sharp	Hybrid LBM–IBM–DEM for magnetic particle microstructures in MR fluids	Magnetorheological fluids	2017
Fukui et al., 2018 [210]	LBM	Sharp	Two-way coupling for particle rotation effects on suspension rheology	Suspension rheology	2018
Barbeau et al., 2022 [211]	FEM	Sharp	High-order FEM–IBM for flow around sphere packings	Fixed-bed reactors	2022
Wu and Chen, 2022 [212]	LBM	Sharp	3D IB–LBM for droplet-particle coating processes	Spray coating	2022
Duprez et al., 2023 [213]	FEM (ϕ-FEM)	Sharp	ϕ-FEM for Stokes flow around particles with optimal convergence	Creeping particle flows	2023
Farooq et al., 2025 [214]	QUICK	Sharp	Radial-basis function for incompressible flow	Bio-inspired moving bodies	2025
Zhao et al., 2021 [215]	FDM (DNS)	Sharp	Sharp-interface IBM–APE for flow-induced noise prediction	Cylinder array acoustics	2021

Table 5. *Cont.*

Reference	Fluid Solver	Boundary Treatment	Summary	Application Area	Year
Gorges et al., 2024 [216]	FVM	Sharp (blocked off)	Comparison of smooth/blocked-off IBM for fixed-bed reactors	Packed bed reactors	2024
Zhou and Balachandar, 2021 [217]	FDM	Sharp (direct forcing)	Analysis of spatiotemporal resolution in IBM with direct forcing	Rigid particulate flows	2021
Xie et al., 2020 [218]	FDM (DRP)	Sharp (ghost cell)	Cartesian grid method for acoustic scattering using ghost-cell IBM	Acoustic scattering	2020
Zhao et al., 2022 [219]	FDTD	Sharp (ghost node)	FDTD–IBM for underwater acoustic scattering	Underwater acoustics	2022
Isoz et al., 2022 [220]	FVM	Sharp (hybrid)	Hybrid fictitious domain–IBM for irregular particle flows	Industrial particle flows	2022
Qin et al., 2022 [221]	FDM	Sharp	Hybrid IBM for particle–complex boundary interactions	Particulate flows	2022
Yu et al., 2023 [127]	FDM/WENO	Sharp (IB–GHPC)	Two-phase wave tank with GHPC pressure solver	Coastal engineering	2023
Giahi and Bergstrom, 2023 [222]	FVM	Sharp (IBS)	Validation of IBS–IBM for fixed/moving bodies	General FSI	2023
Zeng et al., 2021 [223]	FVM	Sharp (multi-sphere)	Lagrangian IBM for proppant transport in hydraulic fractures	Oil/gas industry	2021
Caunt et al., 2023 [224]	FDM	Sharp (Taylor series)	IBM for seismic/acoustic wave propagation over irregular terrain	Geophysics	2023

Zhang et al. (2019) [173] developed a coupled DEM–IB–CLBM framework for erosive particle impacts that accurately captured both particle trajectories and surface wear mechanisms. This approach was extended by Wang et al. (2021) [175] through their polygonal DEM–LBM coupling with an energy-conserving contact algorithm, enabling simulation of arbitrarily shaped particles in viscous flows. For biological suspensions, Yadav and Ghosh (2022) [181] and Panghal and Ghosh (2023) [185] implemented diffuse IBM to study the settling dynamics of permeable planktonic particles, revealing how microscale features influence macroscopic suspension behavior. These implementations demonstrate how diffuse IBM's ability to handle complex particle geometries and interactions makes it indispensable for suspension rheology studies, as further evidenced by Kawaguchi et al.'s (2022) [182] comparative analysis of IBM and VFM for suspension flows.

Wang et al. (2022) [178] developed an IB–LBM framework for elliptical particle deposition in viscous flows, capturing the complex interplay between particle shape and deposition patterns that influence industrial coating processes. This work complemented Fukui and Kawaguchi's (2022) [179] investigation of microscopic particle arrangement effects on suspension rheology in narrow channels, where diffuse IBM's ability to resolve near-wall particle interactions proved crucial for predicting bulk flow properties.

Lattice Boltzmann methods (LBMs) have shown remarkable synergy with diffuse IBMs for particulate and multiphase flow applications, benefiting from their kinetic theory foundation and natural parallelism. Romanus et al. (2021) [177] developed a domain-transferring LBM–IBM for high-Reynolds-number particle settling that maintained accuracy while improving computational efficiency. Cheng and Wachs (2022) [180] implemented an adaptive octree-grid IB–LBM for particle-resolved flows that automatically refined resolution near particles while coarsening elsewhere. Recent advances by Ghosh et al. (2023) [188] produced a stabilized IBM for light particle suspensions (density ratio $\geq$0.04), addressing a longstanding challenge in buoyant particle simulations. These LBM-based

approaches leverage diffuse IBM's ability to handle complex moving boundaries without remeshing—a crucial advantage for systems with numerous interacting particles.

For biological and flexible particle systems, Ghosh and Panghal (2022) [183] implemented a diffuse IBM to study flexible circular particle settling dynamics, revealing deformation mechanisms that affect sedimentation rates in viscoelastic fluids. This framework was extended by Yadav et al. (2023) [187] to examine semi-torus-shaped permeable particle settling, demonstrating how diffuse IBM can handle both geometric complexity and permeability effects in biological sedimentology applications. These studies collectively advanced understanding of non-spherical particle dynamics in viscous flows, with implications for industrial separations and environmental processes.

In multiphase flow simulations, Patel and Natarajan (2018) [189] developed an interpolation-free IBM that robustly handled moving bodies in multiphase environments, while Niu et al. (2022) [191] presented a simple and effective method for simulating multiphase flows with curved boundaries called the "diffuse-interface immersed-boundary scheme." A schematic plot of a droplet's impact on a circular cylinder is shown in Figure 36, and its corresponding shape evolution and velocity field at different time points are shown in Figure 37.

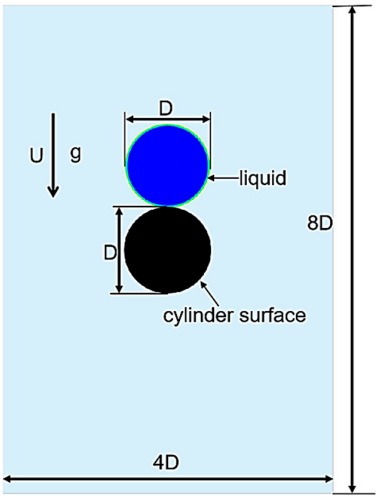


Figure 36. Schematic plot of a droplet's impact on a circular cylinder (Niu et al., 2022) [191].

These implementations addressed critical challenges in maintaining interface sharpness while accommodating complex boundary motion. The method's acoustic applications were advanced by Cheng et al. (2021) [198], who developed a semi-implicit IBM for viscous flow-induced sound generation in computational aeroacoustics. In studies by Hou et al. (2023) [196], a simulation of the entire acoustic field propagation utilizing an immersed-boundary method was carried out, as shown in Figure 38. The study analyzed linear resonance responses for each bubble concerning interior gas velocity. Furthermore, the study delved into a frequency domain analysis of the coupled interactions between the bubbles. Notably, the method accommodated scenarios where two bubbles of different sizes and shapes were in contact with each other, providing insights applicable to underwater scattering targets (Figure 39).

Advanced adaptive implementations have pushed diffuse IBM capabilities further. Zeng et al. (2022) [199] created an adaptive DLM–IBM with subcycling that efficiently handled single and multiphase FSI problems through dynamic resolution control. Yan et al. (2022) [200] (appearing twice in source data) developed an IB–MLBFS with flux correction that improved accuracy for multiphase FSI simulations. These adaptive approaches demonstrate how diffuse IBM continues to evolve to meet the demands of increasingly

complex simulations. Hori et al. (2022) [201] contributed a Eulerian-based IBM with implicit lubrication for dense suspensions, addressing the challenging regime of closely interacting particles.

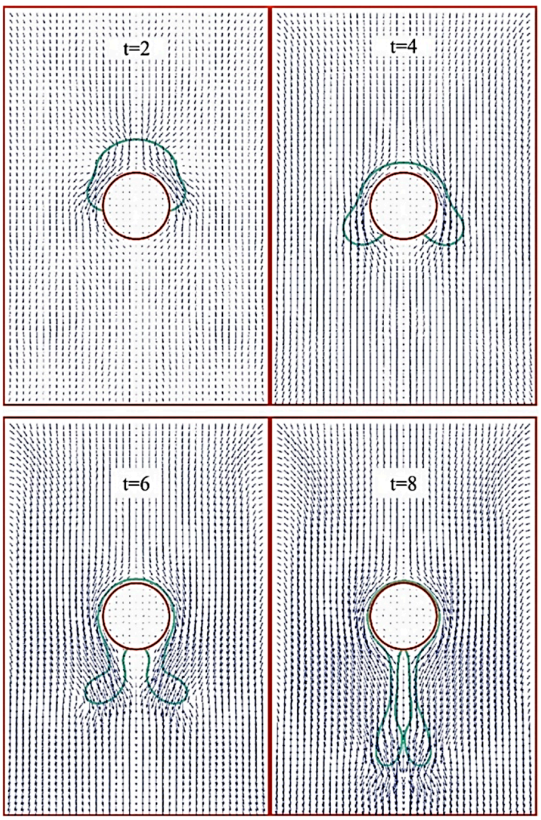


Figure 37. Shape evolution and velocity field for a droplet impacting on a hydrophilic cylinder surface with θ = 60°, Re = 700, We = 10 (Niu et al., 2022) [191].

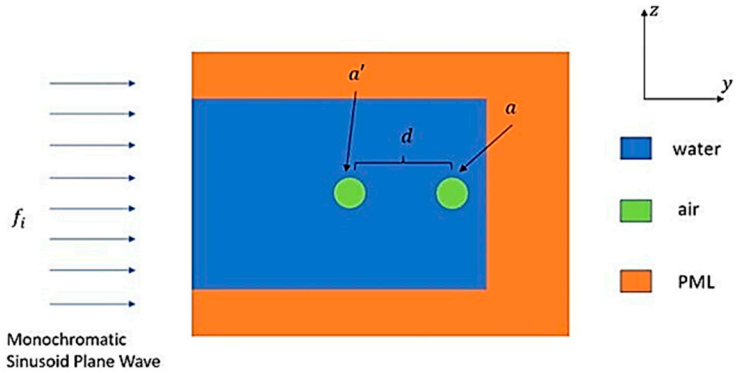


Figure 38. Computational domain for plane waves incident on two spherical bubbles (side view) (Hou et al., 2023) [196].

For multiphase flows with moving interfaces, diffuse IBM has enabled new insights into interfacial phenomena and phase-change processes. Zhang et al. (2021) [176] studied coffee-ring formation in evaporating droplets using IB–LBM, capturing both fluid dynamics and deposition patterns. Zhang et al. (2023) [186] extended this to nanofluid droplet freezing with particle expulsion, demonstrating how diffuse IBM can handle solidification fronts and particle interactions simultaneously. In industrial multiphase applications, Souza et al. (2022) [190] combined adaptive mesh refinement (AMR), volume of fluid (VOF), and IBM for turbulent multiphase FSI, showcasing the method's scalability for

complex industrial flows. These implementations highlight diffuse IBM's versatility in handling multiple interacting physics while maintaining computational tractability.

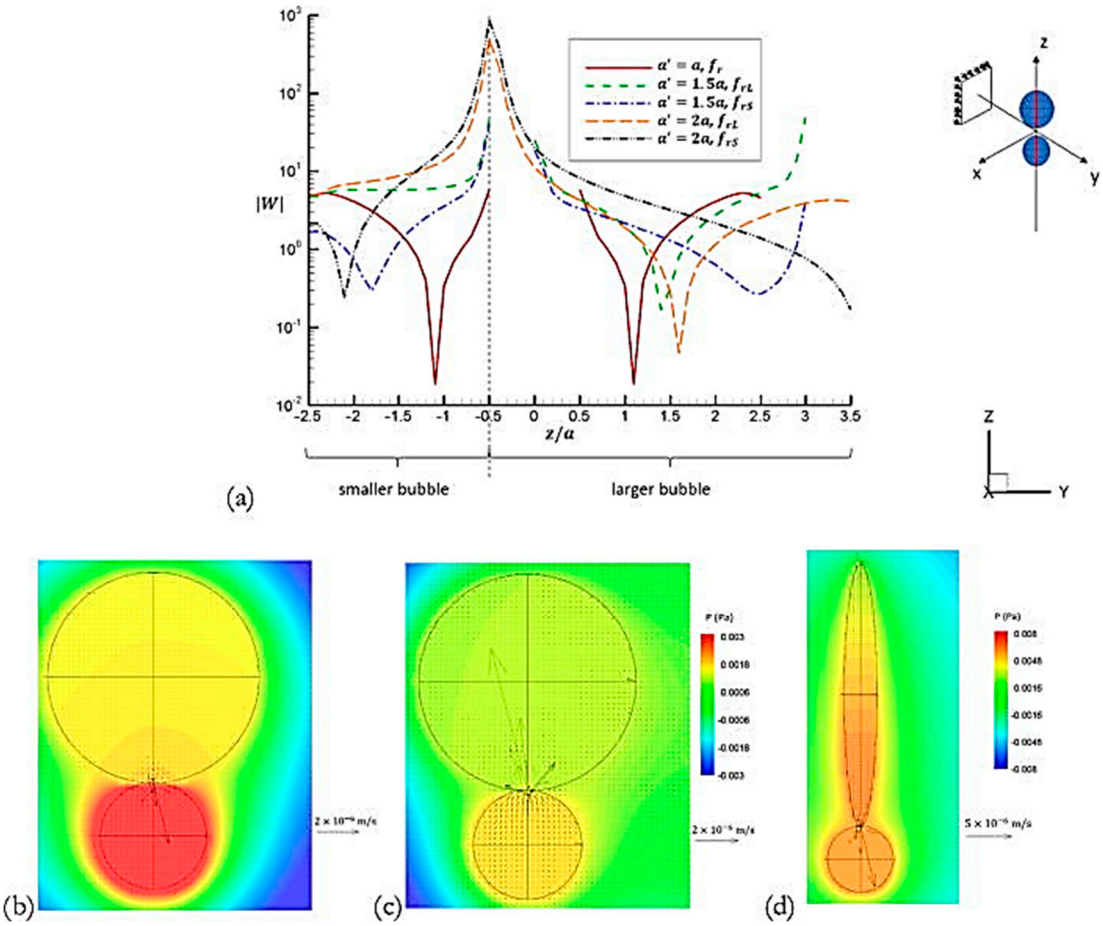


Figure 39. Interior velocity and pressure. Larger bubbles of different radii and a smaller bubble of fixed size (f_i = 2437 Hz, P_0 = 106 Pa, d = 3a = 0.03 m. (**a**) *z*-axis. (**b**) Bubbles in contact (case 7, compression). (**c**) Case 7, expansion. (**d**) Case 8, spherical and prolate (expansion) (Hou et al., 2023) [196].

Bio-inspired applications have driven significant innovations in diffuse IB methodology. Wang and Tian (2019) [136] and Wang and Tian (2020) [135] developed FDM- and FEM-based diffuse IBM frameworks for flapping wing aerodynamics and bioacoustics, revealing how wing flexibility influences both thrust generation and sound production. Wang et al. (2023) [192] created an energy-stable IBM for deformable biological membranes with non-uniform properties, addressing the challenge of simulating heterogeneous elastic structures. These studies demonstrate how diffuse IBM's ability to handle large deformations and complex material properties makes it invaluable for biomechanical and bio-inspired system studies.

In marine and coastal engineering, diffuse IBM implementations have advanced simulation capabilities for free-surface flows and wave–structure interactions. Ye et al. (2020) [193] developed a discrete-forcing IBM with VOF for ship hydrodynamics that accurately captured wave patterns and hull interactions. Zhang et al. (2013) [205] created a two-phase IB–VOF model specifically for ocean engineering problems, demonstrating robust performance in challenging free-surface conditions. Nangia et al. (2019) [197] implemented a distributed Lagrange multiplier (DLM) IBM with AMR for wave–structure interaction (WSI) problems, enabling efficient simulation of high-density ratio systems.

These applications showcase diffuse IBM's ability to handle the combined challenges of free surfaces, moving structures, and turbulent flows that characterize marine environments.

Acoustic applications have benefited from specialized diffuse IBM formulations. Bilbao (2023) [194] and Bilbao (2023) [195] developed 1D and 3D diffuse IBM frameworks for acoustic wave propagation, enabling efficient simulation of sound barriers and irregular boundaries. Cheng et al. (2017) [204] created a compressible IBM for flow-induced noise using influence matrices, while Wang et al. (2020) [24] implemented a high-order IBM for coupled fluid–structure–acoustics in flapping foils. These approaches demonstrate how diffuse IBM can be adapted to handle the unique requirements of aeroacoustic and computational acoustics problems, where precise wave propagation and boundary interactions are crucial.

The finite difference method (FDM) has served as a versatile foundation for many diffuse IBM implementations. Yang et al. (2019) [174] employed FDM-based DNS with diffuse IBM to study optimal perturbations in particle-laden turbulent channel flows. Liu et al. (2017) [202] combined DNS, IBM, and discrete particle methods (DPMs) for sediment transport studies, capturing particle distribution in turbulent boundary layers. For non-Newtonian suspensions, Fazli et al. (2023) [207] implemented an FDM-based diffuse IBM for yield-pseudoplastic particulate flows, demonstrating the method's adaptability to complex rheology. The methodology was first validated using Newtonian benchmark cases, after which it was extended to simulate yield-pseudoplastic cases with stationary or moving particles, as illustrated in Figures 40–42. The extent of yield is shown in blue and unyielded region is shown in red.

Advanced computational techniques have expanded diffuse IBM capabilities for challenging multiscale problems. Jiang et al. (2022) [142] developed a parallel friction-resolved DNS (FR-DNS) with lubrication model for large-scale suspension simulations. Hori et al. (2022) [201] created a Eulerian-based implicit IBM with lubrication treatment for dense suspensions. Chang et al. (2023) [184] implemented a hybrid peridynamic–Eulerian–IBM for FSI that combined the strengths of multiple numerical approaches. These innovative implementations demonstrate how diffuse IBM continues to evolve to address increasingly complex physical systems across scales. In non-traditional application domains, diffuse IBM has shown surprising versatility. The method's adaptability was further highlighted by Bürchner et al. (2023) [206], who employed a γ-scaling IBM for full-waveform inversion in non-destructive testing, enabling high-resolution crack detection through computational wavefield analysis. The study introduced a dimensionless scaling function, γ, inspired by fictitious domain methods (Figure 43), to model void regions in the scalar wave equation problem. Three formulations for mono-parameter FWI and one for two-parameter FWI were developed using γ, as shown Figures 44 and 45.

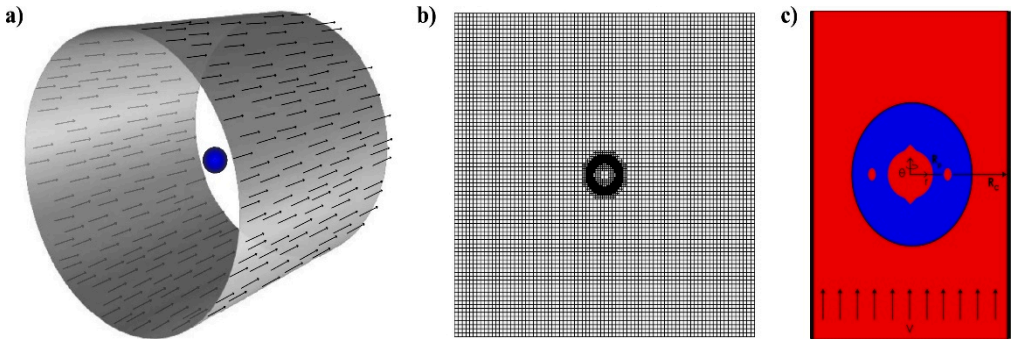


Figure 40. (**a**) Computational domain, (**b**) grid resolution, (**c**) and extent of yielded (blue) and unyielded (red) regions around solid objects (Fazli et al., 2023) [207].

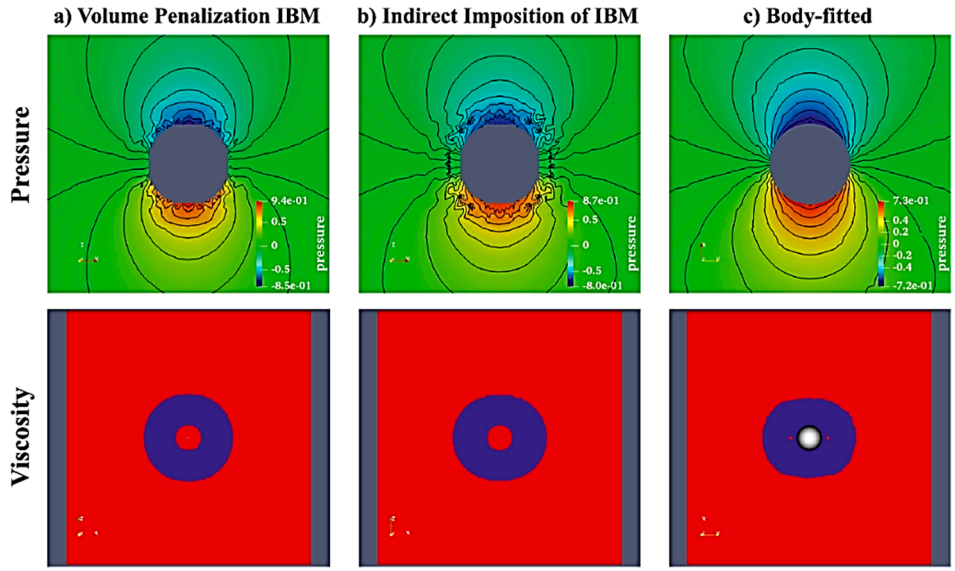


Figure 41. Pressure and viscosity fields for flow over a fixed sphere at Re′ = 0.002956 and Bi* = 0.747 (Fazli et al., 2023) [207].

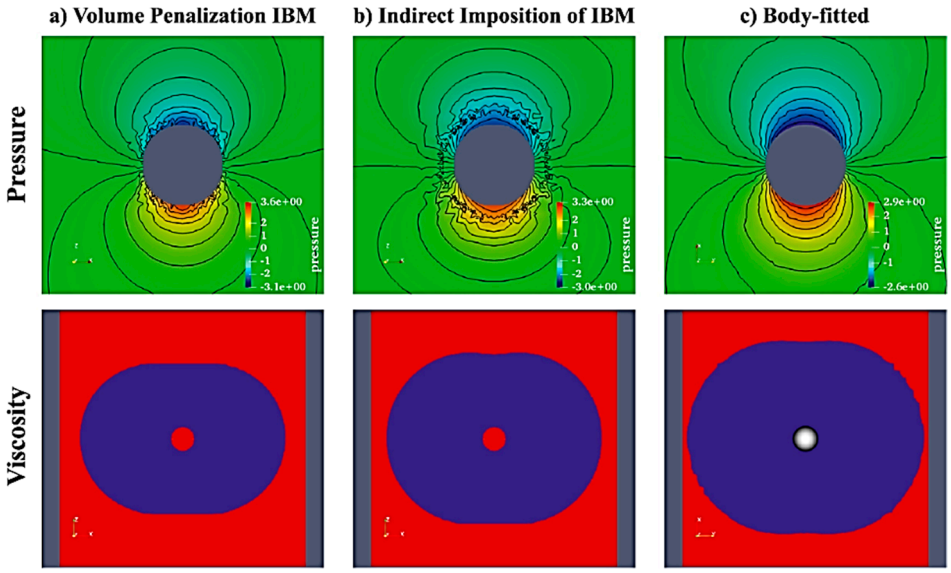


Figure 42. Pressure and viscosity fields for flow over a fixed sphere at Re′ = 0.97 and Bi* = 0.108 (Fazli et al., 2023) [207].

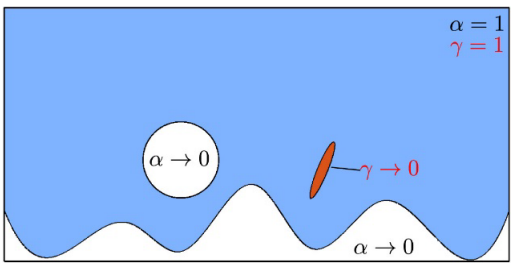


Figure 43. A priori known geometry incorporated by the indicator function α with an unknown void reconstructed by the scaling function γ. Blue: physical domain, red: unknown void domain, white: fictitious domain (Bürchner et al., 2023) [206].

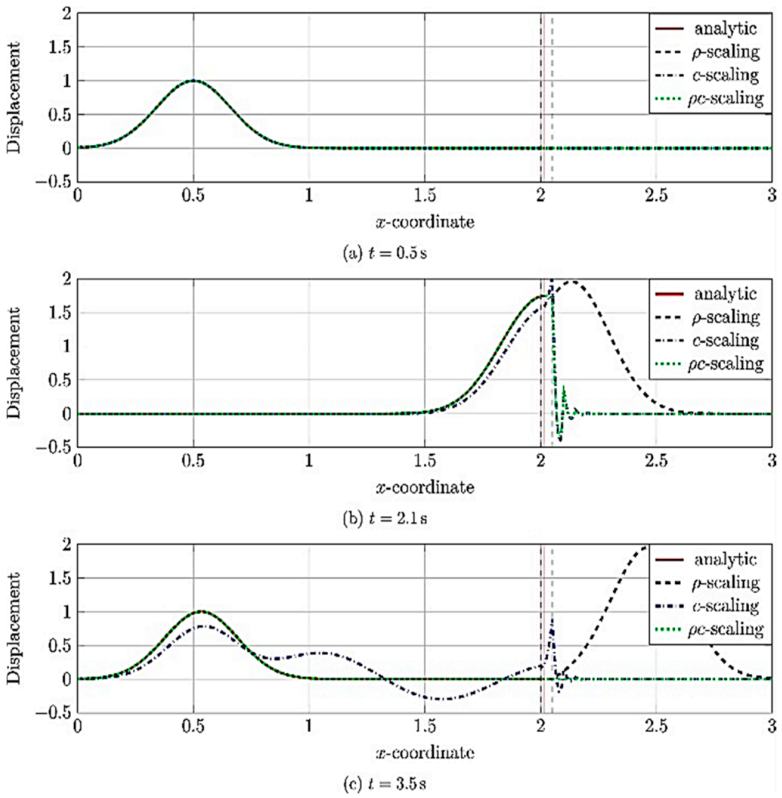


Figure 44. 1D interface problem $p = 2$. The dashed black vertical lines visualize the boundaries of the cut element and the red vertical line the material interface (Bürchner et al., 2023) [206].

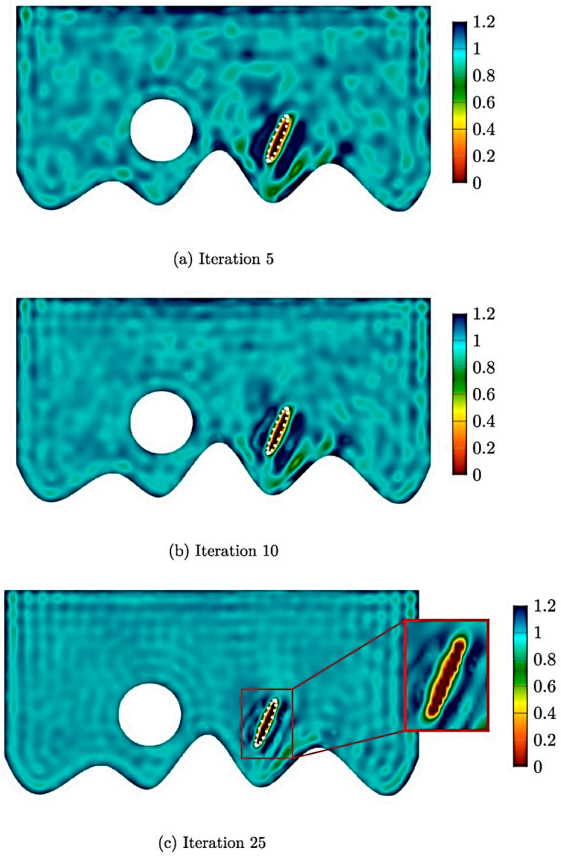


Figure 45. ρ-scaling: reconstructed scaling function γ for an elliptical void. The dotted white ellipse shows the position and size of the real elliptical void (Bürchner et al., 2023) [206].

Wang et al. (2017) [203] developed a compressible multiphase FSI framework with an ANCF structural solver for explosive and impact dynamics, showcasing diffuse IBM's ability to handle extreme deformation scenarios. These unconventional applications highlight the method's fundamental flexibility and robustness.

In dense particulate flows, sharp IBM implementations have provided unprecedented insights into suspension dynamics and industrial processes. Chéron et al. (2023) [208] developed a hybrid sharp–diffuse IBM (HyBM) for dense particle-laden flows that combined the accuracy of sharp interfaces with the robustness of diffuse methods through non-symmetrical operators. The method utilized a direct-forcing formulation and a regularization of the transfer function between Eulerian grid points and Lagrangian markers to satisfy the no-slip condition at particle surfaces, as shown in Figure 46. The instantaneous velocity field of the random arrays of monodispersed spheres for different numerical resolutions are provided in Figure 47.

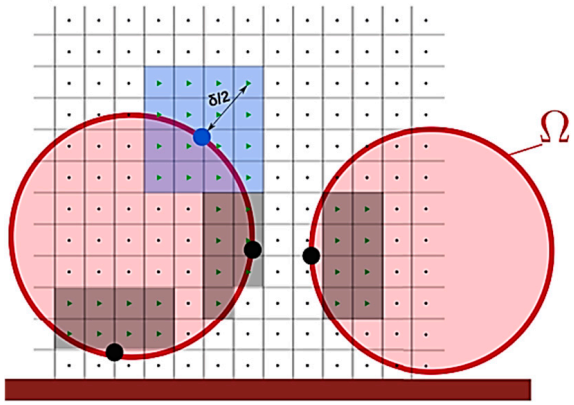


Figure 46. Two IBM particles near a wall over a Cartesian uniform collocated grid (Eulerian framework). Lagrangian markers are shown with filled circles. The non-modified and modified interpolation operator's support at the position of the Lagrangian markers are shown in blue and black, respectively (Chéron et al., 2023) [208].

For magnetorheological fluids, Ido et al. (2017) [209] implemented a hybrid LBM–IBM–DEM framework that captured the formation of magnetic particle microstructures under external fields, demonstrating sharp IBM's ability to handle complex multiphysics coupling. These capabilities were further extended to industrial applications by Zeng et al. (2021) [223], who developed a Lagrangian IBM for proppant transport in hydraulic fractures, addressing critical challenges in oil/gas recovery processes.

The lattice Boltzmann method (LBM) has shown remarkable synergy with sharp IBM for particulate and coating applications. Fukui et al. (2018) [210] established a two-way coupling framework for studying particle rotation effects on suspension rheology, revealing how microscopic dynamics influence macroscopic flow properties. Wu and Chen (2022) [212] advanced these capabilities with a 3D IB–LBM for droplet-particle coating processes that accurately resolved the complex interfacial dynamics in spray coating applications. These implementations leverage LBM's inherent parallelizability while maintaining sharp IBM's precise boundary treatment, enabling large-scale simulations of industrial particulate systems.

Finite element methods (FEMs) have contributed accurate high-order sharp IBM implementations for challenging particulate flows. Barbeau et al. (2022) [211] developed a high-order FEM–IBM for flow around sphere packings in fixed-bed reactors, demonstrating spectral convergence for these geometrically complex systems. Duprez et al. (2023) [213] implemented a ϕ-FEM approach for Stokes flow around particles that achieved optimal

convergence rates in creeping flow regimes. These FEM-based approaches demonstrate how sharp IBM can be combined with advanced spatial discretization to handle particulate systems with complex geometries and tight tolerances for numerical accuracy.

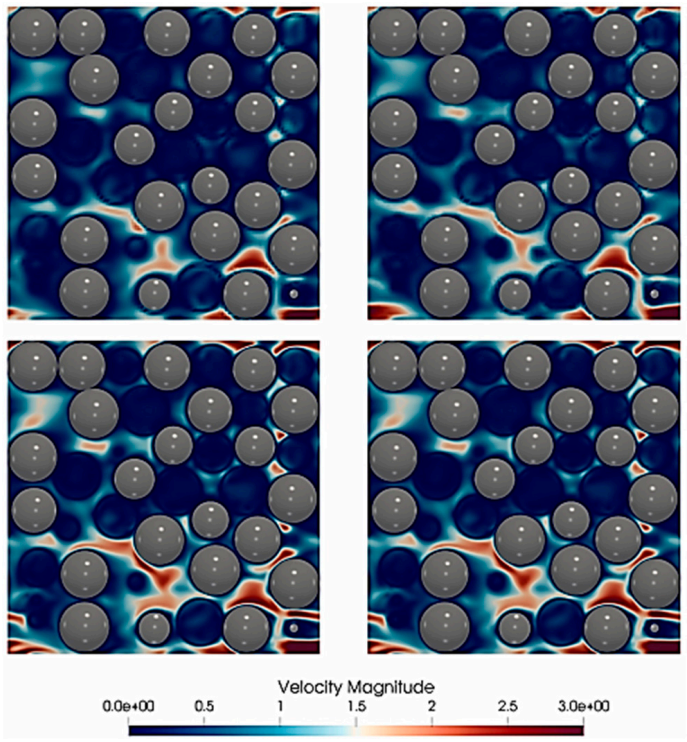


Figure 47. Instantaneous velocity field of the random arrays of monodispersed spheres at solid volume fraction 50% at t = 1.5 s for the standard MLS–IBM and HyBM (left and right, respectively) for two numerical resolutions, $Dp/\Delta x = 16$ and $Dp/\Delta x = 32$ (top and bottom row, respectively) (Chéron et al., 2023) [208].

Acoustic applications have driven significant innovations in sharp IB methodology. Zhao et al. (2021) [215] developed a sharp-interface IBM–APE framework for flow-induced noise prediction in cylinder arrays, capturing both the fluid dynamics and acoustic propagation with high fidelity. Xie et al. (2020) [218] created a Cartesian grid method for acoustic scattering using ghost-cell IBM that maintained accuracy while simplifying grid generation. These were complemented by Zhao et al.'s (2022) [219] FDTD–IBM for underwater acoustic scattering, which demonstrated sharp IBM's versatility across different acoustic modeling paradigms. These implementations showcase how sharp IBM's precise boundary treatment is crucial for accurate wave propagation and scattering simulations.

Industrial applications have motivated the development of robust sharp IBM implementations. Gorges et al. (2024) [216] conducted a comprehensive comparison of smooth and blocked-off IBM for fixed-bed reactors, providing guidelines for method selection in industrial packed bed simulations. The comparison was conducted at particle Reynolds numbers of 300 and 500, with "inline particle image velocimetry (PIV)" measurements serving as a basis for evaluation (Figure 48). Both IBM approaches exhibited excellent agreement with the experimental results, especially at higher resolutions, capturing the more stable flow at a particle Reynolds number of 300. However, differences between the IBM approaches arose in the more unsteady flow at a particle Reynolds number of 500. Contour plots of the y-component of the mean velocity vector field and qualitative cutout of the simulation mesh for the smooth IBM are illustrated in Figure 49 and Figure 50, respectively.

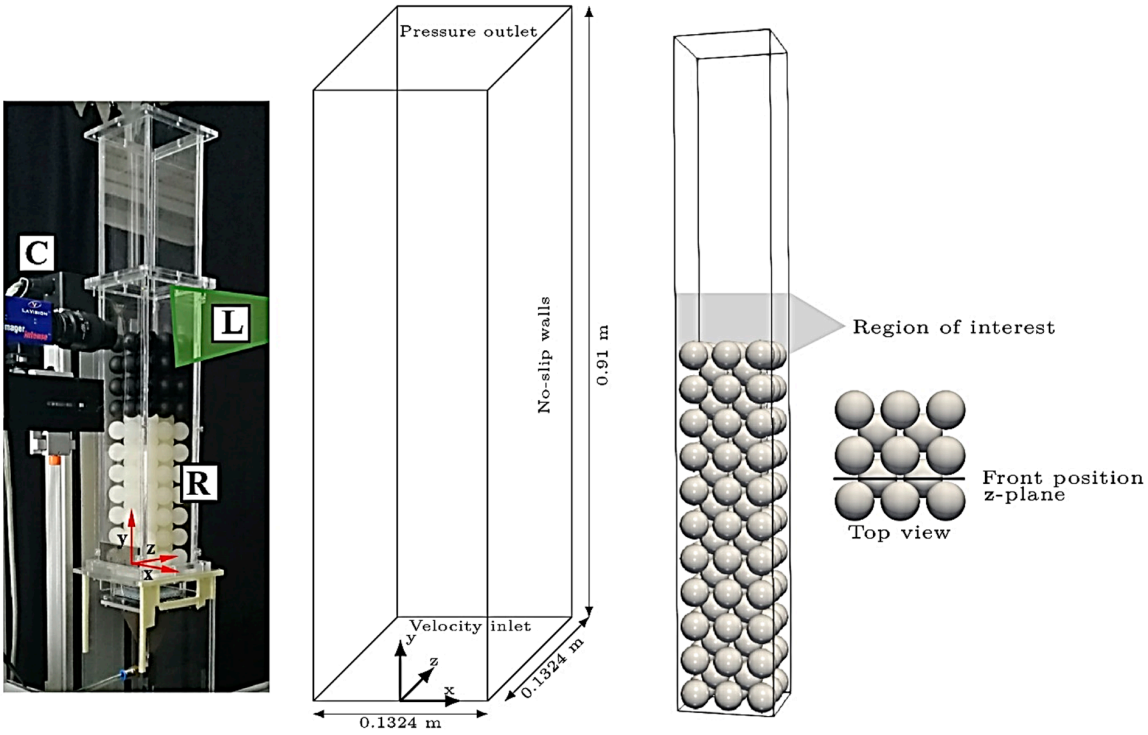


Figure 48. General experimental and numerical setup of the fixed packed bed reactor with 139 spherical particles in body-centered cubic packing (BCC). For the experiments, the packing inside the bulk reactor (R) is illuminated by a laser light sheet (L) and the measurement signal is recorded by the camera (C) (Gorges et al., 2024) [216].

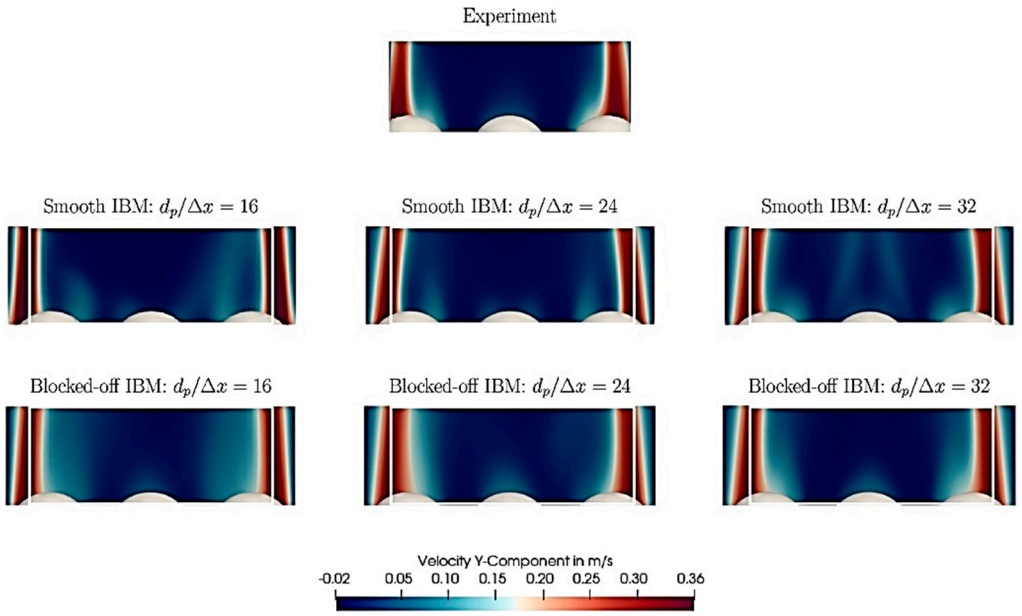


Figure 49. Contour plots of the y-component of the mean velocity vector field in the z = 0.04309 m plane and a height of y = 0.495 m to y = 0.54 m for Re_p = 300 for three different $d_p/\Delta x$ ratios. The white box in the plots of the smooth and blocked-off IBM simulation results corresponds to the measurement area in the experimental results (Gorges et al., 2024) [216].

Isoz et al. (2022) [220] developed a hybrid fictitious domain–IBM for irregular particle flows commonly encountered in industrial processes. These implementations address the dual challenges of geometric complexity and computational efficiency that characterize many industrial applications.

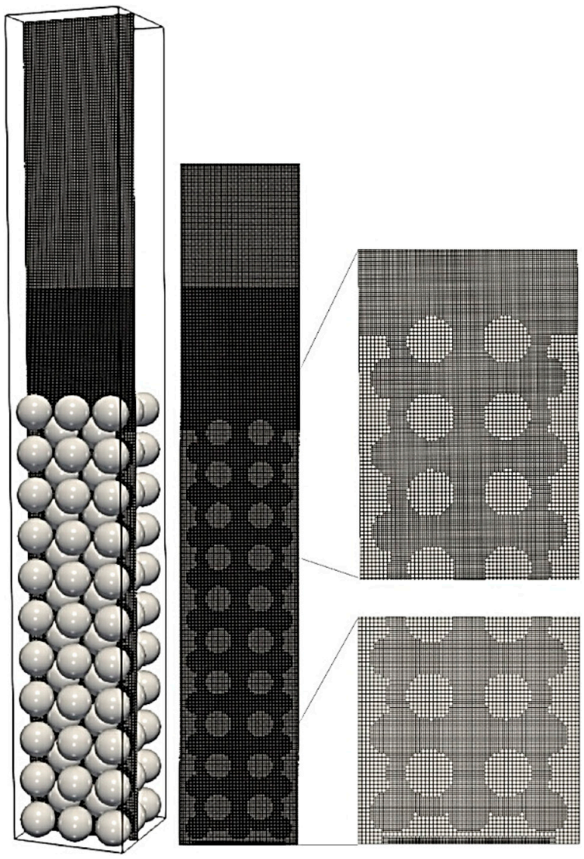


Figure 50. Qualitative cutout of the simulation mesh for the smooth IBM for the z-plane situated at z = 0.04309 m for illustration of the refined mesh for the front-position measuring plane. The bottom zoomed area shows the refined mesh at the velocity-inlet at the bottom of the reactor and the second zoomed area shows the refined mesh around the last layers of particles and the beginning of the region of interest (Gorges et al., 2024) [216].

The finite volume method (FVM) has proven particularly effective for industrial-scale sharp IBM implementations. Giahi and Bergstrom (2023) [222] aimed to verify and validate the immersed boundary methods (IBMs) implemented in FOAM-Extend open-source toolbox versions 4.0 and 4.1. The study comprised five test cases of increasing complexity, wherein simulation results were compared with experimental and numerical data from the literature (Figures 51–53).

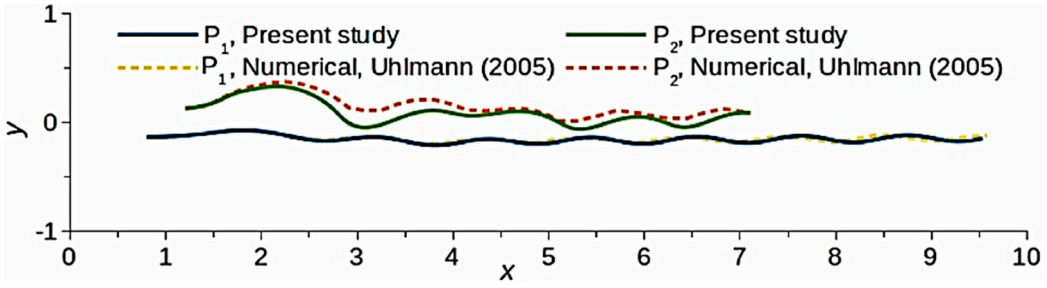


Figure 51. Particle trajectories for two falling particles. The gravity vector acts in the positive x-direction, and thus particles are moving from left to right. P1 is the heavier particle (Giahi and Bergstrom, 2023) [222]. The numerical data is adopted from Uhlmann [225].

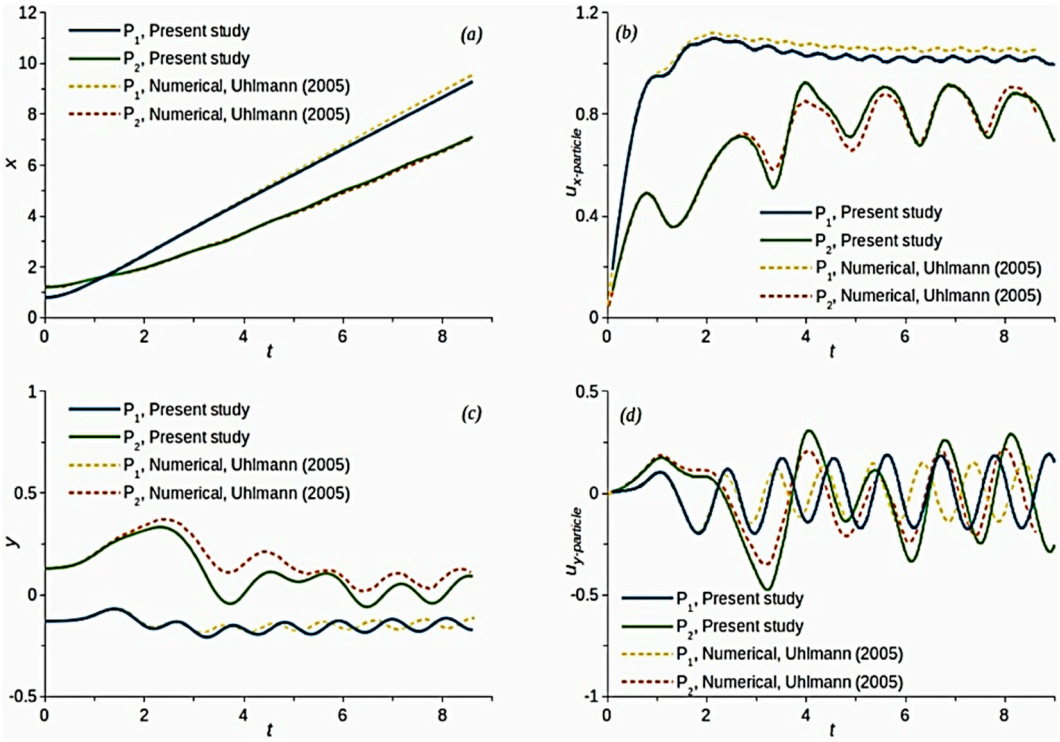


Figure 52. Position and velocity of two falling particles: (**a**) particle x location, (**b**) particle x--component of velocity, (**c**) particle y location, and (**d**) particle y-component of velocity (Giahi and Bergstrom, 2023) [222]. The numerical data is adopted from Uhlmann [225].

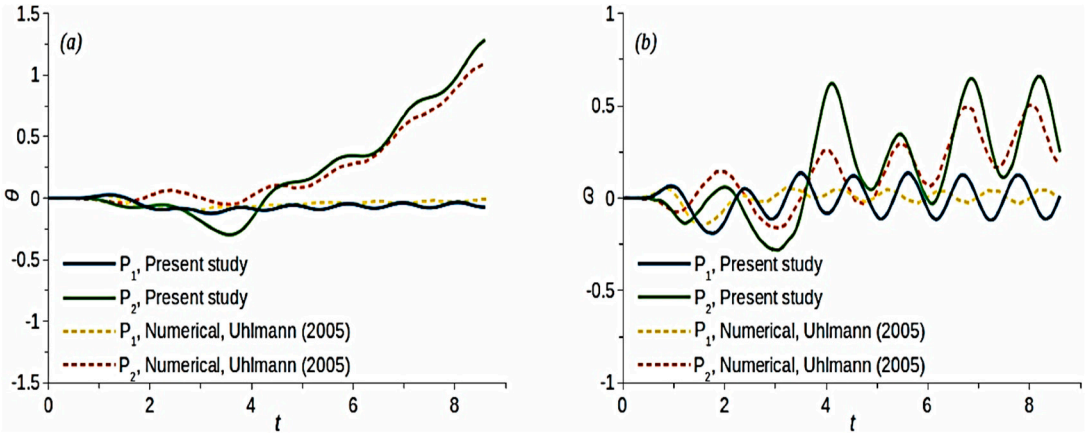


Figure 53. (**a**) Angular position and (**b**) angular velocity of the two falling particles (Giahi and Bergstrom, 2023) [222]. The numerical data is adopted from Uhlmann [225].

Zhou and Balachandar (2021) [217] analyzed spatiotemporal resolution requirements in IBM with direct forcing, providing fundamental insights for particulate flow simulations. These studies establish best practices for applying sharp IBM to industrial fluid–particle systems.

Advanced hybrid approaches have expanded sharp IBM's capabilities for complex boundary interactions. Qin et al. (2022) [221] developed a hybrid IBM for particle–complex boundary interactions that maintained accuracy while handling intricate geometric features. Farooq et al. (2025) [214] implemented a radial-basis function approach within a QUICK scheme for bio-inspired moving bodies, demonstrating sharp IBM's adaptability to different numerical frameworks. These hybrid implementations combine the strengths of multiple numerical techniques to address challenging fluid–structure interaction problems.

In coastal and geophysical applications, sharp IBM has enabled new simulation capabilities. Moreover, the method's success in handling free-surface flows, as demonstrated by Yu et al. (2023) [127] in their two-phase wave tank simulations, underscores its flexibility in coastal and offshore engineering applications (Figures 54–56).

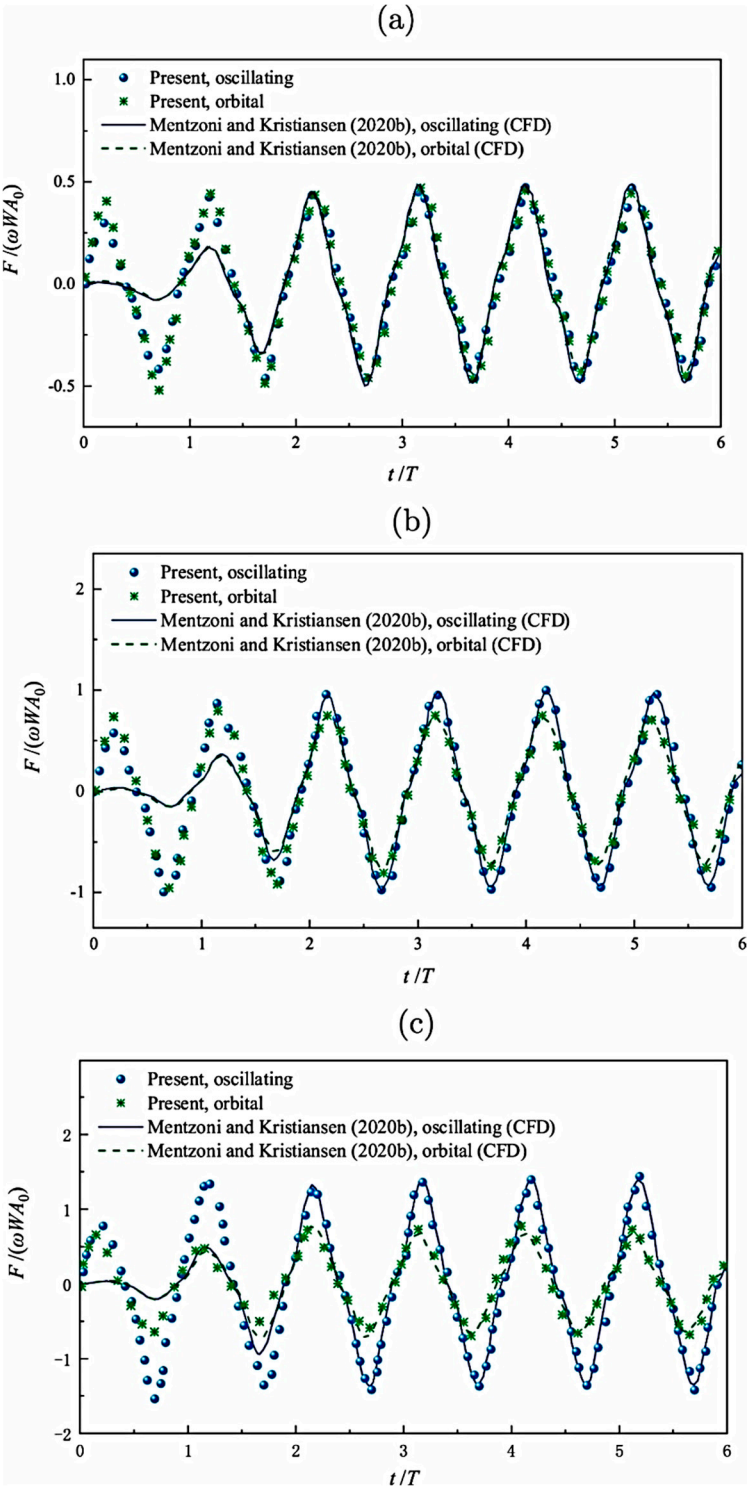


Figure 54. Normalized force on the plate during the first six oscillation cycles of oscillating in oscillating and orbital flows. (**a**) KC = 0.5. (**b**) KC = 1.5. (**c**) KC = 2.5. Present, oscillating: present results for oscillating flow. Present, orbital: present results for orbital flow (Yu et al., 2023) [127]. CFD results for orbital flow simulated by Mentzoni and Kristiansen [226].

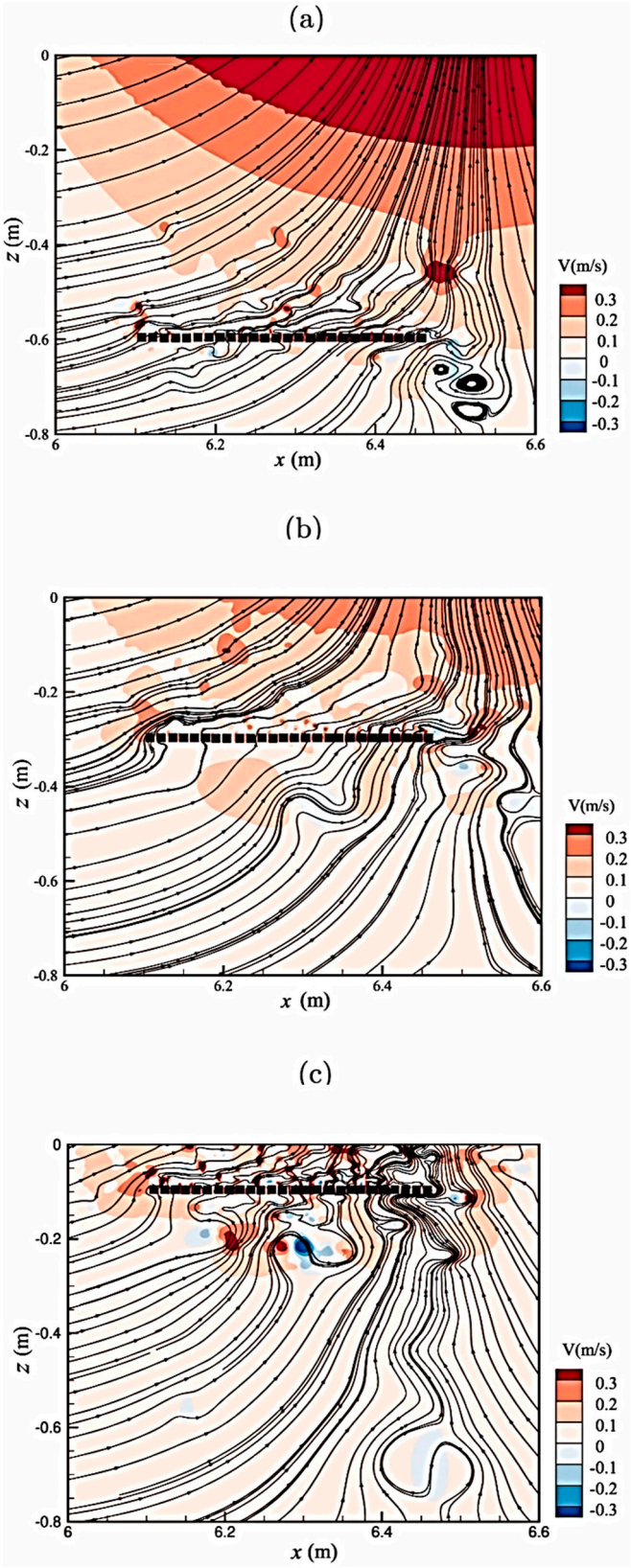


Figure 55. Zoomed-in plot of the streamlines close to the solid part of the perforated plate underneath the free surface. The mean water surface is at Z = 0. The plot is a snapshot at t/T = 11.54 (t = 15 s). Other considered parameters are KC = 0.50 and kD = 0.87 (T = 1.3 s). (**a**) Z/D = 1.67 (Z = −0.6 m); (**b**) Z/D = 0.83 (Z = −0.3 m); (**c**) Z/D = 0.28 (Z = −0.1 m) (Yu et al., 2023) [127].

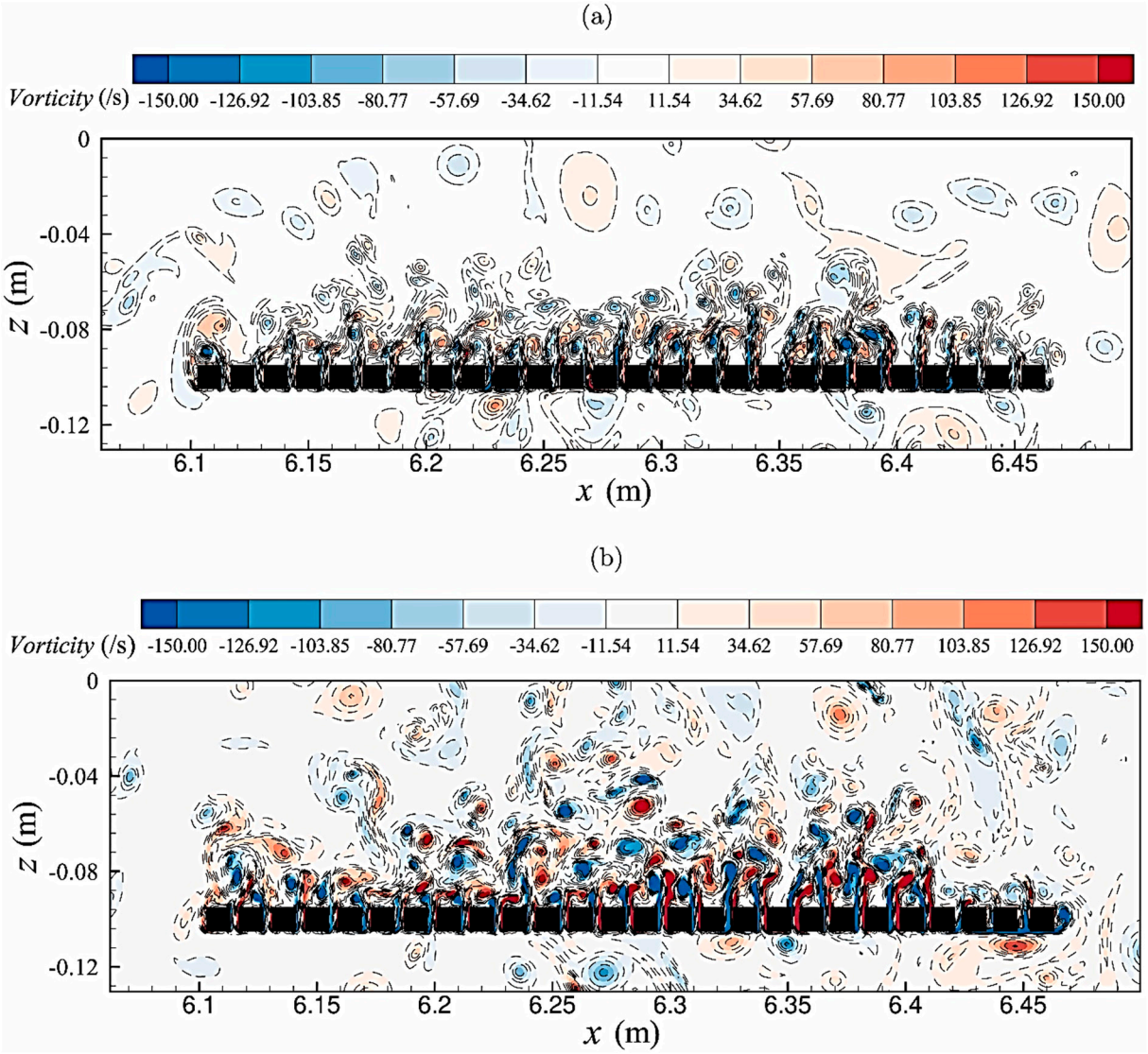


Figure 56. Zoomed-in plot of the vorticity field close to the solid part of the perforated plate underneath the free surface. (**a**) KC = 0.29. (**b**) KC = 0.50. The mean water surface is at Z = 0. The plot is a snapshot at t/T = 11.54 (15 s). Other considered parameters are non-dimensional wave number kD = 0.87 (T = 1.3 s) and non-dimensional submergence Z/D = 0.28 (Z = −0.1 m) (Yu et al., 2023) [127].

Caunt et al. (2023) [224] implemented a Taylor-series based IBM for seismic/acoustic wave propagation over irregular terrain, demonstrating the method's potential for geophysical simulations. These applications showcase sharp IBM's versatility across different physical scales and regimes.

Theoretical foundations and comparative studies have played a crucial role in advancing sharp IB methodology. Haeri and Shrimpton (2012) [227] provided a comprehensive review of IBM and fictitious domain methods for particulate flows, establishing benchmarks for method selection and implementation. This theoretical framework informs contemporary developments in sharp IBM, ensuring rigorous treatment of fundamental numerical aspects.

4.3. Selection of Fluid Solver and Boundary Condition for Multiphysics Applications

The choice of numerical solver for diffuse IBM implementations in thermal-fluid systems reflects careful consideration of physical requirements and computational constraints.

LBM approaches, as demonstrated by Mazharmanesh et al. (2020) [139] and Tong et al. (2020) [140], excel in parallel scalability and complex boundary handling for particulate and energy systems. LBM approaches also excel for particulate and multiphase systems requiring efficient handling of complex boundaries (Zhang et al., 2019; Cheng and Wachs, 2022) [173,180]. FDM implementation offers simplicity and efficiency for fundamental studies of heat transfer enhancement. FEM methods provide geometric flexibility for complex conjugate heat transfer problems, exemplified by Haeri and Shrimpton (2013) [144] and Wang and Tian (2020) [135]. Each framework brings unique advantages that diffuse IBM enhances through its boundary treatment flexibility. Emerging applications continue to push diffuse IBM capabilities in new directions. The method's extension to radiative heat transfer by Abaszadeh et al. (2022) [141] and to rarefied gas flows by Wang et al. (2023) [150] demonstrates its adaptability to diverse physical regimes. Coupling with DEM for particle-laden flows, as shown by Zhang et al. (2016) [151], illustrates how diffuse IBM can integrate with other numerical techniques to handle multiphysics challenges. These cutting-edge applications suggest a bright future for diffuse IBM in increasingly complex thermal-fluid systems.

The selection of numerical solvers for sharp-interface immersed boundary methods (IBM) in multiphysics applications requires careful consideration of accuracy, efficiency, and physical fidelity, with different solvers offering distinct advantages for specific applications. Lattice Boltzmann methods (LBMs) excel in particulate flows due to their natural handling of complex boundaries and parallelizability (Fukui et al., 2018; Wu and Chen, 2022) [210,212], while finite difference methods (FDMs) and finite-difference time-domain (FDTD) approaches are preferred for acoustic applications requiring precise wave propagation (Zhao et al., 2021; Zhao et al., 2022) [215,219]. Finite element methods (FEMs) provide geometric flexibility for industrial packed-bed reactors (Barbeau et al., 2022) [211], and finite volume methods (FVMs) remain crucial for robust conservation in industrial applications (Giahi and Bergstrom, 2023) [222]. Emerging hybrid approaches like Chéron et al.'s (2023) [208] HyBM and Farooq et al.'s (2025) [214] radial-basis–QUICK combination strategically blend solver strengths to address limitations of individual methods, guided by resolution analyses like those of Zhou and Balachandar (2021) [217]. Current trends show increasing specialization based on dominant physics, scale requirements, and interface complexity, with comparative studies (Haeri and Shrimpton, 2012) [227] informing a pragmatic "horses for courses" approach that matches solvers to specific application needs while employing hybrid methods to bridge gaps, ensuring sharp IBM remains effective for increasingly complex multi-physics challenges.

5. Evolution of IBM

The immersed boundary (IB) method has seen extensive application across multiple domains, including biological systems, vortex-induced vibration (VIV), fluid–structure interactions (FSIs), and engineering simulations. Since 2007, notable advancements have emerged, beginning with the formulation of an adaptive IB scheme for cardiac mechanics, which exhibited second-order accuracy in simulating blood flow around heart valves. In the ensuing years, research concentrated on vesicle–fluid interactions, bacterial flagellar dynamics, and biofilm behavior, facilitating applications in modeling stress and deformation in biological entities, such as eye blast overpressure effects and plankton settling in aquatic ecosystems. Between 2010 and 2012, refinements to IB methodologies were made to investigate hair–fluid interactions employing penalty IB techniques (modified version of immersed boundary method that considers mass of boundary), alongside extensions to model flexible structures in maritime applications. From 2014 to 2017, innovative IB-based FSI solvers were developed for compressible fluid dynamics, thin fluid layers, and vortex

interactions, thereby enhancing the understanding of the impact of hydrodynamic forces on vibration amplitudes and energy harvesting mechanisms in structures such as hydrofoils and flags. From 2018 to 2020, IB methods were adeptly adapted for wave–structure interactions, underwater vehicle dynamics, and coastal infrastructure applications, with investigations illustrating their capability to simulate flow-induced vibrations in wind turbines and facilitate energy extraction from wave-induced motions through model predictive control (MPC) techniques. By 2021–2023, research had pivoted towards hybrid IB models that integrated lattice Boltzmann methods and artificial intelligence-driven methodologies to augment numerical precision, turbulence representation, and complex solid–fluid interactions. Recent developments in 2024 focused on IB applications pertaining to highly flexible structures, real-time engineering simulations, and fluid–structure interaction challenges involving intricate geometries. Throughout these advancements, significant innovations have encompassed adaptive mesh refinement, high-order accuracy methodologies, and novel IB strategies aimed at addressing real-world challenges across biological, mechanical, and marine engineering domains.

6. Future Scope and Concluding Remarks

Immersed boundary methods (IBMs) have emerged as indispensable computational tools across diverse fields, with their selection—sharp or diffuse interface—dictated by application-specific demands. In biological fluid–structure interaction (FSI), sharp-interface IBM excels in capturing precise boundary conditions for vocal fold dynamics, cardiovascular flows, and cellular mechanics, while diffuse-interface methods prove advantageous for complex geometries, large deformations, and multi-scale phenomena, despite ongoing challenges in turbulent and high-Reynolds-number flows. For vortex-induced vibrations (VIVs) and flexible body interactions, sharp-interface IBM enables high-fidelity analysis of wake dynamics and industrial applications, whereas diffuse methods offer robustness in handling bio-inspired propulsion and renewable energy systems. In heat transfer, sharp-interface techniques provide accuracy for conjugate, radiative, and multiphase thermal problems, while diffuse methods facilitate simulations of moving boundaries and microscale systems. The integration of IBM with emerging technologies—such as machine learning for adaptive boundary treatments, high-performance computing for large-scale systems, and multiphysics coupling for energy and biomedical applications—highlights its evolving potential. As computational capabilities advance, both sharp and diffuse IBM are poised to address increasingly complex challenges, from turbulent flows and non-Newtonian suspensions to aeroacoustics and microelectronics cooling, solidifying their role as transformative tools in computational science and engineering.

The latest studies indicate several solid strategies that can enhance immersed boundary (IB) frameworks to outperform their current caliber. One potentially effective direction is the incorporation of IB methods with physics-informed neural networks (PINNs), as established by Farea et al. (2025) [228], who proposed an adaptive B-spline PINN architecture for fluid–structure interaction challenges. Their approach, which mitigates mesh deformation whilst resolving complicated boundary behavior, allows accurate modeling of dynamic interfaces such as bioinspired motion and vortex shedding. For high-Reynolds-number flows, Zuber et al. (2025) [229] successfully integrated wall-modeled large-eddy simulation (WMLES) with an immersed boundary solver to simulate full-scale rotorcraft aerodynamics, obtaining realistic vortex dynamics utilizing GPU acceleration. This evaluates the use of hybrid IB–turbulence models in aerospace and energy applications where wall-resolved LES is impractical. Simultaneously, Gopakumar et al. (2025) [230] introduced a conformal prediction framework for uncertainty quantification in neural PDE solvers, which can be expanded to IB–PINN formulations to facilitate statistically sound error bounds without ne-

cessitating labeled data—specifically valuable in environmental and biomedical modeling. Lastly, learning-based interfacial flux modeling is rising as a robust and versatile substitute to physics-based closures. The features-embedded learning immersed boundary (FEL-IB) model (Zhou et al., 2025) [231] makes use of neural networks to estimate interfacial momentum exchange, providing enhanced fidelity in multiphase turbulence and thermofluid challenges. Collectively, these techniques mark a shift toward intelligent, scalable, and reliable IB simulations grounded in data-driven and hybrid computational physics. A more systematic discussion of the above can also be found in our previous review (Powar et al., 2025) [232].

The potential for advancing immersed boundary (IB) frameworks lies in addressing key challenges and exploring emerging opportunities. Future research could focus on creating more efficient numerical techniques to handle large-scale simulations involving intricate geometries and dynamically evolving interfaces. Enhancing the precision of these methods for accurately capturing complex behaviors, such as turbulent flows, deformable structures, and multiphase systems, is another critical area of exploration. The integration of IB approaches with machine learning and data-driven strategies could lead to real-time simulation capabilities and predictive analytics. Furthermore, leveraging advances in high-performance computing, such as quantum and exascale platforms, can significantly improve the scalability and resolution of simulations, expanding the scope of applications in fields like energy, medicine, and environmental science.

Alongside methodological and computational advancements, the experimental validation and reproducibility of IB methods remain inadequately addressed, yet are indispensable for the community. Amidst the rising interest in IBM across multiple disciplines, a significant share of the literature is devoid of rigorous grid-independence studies, open-source code repositories, or validation against experimental benchmarks. These exclusions curb the ability to verify results, establish standardized comparisons across different modeling frameworks, and reproduce studies. Future work should primarily focus on the inclusion of these reproducibility metrics and ensure transparency in model development and simulation protocols. Doing so would influentially improve the robustness, credibility, and translational potential of IB research.

Building on the foundational discussions of earlier work, this article has examined the latest progress and implementations of IB methods across disciplines, including biology, VIV, heat transfer, particle interactions, multi-fluid systems, and acoustic phenomena. By addressing theoretical advancements and practical challenges, it highlights the evolving role of IB methodologies in solving sophisticated interaction problems. As the field continues to innovate and expand, IB techniques are poised to play a pivotal role in advancing computational tools and addressing critical challenges across a broad spectrum of scientific and engineering domains.

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